

Masters

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Computer graphics (cs602) Final term mcqs fall 2013 “Libriansmine”

Question # 1 Total Marks: 1

Consider the following problem from lighting: A point (P1) is at (0, 0, 0) with normal equal to $1/(2*\sqrt{2})*(\sqrt{2}, 2, \sqrt{2})$. The light is located at (0, 1, 0). The viewer is located at (1, 1, 1). The specular and diffuse coefficients for the light are both (1, 1, 1), with the ambient portion as (.2, .2, .2). Complete the reflection vector. [-----] $r = [0] [1/\sqrt{2}]$ Choose from the values below:

Select correct option:

0

1

-1

1/sqrt(2) ✓

Question # 2 Total Marks: 1

Dark lights are nothing more than lights in which one or more of the color values are_____.

Select correct option:

Unknown

Negative ✓ pg 232

Null

Positive

Question # 3 Total Marks: 1

The Phong reflection model simplifies light-matter interactions essentially into 4 vectors and a number of constants. Which piece of the Phong model is responsible for giving spheres their bright white spots?

Select correct option:

Specular ✓ page 289

Diffuse

Ambient

Ambient and Diffuse

Question # 4 Total Marks: 1

We want our scene to look more realistic, we should use _____ lights.

Select correct option:

Point

Parallel

Spot

None of the given ✓

Question # 5 Total Marks: 1

This is a simple example of line clipping: the display window is the canvas and also the default -----, thus all line segments inside the canvas are drawn.

Select correct option:

✓ **Clipping Rectangle** page 244

Clipping Circle

Clipping Polygon

Clipping Angle

Question # 6 Total Marks: 1

One problem with Gouraud shading is that the ----- intensities can never be greater than the intensities at the edges.

Select correct option:

✓ **Triangles** page 243

Squares

Rectangles

Polygons

Question # 7 Total Marks: 1

There is more penetration of light in case of _____ surfaces.

Select correct option:

Conductor (like metals)

Nonconductor (like dielectrics) ✓ page 237

Both conductor and nonconductor

None of the given

Question # 8 Total Marks: 1

_____ lights should be avoided because they are not for real time environment.

Select correct option:

Point

Parallel

✓ **Spot** page 241

None of the given

Question # 9 Total Marks: 1

The physical range of colours a device can display is called _____ of the device.

Select correct option:

Sharpness

✓ **Gamut** page 226

Colouring

Colouring with Sharpness

Question # 10 Total Marks: 1

A polygon is usually defined by a sequence of vertices and -----

Select correct option:

✓ **Edges** page 245

Vertices

Points

Ending lines

Question # 11 Total Marks: 1

Which of the following affine transforms does NOT affect vectors?

Scale ✓

Rotation

Shear

Translation

Question # 12 Total Marks: 1

_____ is simply the calculation of color reflected by the surface.

Select correct option:

Shading ✓ page 237

Clamping

Scaling

None of the given

Question # 13 Total Marks: 1

This projection technique has the direction of projection perpendicular to the viewing plane, and the viewing direction is perpendicular to one of the principle faces.

Select correct option:

Axonometric Parallel Projection

Oblique Parallel Projection

✓ **Orthographic Parallel Projection** page 191

Oblique Projection

Question # 14 Total Marks: 1

When obtaining normals for a triangle, which of the following mathematical constructs is NOT used?

Select correct option:

Vector normalization

Vector cross products

Vector dot products ✓

Point-Point subtraction

Question # 15 Total Marks: 1

Loosely, the alpha component of the RGBA quad represents the _____ of a surface.

Select correct option:

✓ **Opacity** page 224

Light

Darkness

Shine

Question # 16 Total Marks: 1

Which is the oldest in given types of the shading?

Select correct option:

Flat Shading ✓

Phong Shading

Gouraud Shading

None of all

Question # 17 Total Marks: 1

The _____ technique has the direction of projection perpendicular to the viewing plane, but the viewing direction is NOT perpendicular to one of the principle faces.

Select correct option:

Oblique Parallel Projection

Axometric Parallel Projection ✓

Orthographic Parallel Projection

Oblique Projection

Question # 18 Total Marks: 1

The ColorSpace tool is a handy tool that we can use to interactively add two colours together to see the effects of the various strategies for handling oversaturated colours.

Select correct option:

True ✓

False

Question # 19 Total Marks: 1

An algorithm that clips a polygon must deal with many ----- cases. The case is particularly note worthy in that the concave polygon is clipped into ----- isolate polygons.

Select correct option:

Similar, three

Different, two ✓

Different, three

Similar, two

Question # 20 Total Marks: 1

If we are not worried about computational resources, we should use _____ lights.

Select correct option:

Point ✓

Parallel

Spot

None of the given

Question # 21 Total Marks: 1

_____ lighting is not dependent on any source.

Select correct option:

Ambient ✓

Diffuse

Specular

Emissive

Question # 22 Total Marks: 1

In order to get a more realistic representation of lighting, we'll need to understand how light passes through a medium and how hitting the boundary layer at the ----- of two media can affect light's properties.

Intersection ✓ page 293

Union

Endpoints

Edges

Question # 23 Total Marks: 1

Lambertian shading was used mostly back when computers weren't fast enough to do _____ in real time.

Select correct option:

Phong shading

Processing

Shading

✓ **Gouraud shading** page 242

Question # 24 Total Marks: 1

In Perspective Projection the point of View (POV) must lie on the _____.

Select correct option:

All axis

✓ **Z axis** page 196

X axis

Y axis

Question # 25 Total Marks: 1

If we want any object to glow, we should use _____ lights.

Select correct option:

Ambient

Diffuse

Specular

Emissive ✓ page 237

Question # 26 Total Marks: 1

An object's _____ determine its orientation relative to the light sources. For each vertex, OpenGL uses the assigned normal to determine how much light that particular vertex receives from each light source.

Unit

✓ **Normal** page 397

None of given

Question # 27 Total Marks: 1

The ColorSpace tool is a handy tool that we can use to interactively add two colours together to see the effects of the various strategies for handling oversaturated colours.

Select correct option:

✓ **True**

False

Question # 28 Total Marks: 1

_____ is simply the calculation of color reflected by the surface.

Select correct option:

Shading ✓

Clamping

Scaling

None of the given

Question # 29 Total Marks: 1

When you hit a surface in ray tracing, generally shadow rays are tested against all objects in a scene. If these rays come back saying they hit an object in the scene, which of the following do you do?

Select correct option:

add all components (i.e. ambient, diffuse and specular) from that light source to the object.

add all EXCEPT the ambient light from that light source to the object (i.e. diffuse and specular)

add only the ambient light from that light source to the object

add none of the light from that light source to the object ✓

Question # 30 Total Marks: 1

There are not many different ways of representing the intensity of a particular colour element.
Select correct option:

True

False

Question # 31 Total Marks: 1

In Perspective Projection the screen plane must be parallel to the_____.
Select correct option:

Y-Z plane

X-Y plane pg 198

Z-Y plane

X-Z plane

Question # 32 Total Marks: 1

_____ light is reflected in all directions from surface.
Select correct option:

Ambient

Diffuse pg 236

Specular

Emissive

Question # 33 Total Marks: 1

The reflected light wave turns out to be a / an _____ case since light is reflected at the same angle as the incident wave (when the surface is smooth and uniform, as we'll assume for now).

Unknown

Simple pg 293

Complex

Abnormal

Question # 34 Total Marks: 1

_____ sets the reshape callback for the current window. The reshape callback is triggered when a window is reshaped.

glutMainLoop

glutIdleFunc

glutReshapeFunc pg 309

glutDisplayFunc

Question # 35 Total Marks: 1

A space curve can be confined to a plane.

Yes

No

Question # 36 Total Marks: 1

To convert the information in the A matrix into that required for the P matrix, we do some simple matrix algebra, First we have $UA=UNP$ then Simply $A = \text{-----}$

UP

NP pg 330

UN

None

Question # 37 Total Marks: 1

Perspective projection is specified with the function `glFrustum()`.

Yes

No

Question # 38 Total Marks: 1

Choose a camera lens or adjust the zoom

projection transformation pg 369

viewport transformation

modeling transformation'

viewing transformation

Question # 39 Total Marks: 1

Using a lighting model based upon the Blinn Phong model means that we'll always get a uniform specular highlight based upon the colour of the ----- light and material, which means that all reflections based on this model, will be reminiscent of plastic.

Union

Refracting

Intersection

Reflecting ✓ page 293

Question # 40 Total Marks: 1

Refractive index is a function of temperature, mostly due to density changes in materials with changes in temperature.

True

False

Question # 41 Total Marks: 1

To convert the information in the A matrix into that required for the P matrix, we do some simple matrix algebra. First we have $GP = \text{-----}$

Select correct option:

NP

NU

UP

✓ **UNP** pg 330

Question #42 Total Marks: 1

Length L depends on the angle alpha and the z coordinate of the point to be projected and L can be represented by -----.

Select correct option:

✓ **$z * 1 / \tan(\alpha)$**

$z * L^2$

$z * 1 / \tan(\beta)$

$z * 1 / \tan(\gamma)$

Question # 43 Total Marks: 1

The traditional approach in real-time computer graphics has been to calculate lighting at a vertex as a sum of the _____ light.

Select correct option:

Ambient

Ambient, diffuse, and specular ✓

Specular

Diffuse, and specular

Question # 44 Total Marks: 1

Another way to define a space curve by using intermediate points and the tangents at each end for making the curve

Select correct option:

Yes

No ✓

Question # 45 Total Marks: 1

An independent consortium, the OpenGL Architecture Review Board, guides the OpenGL specification. With broad industry support, OpenGL is the only truly open, vendor-neutral, ----- graphics standard.

Select correct option:

Tertiary

Binary

Single platform

✓ **Multiplatform** page 298

Question # 46 Total Marks: 1

glutReshapeWindow requests a change in the size of the current window. The width and height parameters are size extents in pixels. The width and height must be ----- values.

Select correct option:

Neutral

Negative

Positive pg 308

Non of the given

Question # 47 Total Marks: 1

A space curve is not confined to a plane. It is free to twist through space. To define a space curve we must use parametric functions that are -----.

Select correct option:

Binary polynomials

Mono polynomials

Quadratic polynomials

Cubic polynomials pg 328

Question # 48 Total Marks: 1

Refractive index is a function of temperature, mostly due to changes in ----- of materials with changes in temperature. A simple correction can be applied in most circumstances to allow us to use a value given at one temperature at another.

Select correct option:

density page 297

pressure

nature

volume

Question # 49 Total Marks: 1

If we assign a different value to the parametric variable for the intermediate point, then we obtain different values for the coefficients. This, in turn, means that a different curve is produced, although it passes through the ----- three points.

Select correct option:

isolate

different

same pg 325

none

Question # 50 Total Marks: 1

Signed integer color components, when specified, are linearly mapped to floating-point values such that the most positive representable value maps to 1.0, and the most negative representable value maps to ----- . Floating-point values are mapped directly.

Select correct option:

-1.0

0.0

2.0

1.0 ✓

Question # 51 Total Marks: 1

The attenuation formula is $f = \frac{1}{C + Ld + Qd^2}$, where C, L and Q are the constant, linear and quadratic attenuation factors and d is the distance between the vertex being lit and the light source.

$\frac{1}{C + Ld + Qd^2}$ ✓

$\frac{1}{C + Ld + Qd}$

$\frac{1}{C + L + d + Qd^2}$

$\frac{1}{Cd + Ld + Qd^2}$

Question # 52 Total Marks: 1

End points and an intermediate point on the curve, then we now ----- quantities that we can express in terms of these coefficients (3 points x 3 coordinates each), and we can use these three points to define a unique curve.

Select correct option:

Six

Three

Two

Nine ✓

Question # 53 Total Marks: 1

Choose a camera lens or adjust the zoom

Select correct option:

projection transformation ✓

viewport transformation

modeling transformation

viewing transformation

Question # 54 Total Marks: 1

_____ OpenGL function is used for aiming and positioning the camera towards the object

Select correct option:

glLoadIdentity()

gluLookAt() ✓

glFrustum()

None of Above

Question # 55 Total Marks: 1

Question # 6 of 10 (Start time: 05:34:29 AM) Total Marks: 1

Bezier curve is tangent to the lines connecting _____.

Select correct option:

First two points

Last two points

First two points and last two point ✓

None of the given

Question # 56 Total Marks: 1

A parametric curve is one whose defining equations are given in terms of a -----, common, independent variable called the parametric variable.

Select correct option:

Triple

Double

Single ✓

None of the given

Question # 57 Total Marks: 1

_____ sets the global idle call back to be 'func' so a GLUT program can perform background processing tasks or continuous animation when window system events are not being received.

Select correct option:

glutIdleFunc ✓

glutMainLoop

glutDisplayFunc

glutReshapeFunc

Question # 58 Total Marks: 1

The reflection coefficients are in the ----- range and are specified as part of the material property. However, they are strictly empirical and since they simply adjust the overall intensity of the material colour, the material colour values are usually adjusted so the colour intensity varies rather than using a reflection coefficient.

Select correct option:

[0, 10]

[0, 1] ✓

[0, 5]

[0, 2]

Question # 59 Total Marks: 1

Another way to define a space curve by using intermediate points and the tangents at each end for making the curve

Select correct option:

Yes

No ✓

Question # 60 Total Marks: 1

What makes this really challenging to model is that the index of refraction for most materials is a function of the ----- of the light. This means that not only is there a shift in the angle of refraction, but that the shift is different for differing ----- of light.

Select correct option:

Reflecting angle, Reflecting angle

Refracting angle, Refracting angle

Frequency, Frequency

Wavelength, Wavelength ✓

Question # 61 Total Marks: 1

To ensure a smooth transition from one section of a piecewise _____ to the next, we can impose various continuity conditions at the connection points

Select correct option:

non parametric curve

parametric curve

polygon vector

Non of the these ✓

Question # 62 Total Marks: 1

The curve is always contained within the _____ of the control points

Select correct option:

Tangents

Convex Hull ✓

Subdivision

None of Above

Question # 63 Total Marks: 1

Signed integer color components, when specified, are linearly mapped to floating-point values such that the most positive representable value maps to 1.0, and the most negative representable value maps to ----- . Floating-point values are mapped directly.

Select correct option:

-1.0

0.0

2.0

1.0 ✓

Question # 64 Total Marks: 1

As f represents the attenuation of light, then f only approaches zero, when d approaches at distance -----.

Select correct option:

Zero

Infinity ✓

$1/n$

$1/2$

Question # 65 Total Marks: 1

Projection can be defined as a mapping of point $P(x, y, z)$ onto its image $P'(x', y', z')$ in the -----, which constitutes the display surface. The mapping is determined by a projection line called the projector that passes through P and intersects the -----.

Select correct option:

Two Coordinate Planes

View plane or projection plan ✓

Three Coordinate Planes

Mapping plane

Question # 66 Total Marks: 1

Determine how large we want the final photograph to be - for example, we might want it enlarged

Select correct option:

projection transformation

viewport transformation ✓

modeling transformation

viewing transformation

Question # 67 Total Marks: 1

Ambient light is the light that comes from -----directions, thus all surfaces are illuminated equally regardless of orientation. However, this is a big hack in traditional lighting calculations since "real" ambient light really comes from the light reflected from the "environment."

Select correct option:

All ✓

Opposite

Same

Four different

Question # 68 Total Marks: 1

Silhouette edges occur when dot product of surface normal vector and the view vector is -----.

Select correct option:

Zero ✓

One

Both zero and one

Question # 69 Total Marks: 1

If the current matrix (according to `glMatrixMode`) is multiplied by the translation matrix, with the product replacing the current matrix. That is, if M is the current matrix and T is the translation matrix, then M is replaced with -----.

Select correct option:

$M-T$

$M+T$

M/T

$M*T$ ✓

Question # 70 Total Marks: 1

Arrange the scene to be photographed into the desired composition

Select correct option:

projection transformation

viewport transformation

modeling transformation ✓

viewing transformation

Question # 71 Total Marks: 1

In the forms of texture mapping, Image to world space and world space to image, each suffers from different problems related to minification and magnification. Which of the two shows the following problem: When the texture is larger than the screen space it maps to, many texture units (texels) are never sampled?

Select correct option:

Image to world space

World space to image ✓

X-axis

Y-axis

Question # 72 Total Marks: 1

Imagine a curve in three-dimensional space, each point on the curve has a unique set of coordinates: a specific x value, y value, and z value. Each coordinate is controlled by a -----
-- parametric equation.

Select correct option:

Opposite

Similar

Separate ✓

Question # 73 Total Marks: 1

We allow the parametric variable to take on values only in the interval -----.

Select correct option:

-1 <= u <= 0

0 <= u <= 2

0 <= u <= 1 ✓

-1 <= u <= 1

Question # 74 Total Marks: 1

Bezier curve can represent the more complex piecewise _____ curve.

Select correct option:

Polynomial ✓

Exponential

Cubic

None of above

Question # 75 Total Marks: 1

NURBS stands for -----.

Select correct option:

Non Universal Rational Binary Spline

Non Uniform Rational Binary Splines

Non Uniform Rational Beta Splines ✓

Non Universal Rational Beta Splines

Question # 76 Total Marks: 1

A fractal is generally a property called _____.

Select correct option:

Fractal Dimension

Self-similarity ✓

Koch Curve

None of above

Question # 77 Total Marks: 1

Normalized cross product of two vectors on that surface provides normal vector

Select correct option:

Yes ✓

No

Question # 78 Total Marks: 1

Every point on a curve has a straight line associated with it called the _____

Select correct option:

State line

tangent line ✓

curved line

None of the given

Question # 79 Total Marks: 1

The value returned is a unique small integer identifier for the window. The range of allocated identifiers starts at ----- . This window identifier can be used when calling glutSetWindow.

Select correct option:

Three

Two

One ✓

Zero

Question # 80 Total Marks: 1

Curve and surface equations can be expressed in either a parametric or a nonparametric form.

Select correct option:

True ✓

False

Question # 81 Total Marks: 1

Bézier polynomial functions are the basic functions of _____ curves.

Select correct option:

NURBS

Bézier ✓

Both NURBS and Bézier

None of the given

Question # 82 Total Marks: 1

Geometric patterns that is repeated at ever smaller scales to produce irregular shapes and surfaces are called _____

Select correct option:

Geometric patterns

Fractals ✓

Animated components

Segments

Question # 83 Total Marks: 1

Which of the following is NOT a modern application for Computer Graphics-----?

Stop-motion animation ✓

Computer Aided Geometric Design

Video Games

Scientific Visualization

Question # 84 Total Marks: 1

TV series are made as simply as possible from the animation point of view. This approach is generally known as -----.

Full animation

Limited animation ✓

Low animation

High resolution

Question # 85 Total Marks: 1

An eight frame run cycle that ----- frame/frames to each step gives a fast and vigorous dash. At this speed the successive leg positions are quite widely separated and may need dry brush or speed lines to make the movement flow.

Two

One

Three

Four ✓

Question # 86 Total Marks: 1

reflection is the effect of reflecting light toward the direction from which it came, no matter the orientation of the surface.

Forward scattering
Diffuse Lambertian
Backscattering

Retro ✓

Question # 87 Total Marks: 1

Tessellation can be adaptive to the _____ degree of curvature of a surface.

Local ✓

Static
Global
Variable

Question # 88 Total Marks: 1

An orthogonal set of vectors-----

Must be a set of linearly independent vectors ✓

Must be a set of linearly dependent vectors
Must be made up of the basis vectors (e_1 , e_2 , and e_3)
Can be made up of any set of vectors

Question # 89 Total Marks: 1

Bezier curve is numerically the ----- of all the polynomial-based curves used in these applications.

None of the given
Most stable ✓
Less stable
Most unstable

Question # 90 Total Marks: 1

Bezier curve is the ideal standard for representing the ----- piecewise polynomial curves.

None of the given

Non complex

Most complex

More complex ✓

Question # 91 Total Marks: 1

Keep polygon orientations consistent to make sure that when viewed from the outside, all the polygons on the surface are oriented in the same direction.

None of the given

Neither

Different

Same ✓

Question # 92 Total Marks: 1

The ----- is most simple example that exhibits the property self similarity.

Mosse

Fern ✓

None of the given

Thohar

Question # 93 Total Marks: 1

A common mistake people make when creating three-dimensional graphics is to start thinking too soon that the final image appears on a flat, two-dimensional screen. Avoid thinking about which pixels need to be drawn, and instead try to visualize ----- space.

Multi-dimensional

One-dimensional

Two-dimensional

Three-dimensional ✓

Question # 94 Total Marks: 1

Which of the following properties of rational Bezier curves fails if the weight assigned to a control point is negative?

End-point interpolation

Variation Diminishing

Symmetry

Convex-Hull ✓

Question # 95 Total Marks: 1

In the Phong reflection model, there are 3 constants (a, b, c) which are used to describe the qualities of which of the following phenomena?

The attenuation of a point light source with distance ✓

The size (in each dimension) which the light is assumed to have

The amount to perturb reflection vectors as they are mirrored across the normal

The material reaction to ambient, diffuse and specular light (respectively)

Question # 96 Total Marks: 1

When you hit a surface in ray tracing, generally shadow rays are tested against all objects in a scene. If these rays come back saying they hit an object in the scene, which of the following do you do?

add all components (i.e. ambient, diffuse and specular) from that light source to the object.

add all EXCEPT the ambient light from that light source to the object (i.e. diffuse and specular)

add only the ambient light from that light source to the object

add none of the light from that light source to the object ✓

Question # 97 Total Marks: 1

The ColorSpace tool is a handy tool that we can use to interactively add two colours together to see the effects of the various strategies for handling oversaturated colours.

False

True ✓

Question # 98 Total Marks: 1

A polygon is usually defined by a sequence of ----- and Edges.

Ending lines

Points

Vertices ✓

Edges

Question # 99 Total Marks: 1

Which of the following properties of Bezier curves guarantees that a line passes through the control polygon as many times or more times than the line passes through the Bezier curve itself?

End-point interpolation

Variation Diminishing ✓

Symmetry

Convex-Hull

Question # 100 Total Marks: 1

Parity is a concept used to determine which _____ lie within a polygon. (Choose best suitable answer)

Edge

Vertices

Pixels ✓

None of the given

Question # 101 Total Marks: 1

The actual filling process in boundary filling algorithm begins when a point _____ of the figure is selected.

Outside the boundary

Inside the boundary ✓

At boundary

None of the given

Question # 102 Total Marks: 1

Weiler-Atherton Polygon Clipping technique modify the vertex-processing procedures for window boundaries so that _____ polygons are displayed correctly.

Convex

Concave ✓

Complex

None of the given

Question # 103 Total Marks: 1

If a line connecting any two points within a polygon does not intersect any edge, then it will be a _____ polygon.

Convex ✓

Concave

Complex

None of the given

Question # 104 Total Marks: 1

_____ can be defined as a mapping of point $P(x, y, z)$ onto its image $P'(x', y', z')$ in the view plane which constitutes the display surface.

Mapping plane

Three Coordinate Planes

View plane

Projection ✓

Question # 105 Total Marks: 1

Computer graphics and computer vision are _____ disciplines.

Related

Interrelated

Same

Different ✓

Question # 106 Total Marks: 1

OpenGL has become the industry's most widely used and supported _____ graphics application programming interface (API), bringing thousands of applications to a wide variety of computer platforms.

2-Dimensional

3-Dimensional

2-Dimensional and 3-Dimensional ✓

Question # 107 Total Marks: 1

_____ sets the reshape callback for the current window.

glutIdle function

glutKeyboardFunc

glutReshapeFunc ✓

glutDisplayFunc

Question # 108 Total Marks: 1

$X^2/a^2 - y^2/b^2 = 1$ is an equation of

Hyperbola

Parabola ($4px = y^2$)

None of given

Ellipse ($X^2/a^2 + y^2/b^2 = 1$)

Question # 109 Total Marks: 1

To modify the object shape, shearing transformation can not be used.

False

True

Question # 110 Total Marks: 1

Rotation is performed around a fixed point called _____.

Pivot point rotation ✓

Question # 111 Total Marks: 1

Computer graphics is very helpful in producing graphical representations for scientific visualization and analysis

True ✓

False

Question # 112 Total Marks: 1

Save a line with both endpoints inside all clipping boundaries.

Trivial Reject (decline)

Trivial Accept ✓

None of given

Question # 113 Total Marks: 1

Dark lights are nothing more than lights in which one or more of the color values are ____.

Unknown

Negative ✓

Positive

Zero

Question # 114 Total Marks: 1

A series of _____ computer operations convert an object's three-dimensional coordinates to pixel positions on the screen. Transformations, which are represented by matrix multiplication, include modeling, viewing, and projection operations. Such operations include rotation, translation, scaling, reflecting, orthographic projection, and perspective projection.

Three ✓

Two

Four

Ten

Question # 115 Total Marks: 1

At a physical surface, our eye's perception of the color depends on the distribution of photon energies that arrive and trigger our _____ cells.

Eye

Retin

Cone ✓

Question # 116 Total Marks: 1

In the Phong Reflection model, _____ light is the same everywhere.

Ambient ✓

Diffuse

Speculares

Emissive

Question # 117 Total Marks: 1

A plane is two dimensional since in order to uniquely define any point on its surface we require _____ numbers.

Two ✓

Three

Four

Five

Question # 118 Total Marks: 1

In perspective projection, for your view to come out correctly, you will also want the _____ to pass through the middle of the screen.

X axis

Y axis

Z axis ✓

None

Question # 119 Total Marks: 1

Neither floating-point nor signed integer values are clamped to the range _____ before updating the current color.

0 , -1.0

-1 , 1

1 , -1

0, 1

Question # 120 Total Marks: 1

Bezier curve is the ideal standard for representing the _____ piecewise polynomial curves.

Most complex

Less complex

None of given

More complex ✓

Question # 121 Total Marks: 1

In the forms of texture mapping, Image to world space and world space to image, each suffers from different problems related to minification and magnification. Which of the two shows the following problem: When the texture is larger than the screen space it maps to, many texture units (texels) are never sampled?

Select correct option:

Image to world space ✓

World space to image

X-axis

Y-axis

Question # 122 Total Marks: 1

Both Boundary Filling and Flood filling algorithms are non-recursive techniques,

False

True

Question # 123 Total Marks: 1

The Phong reflection model simplifies light-matter interactions into (essentially) 4 vectors and a number of constants. Each piece of the Phong model uses different vectors and constants.

Which portion does NOT include taking a dot product?

Ambient ✓

Diffuse

Specular

Good Luck Guys
by
V4

Special thanks to mohid mirza, Sagittarian malik and ch. meo

Solved by: Well Wisher (Sahar) Class BSCS 6 th Semester			
Subject	Cs602 (computer graphics)	Solution Type:	Final Term Paper + Quizez
Institute:	Virtual University of Pakistan		

1. Computer graphics and computer vision are _____ disciplines.

Related

Interrelated

Same

Different ✓

Ref : Computer vision is, in some ways, the inverse of computer graphics. While computer graphics produces image data from 3D models, computer vision often produces 3D models from image data.

http://en.wikipedia.org/wiki/Computer_vision

2. OpenGL has become the industry's most widely used and supported _____ graphics application programming interface (API), bringing thousands of applications to a wide variety of computer platforms.

Dimensional

Dimensional

2 Dimensional and 3-Dimensional ✓

Ref: <http://www.opengl.org/about/>

3. _____ sets the reshape callback for the *current window*.

glutIdle function

glutKeyboardFunc

glutReshapeFunc ✓

glutDisplayFunc

Ref:

<https://developer.apple.com/library/mac/documentation/Darwin/Reference/ManPages/man3/glutReshapeFunc.3.html>

4. _____ sets the global idle callback to be 'func' so a GLUT program can perform background processing tasks or continuous animation when window system events are not being received.

glutIdle function ✓

glutKeyboardFunc

glutReshapeFunc

glutDisplayFunc

Ref: <http://www.cs.uccs.edu/~ssemwal/man.html>

5. $X^2/a^2 - y^2/b^2 = 1$ is an equation of

Hyperbola ✓ **Page no : 70**

Parabola

None of given

Ellipse

6. To modify the object shape, shearing transformation cannot be used.

False ✓ **Page no : 124**

True

7. Rotation is performed around a fixed point called ____.

Pivot point rotation ✓ **Page no : 114**

8. Computer graphics is very helpful in producing graphical representations for scientific visualization and analysis

True ✓

False

9. Save a line with both endpoints inside all clipping boundaries.

Trivial Reject

Trivial Accept ✓ **Page no : 137**

None of given

10. Dark lights are nothing more than lights in which one or more of the color values are ____.

Unknown

Negative ✓ **Page no : 230**

Positive

Zero

11. A series of _____ computer operations convert an object's three-dimensional coordinates to pixel positions on the screen. Transformations, which are represented by matrix multiplication, include modeling, viewing, and projection operations. Such operations include rotation, translation, scaling, reflecting, orthographic projection, and perspective projection.

Three ✓

Page no : 366

Two

Four

Ten

At a physical surface, our eye's perception of the color depends on the distribution of photon energies that arrive and trigger our _____ cells.

Eye

Retina

Cone ✓

Page no : 393

12. This projection technique has the direction of projection perpendicular to the viewing plane, but the viewing direction is NOT perpendicular to one of the principle faces.

Orthographic Parallel Projection

Axonometric Parallel Projection ✓ Page no : 189

Oblique Parallel Projection

13. The Phong reflection model simplifies light-matter interactions into (essentially) 4 vectors and a number of constants. Which piece of the Phong model is responsible for giving spheres their bright white spots?

Specular ✓ Page no : 234

Ambient

Diffuse

14. In the Phong Reflection model, _____ light is the same everywhere.

a. Ambient ✓

b. Diffuse

c. Specular

d. Emissive

e.

Ref: www.cs.unc.edu/~jwendt/classes/COMP136/quizzes/quiz3Answers.doc

15. A plane is two dimensional since in order to uniquely define any point on its surface we require _____ numbers.

Two ✓

Three

Four

Five

16. In perspective projection, for your view to come out correctly, you will also want the _____ to pass through the middle of the screen.

X axis

Y axis

Z axis ✓

Page no : 195

None

17. Neither floating-point nor signed integer values are clamped to the range _____ before updating the current color.

0 , -1.0

-1 , 1

1 , -1

0, 1 ✓

Page no : 316

18. Bezier curve is the ideal standard for representing the _____ piecewise polynomial curves.

Most complex

Less complex

None of given

More complex ✓

Page no : 333

19. An object's _____ determine its orientation relative to the light sources. For each vertex, OpenGL uses the assigned normal to determine how much light that particular vertex receives from each light source.

Unit

Normal ✓

Page no : 395

None of given

20. Which was the most oldest shading model?

- a. **Flat Shading** ✓
- b. Phong Shading
- c. Gouraud Shading

21. Which of the following affine transforms does NOT affect vectors?

- a) Scale
- b) Rotation
- c) Shear
- d) **Translation** ✓ **Page no : 113**

22) This projection technique does NOT have the direction of projection perpendicular to the viewing plane.

- a) Orthographic Parallel Projection
- b) Axonometric Parallel Projection
- c) **Oblique Parallel Projection** ✓ **Page no : 189**

23) This projection technique has the direction of projection perpendicular to the viewing plane, and the viewing direction is perpendicular to one of the principle faces.

- a) **Orthographic Parallel Projection** ✓ **Page no : 189**
- b) Axonometric Parallel Projection
- c) Oblique Parallel Projection

24) In OpenGL, there are several different matrices. We have discussed two of them in class. Which one of the below would be used in conjunction with a glRotatef function call?

- a) **GL_MODELVIEW** ✓ **Page no : 388**
- b) GL_PROJECTION

25) In OpenGL, there are several different matrices. We have discussed two of them in class. Which one of the below would be used in conjunction with glFrustum?

- a) GL_MODELVIEW
- b) **GL_PROJECTION** ✓ **Page no : 369**

26) Which of the following is the order that geometry operations are performed in OpenGL (where we read the order from left to right)?

- a) GL_PROJECTION → GL_MODELVIEW → Perspective division
- b) **GL_MODELVIEW → GL_PROJECTION → Perspective division** ✓
- c) Perspective division → GL_PROJECTION → GL_MODELVIEW
- d) GL_MODELVIEW → Perspective division → GL_PROJECTION
- e) GL_PROJECTION → Perspective division → GL_MODELVIEW

27) The Phong reflection model simplifies light-matter interactions into (essentially) 4 vectors and a number of constants. Each piece of the Phong model uses different vectors and constants. Which portion does NOT include taking a dot product?

- a) **Ambient** ✓ **Page no : 234**
- b) Diffuse
- c) Specular

FINAL TERM PAPER 2010

Question No: 1 (Marks: 1) - Please choose one

Which of the following is NOT a modern application for Computer Graphics-----

- ▶ **Stop-motion animation** ✓ Page no : 6
- ▶ Computer Aided Geometric Design
- ▶ Video Games
- ▶ Scientific Visualization

Question No: 2 (Marks: 1) - Please choose one

Both Boundary Filling and Flood filling algorithms are non-recursive techniques,

- ▶ **False** ✓ Page no : 97
- ▶ True

Question No: 3 (Marks: 1) - Please choose one

TV series are made as simply as possible from the animation point of view. This approach is generally known as -----.

- ▶ Full animation
- ▶ **Limited animation** ✓ Page no : 423
- ▶ Low animation
- ▶ High resolution

Question No: 4 (Marks: 1) - Please choose one

An eight frame run cycle that ----- frame/frames to each step gives a fast and vigorous dash. At this speed the successive leg positions are quite widely separated and may need dry brush or speed lines to make the movement flow.

▶ Two

▶ One

▶ Three

▶ **Four** ✓ **Page no :432**

Question No: 5 (Marks: 1) - Please choose one

Reflection is the effect of reflecting light toward the direction from which it came, no matter the orientation of the surface.

▶ Forward scattering

▶ Diffuse Lambertian

▶ Backscattering

▶ **Retro** ✓ **Page no : 288**

Question No: 6 (Marks: 1) - Please choose one

What makes this really challenging to model is that the index of refraction for most materials is a function of the----- of the light. This means that not only is there a shift in the angle of refraction, but that the shift is different for differing -----of light.

▶ Reflecting angle, Reflecting angle

▶ Refracting angle, Refracting angle

▶ Frequency, Frequency

▶ **Wavelength, Wavelength** ✓ **Page no : 293**

Question No: 7 (Marks: 1) - Please choose one

The reflected light wave turns out to be a -----case since light is reflected at the same angle as the incident wave (when the surface is smooth and uniform, as we'll assume for now).

▶ Abnormal

▶ Complex

▶ Simple ✓ Page no : 291

▶ Unknown

Question No: 8 (Marks: 1) - Please choose one

Tessellation can be adaptive to the _____ degree of curvature of a surface.

▶ Local ✓ Mid Page no : 165

▶ Static

▶ Global

▶ Variable

Question No: 9 (Marks: 1) - Please choose one

_____ sets the reshape call back for the *current window*. The reshape callback is triggered when a window is reshaped.

▶ glutMainLoop

▶ glutIdleFunc

▶ glutReshapeFunc ✓ Page no : 307

▶ glutDisplayFunc

Question No: 10 (Marks: 1) - Please choose one

Signed integer colour components, when specified, are linearly mapped to floating-point values such that the most positive represent able value maps to 1.0, and the most negative represent able value maps to ----- . Floating-point values are mapped directly.

▶ -1.0

▶ 0.0

▶ 2.0

▶ 1.0 ✓ Page no : 315

Question No: 11 (Marks: 1) - Please choose one

NURBS stands for-----.

- ▶ Non Universal Rational Binary Spline
- ▶ Non Uniform Rational Binary Splines
- ▶ **Non Uniform Rational Beta Splines** ✓ Page no :320
- ▶ Non Universal Rational Beta Splines

Question No: 12 (Marks: 1) - Please choose one

An orthogonal set of vectors-----

- ▶ **Must be a set of linearly independent vectors** ✓
- ▶ Must be a set of linearly dependent vectors
- ▶ Must be made up of the basis vectors (e_1 , e_2 , and e_3)
- ▶ Can be made up of any set of vectors

Ref :

<http://www.ualberta.ca/dept/math/gauss/fcm/LinAlg/InRn/SbVctrSpc/OrthgnlLnrllyIndpndnt.htm>

Question No: 13 (Marks: 1) - Please choose one

Bezier curve is numerically the ----- of all the polynomial-based curves used in these applications.

- ▶ None of the given
- ▶ **Most stable** ✓
- ▶ Less stable
- ▶ Most unstable

Ref :

<http://books.google.com.pk/books?id=YmQy799fIPkC&pg=PA264&lpg=PA264&dq=Bezier+curve+is+numerically+the+-----+of+all+the+polynomial-based+curves+used+in+these+applications.&source=bl&ots=MHnr87FLlQ&sig=wG0oXl00vxWtEY7RnfNnOXJrc08&hl=en&sa=X&ei=nTroUaDdL6eB4ASioIHgBA&ved=0CCoQ6AEwAA#v=onepage&q=Bezier%20curve%20is%20numerically%20the%20----->

Question No: 14 (Marks: 1) - Please choose one

Bezier curve is the ideal standard for representing the ----- piecewise polynomial curves.

- ▶ None of the given
- ▶ Non complex
- ▶ Most complex
- ▶ **More complex** ✓ **repeated**

Question No: 15 (Marks: 1) - Please choose one

Keep polygon orientations consistent to make sure that when viewed from the outside, all the polygons on the surface are oriented in the ____ direction.

- ▶ None of the given
- ▶ Neither
- ▶ Different
- ▶ **Same** ✓ **Page no : 340**

Question No: 16 (Marks: 1) - Please choose one

The ----- is most simple example that exhibits the property self similarity.

- ▶ Mosse
- ▶ **Fern** ✓ **Page no : 350**
- ▶ None of the given
- ▶ Thohar

Question No: 17 (Marks: 1) - Please choose one

A common mistake people make when creating three-dimensional graphics is to start thinking too soon that the final image appears on a flat, two-dimensional screen. Avoid thinking about which pixels need to be drawn, and instead try to visualize ----- space.

- ▶ Multi-dimensional
- ▶ One-dimensional
- ▶ Two-dimensional
- ▶ **Three-dimensional** ✓ **Page no: 366**

Question No: 18 (Marks: 1) - Please choose one

Which of the following properties of rational Bezier curves fails if the weight assigned to a control point is negative?

- ▶ End-point interpolation
- ▶ Variation Diminishing
- ▶ Symmetry
- ▶ **Convex-Hull** ✓ **page no : 335**

Question No: 19 (Marks: 1) - Please choose one

In the Phong reflection model, there are 3 constants (a, b, c) which are used to describe the qualities of which of the following phenomena?

- ▶ **The attenuation of a point light source with distance** ✓ **Page no : 285**
- ▶ The size (in each dimension) which the light is assumed to have
- ▶ The amount to perturb reflection vectors as they are mirrored across the normal
- ▶ The material reaction to ambient, diffuse and specular light (respectively)

Question No: 20 (Marks: 1) - Please choose one

The Phong reflection model simplifies light-matter interactions into (essentially) 4 vectors and a number of constants. Which piece of the Phong model is responsible for giving spheres their bright white spots?

- ▶ **Specular** ✓ **repeated**
- ▶ Diffuse
- ▶ Ambient

Question No: 21 (Marks: 1) - Please choose one

When you hit a surface in ray tracing, generally shadow rays are tested against all objects in a scene. If these rays come back saying they hit an object in the scene, which of the following do you do?

- ▶ add all components (i.e. ambient, diffuse and specular) from that light source to the object.
- ▶ add all EXCEPT the ambient light from that light source to the object (i.e. diffuse and specular)
- ▶ **add only the ambient light from that light source to the object**
- ▶ add none of the light from that light source to the object ✓

Question No: 22 (Marks: 1) - Please choose one

The Color Space tool is a handy tool that we can use to interactively add two colours together to see the effects of the various strategies for handling oversaturated colours.

- ▶ False
- ▶ **True** ✓ **page no : 230**

Question No: 23 (Marks: 1) - Please choose one

A polygon is usually defined by a sequence of ----- and Edges.

▶ Ending lines

▶ Points

▶ **Vertices** ✓ **Page no : 243**

▶ Edges

Question No: 24 (Marks: 1) - Please choose one

Which of the following properties of Bezier curves guarantees that a line passes through the control polygon as many times or more times than the line passes through the Bezier curve itself?

▶ End-point interpolation

▶ Variation Diminishing

▶ Symmetry

▶ **Convex-Hull** ✓

Ref: <http://cagd.cs.byu.edu/~557/text/ch2.pdf>

Question No: 25 (Marks: 1) - Please choose one

Parity is a concept used to determine which _____ lie within a polygon. (Choose best suitable answer)

▶ Edge

▶ Vertices

▶ **Pixels** ✓ **Page no : 80**

▶ None of the given

Question No: 26 (Marks: 1) - Please choose one

The actual filling process in boundary filling algorithm begins when a point _____ of the figure is selected.

▶ Outside the boundary

▶ **Inside the boundary** ✓

▶ At boundary

▶ None of the given

Question No: 27 (Marks: 1) - Please choose one

Weiler-Atherton Polygon Clipping technique modify the vertex-processing procedures for window boundaries so that _____ polygons are displayed correctly.

- ▶ Convex
- ▶ Concave ✓ Page no : 245
- ▶ Complex
- ▶ None of the given

Question No: 28 (Marks: 1) - Please choose one

If a line connecting any two points within a polygon does not intersect any edge, then it will be a _____ polygon.

- ▶ Convex Page no : 78
- ▶ Concave
- ▶ Complex
- ▶ None of the given

Question No: 29 (Marks: 1) - Please choose one

_____ can be defined as a mapping of point $P(x, y, z)$ onto its image $P'(x', y', z')$ in the view plane which constitutes the display surface.

- ▶ Mapping plane
- ▶ Three Coordinate Planes
- ▶ View plane Repeated
- ▶ Projection ✓ pg 190

Question No: 30 (Marks: 1) - Please choose one

The reflected light wave turns out to be a / an _____ case since light is reflected at the same angle as the incident wave (when the surface is smooth and uniform, as we'll assume for now).

► Unknown

► **Simple** ✓ Page no: 291

► Complex

► Abnormal

Which of the following is NOT true about quaternions?

- a) They are made up of 4 numbers
- b) They should always be normalized to length 1
- c) They can be used to represent all affine transforms** ✓
- d) They can be used to define the rotation of an object

FINAL TERM QUIZES

1) In class, we discussed three forms of shading for “Utah” graphics. Which was the first to use per vertex normals?

- a) Flat Shading
- b) Phong Shading
- c) Gouraud Shading** ✓ Page no : 240

2) Given any implicit equation, which of the following is true for all (x, y, z) that make the equation exactly zero?

- a) All those points are inside the object defined by the implicit equation
- b) All those points are on the surface of the object defined by the implicit equation** Page no : 205 ✓
- c) All those points are outside the object defined by the implicit equation
- d) You can't know anything without knowing what the implicit equation is

3) When solving ray-sphere intersections using the implicit equation for a sphere, you must solve the quadratic equation. Which of the following do you know if the $B^2 - 4AC$ (i.e. the part under the square root) is negative?

- a) The ray intersects the sphere at a negative t ... discard this result
- b) The ray intersects the sphere at a positive t ... continue to the solution
- c) The ray does not intersect the sphere... discard this result** ✓ Page no : 265
- d) The ray begins inside the sphere... this is a special case

4)

_____ sets the global idle call back to be 'func' so a GLUT program can perform background processing tasks or continuous animation when window system events are not being received.

Select correct option:

glutIdleFunc ✓

glutMainLoop

glutDisplayFunc

glutReshapeFunc

Ref: <http://www.opengl.org/resources/libraries/glut/spec3/node63.html>

5)

A space curve can be confined to a plane.

Select correct option:

True

False ✓ Page no : 326

6)

A tangent vector certainly defines the slope at one end of the curve, but a vector has characteristics of.....

Select correct option:

direction

magnitude

both direction and magnitude ✓ Page no : 331

None of the given

7)

We allow the parametric variable to take on values only in the interval -----.

Select correct option:

$-1 \leq u \leq 0$

$0 \leq u \leq 2$

$0 \leq u \leq 1$ ✓ Page no : 321

$-1 \leq u \leq 1$

8)

The degree of a Bezier curve is equal to $n-1$, where n is the number of control points

Select correct option:

Yes ✓ Page no : 334

No

9)

Brenstien polynomial functions are the basic functions of _____ curves.

Select correct option:

NURBS

Bezier ✓ Page no : 337

Both NURBS and Bazier

None of the given

10)

A parametric curve is one whose defining equations are given in terms of a -----, common, independent variable called the parametric variable.

Select correct option:

Triple

Double

Single ✓ Page no : 320

None of the given

11)

Bit mask to select a window with multisampling support. If multisampling is not available, a ----- window will automatically be chosen.

Select correct option:

Non-multisampling ✓ Page no : 305

Multisampling

Mono-multisampling

Di-multisampling

12)

Bezier curve is tangent to the lines connecting _____.

Select correct option:

First two points
Last two points
Fist two points and last two point ✓
None of the given

13)

OpenGL is well structured with an intuitive design and logical commands. Efficient OpenGL routines typically result in applications with fewer lines of code than those that make up programs generated using other graphics libraries or packages. In addition, OpenGL drivers -----
----- information about the underlying hardware, freeing the application developer from having to design for specific hardware features.

Select correct option:

Encapsulate ✓ **Page no : 297**

Shows
Hibernates
None of the given

14)

A space curve is not confined to a plane. It is free to twist through space. To define a space curve we must use parametric functions that are -----.

Select correct option:

Binary polynomials
Mono polynomials
Quadratic polynomials ✓

Cubic polynomials ✓ **Page no : 326**

15)

End points and an intermediate point on the curve, then we now ----- quantities that we can express in terms of these coefficients (3 points x 3 coordinates each), and we can use these three points to define a unique curve.

Select correct option:

Six
Three
Two

Nine ✓ **Page no : 321**

16)

To convert the information in the A matrix into that required for the P matrix, we do some simple matrix algebra, First we have $UA=UNP$ then Simply $A = \text{-----}$

Select correct option:

UP

NP ✓

UN

None of the given

17)

If we assign a different value to the parametric variable for the intermediate point, then we obtain different values for the coefficients. This, in turn, means that a different curve is produced, although it passes through the ----- three points.

Select correct option:

Isolate

Different

Same ✓ Page no : 323

None of the given

16)

In order to get a more realistic representation of lighting, we'll need to understand how light passes through a medium and how hitting the boundary layer at the ----- of two media can affect light's properties.

Select correct option:

Intersection ✓ Page no : 295

Union

Endpoints

Edges

17)

To ensure a smooth transition from one section of a piecewise _____ to the next, we can impose various continuity conditions at the connection points

Select correct option:

non parametric curve

parametric curve

polygon vector

Non of the these ✓

Ref : www.mrl.snu.ac.kr/courses/CourseGraphics/Splines.ppt

18)

Bezier curve can represent the more complex piecewise _____ curve.

Select correct option:

Polynomial ✓ Page no : 33

Exponential

Cubic

None of above

19)

Curve and surface equations can be expressed in either a parametric or a non parametric form.

Select correct option:

True

False Page no : 333

20)

Using a lighting model based upon the Blinn Phong model means that we'll always get a uniform specular highlight based upon the colour of the ----- light and material, which means that all reflections based on this model, will be reminiscent of plastic.

Select correct option:

Union

Refracting Page no : 291

Intersection

Reflecting ✓ pg 293

21)

If the current matrix (according to glMatrixMode) is multiplied by the translation matrix, with the product replacing the current matrix. That is, if M is the current matrix and T is the translation matrix, then M is replaced with -----.

Select correct option:

M-T

M+T

M/T

M*T ✓

22)

With similar expressions for $y(u)$ and $z(u)$. Again the a , b , c and d terms are constant coefficients. As we did with Equation for a plane curve, we combine the $x(u)$, $y(u)$, and $z(u)$ expressions into a single vector equation $P(u) = \text{-----}$.

Select correct option:

Au^2+bu^1+cu+d

$Au^4+bu^3+cu^2+d^1$

$Au^3+bu^2+cu^2+d$

Au^3+bu^2+cu+d ✓ Page no : 326

23)

The matrix generated by `gluPerspective` is multiplied by the current matrix, just as if `glMultMatrix` were called with the generated matrix. To load the perspective matrix onto the current matrix stack instead, precede the call to `gluPerspective` with a call to -----.

Select correct option:

`glRotated`

`gluPerspective`

`glTranslated`

`glLoadIdentity` ✓ Page no : 313

24)

Each number that makes up a matrix is called an _____ of the matrix.

- **Element** ✓ Page no : 101
- Variable
- Value
- Component

25)

Which one of the following step is not involved to write pixel using video BIOS services.

- Setting desired video mode
- Using BIOS service to set color of a screen pixel
- Calling BIOS interrupt to execute the process of writing pixel.
- **Using OpenGL service to set color of a screen pixel** ✓

26)

Shadow mask methods can display a _____ range of colors.

- Small
- **Wide** ✓ **Page no : 29**
- Random
- Crazy

27)

Using Cohen-Sutherland line clipping, it is impossible for a vertex to be labeled 1111.

- **True** ✓
- False

28)

Intensity of the electron beam is controlled by setting _____ levels on the control grid, a metal cylinder that fits over the cathode.

- Amplitude
- Current
- **Voltage** **Page no : 26**
- Electron



CS602- Computer Graphics

Solved MCQS
From Finalterm Papers

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FINALTERM EXAMINATION Spring 2010 CS602- Computer Graphics (VISIT VURANK FOR MORE)

Question No: 1 (Marks: 1) - Please choose one

Which of the following is NOT a modern application for Computer Graphics-----

- ▶ **Stop-motion animation (Page 6)**
- ▶ Computer Aided Geometric Design
- ▶ Video Games
- ▶ Scientific Visualization

Question No: 2 (Marks: 1) - Please choose one

Both Boundary Filling and Flood filling algorithms are non-recursive techniques,

- ▶ **False [click here 4 detail](#)**
- ▶ True

Question No: 3 (Marks: 1) - Please choose one

TV series are made as simply as possible from the animation point of view. This approach is generally known as -----.

- ▶ Full animation
- ▶ **Limited animation (Page 428)✓**
- ▶ Low animation
- ▶ High resolution

Question No: 4 (Marks: 1) - Please choose one

An eight frame run cycle that ----- frame/frames to each step gives a fast and vigorous dash. At this speed the successive leg positions are quite widely separated and may need dry brush or speed lines to make the movement flow.

- ▶ Two
- ▶ One
- ▶ Three

▶ **Four (Page 437)** ✓

Question No: 5 (Marks: 1) - Please choose one

----- reflection is the effect of reflecting light toward the direction from which it came, no matter the orientation of the surface.

- ▶ Forward scattering
- ▶ Diffuse Lambertian
- ▶ Backscattering

▶ **Retro (Page 293)**

Question No: 6 (Marks: 1) - Please choose one

What makes this really challenging to model is that the index of refraction for most materials is a function of the----- of the light. This means that not only is there a shift in the angle of refraction, but that the shift is different for differing -----of light.

- ▶ Reflecting angle, Reflecting angle
- ▶ Refracting angle, Refracting angle
- ▶ Frequency, Frequency

▶ **Wavelength, Wavelength (Page 229)**

Question No: 7 (Marks: 1) - Please choose one

The reflected light wave turns out to be a -----case since light is reflected at the same angle as the incident wave (when the surface is smooth and uniform, as we'll assume for now).

- ▶ Abnormal
- ▶ Complex

▶ **Simple (Page 296)**

- ▶ Unknown

Question No: 8 (Marks: 1) - Please choose one

Tessellation can be adaptive to the _____ degree of curvature of a surface.

▶ **Local (Page 170)**

- ▶ Static
- ▶ Global
- ▶ Variable

Question No: 9 (Marks: 1) - Please choose one

_____ sets the reshape callback for the *current window*. The reshape callback is triggered when a window is reshaped.

- ▶ glutMainLoop
- ▶ glutIdleFunc
- ▶ **glutReshapeFunc (Page 312)** ✓
- ▶ glutDisplayFunc

Question No: 10 (Marks: 1) - Please choose one

Signed integer color components, when specified, are linearly mapped to floating-point values such that the most positive representable value maps to 1.0, and the most negative representable value maps to -----
 -. Floating-point values are mapped directly.

- ▶ -1.0
- ▶ 0.0
- ▶ 2.0
- ▶ **1.0 (Page 320)** ✓

Question No: 11 (Marks: 1) - Please choose one

NURBS stands for-----.

- ▶ Non Universal Rational Binary Spline
- ▶ Non Uniform Rational Binary Splines
- ▶ **Non Uniform Rational Beta Splines (Page 325)** ✓
- ▶ Non Universal Rational Beta Splines

Question No: 12 (Marks: 1) - Please choose one

An orthogonal set of vectors-----

- ▶ Must be a set of linearly independent vectors
- ▶ Must be a set of linearly dependent vectors
- ▶ **Must be made up of the basis vectors (e1, e2, and e3) [click here 4 details](#)** ✓
- ▶ Can be made up of any set of vectors

Question No: 13 (Marks: 1) - Please choose one

Bezier curve is numerically the ----- of all the polynomial-based curves used in these applications.

- ▶ None of the given
- ▶ **Most stable (Page 338)** ✓
- ▶ Less stable
- ▶ Most unstable

Question No: 14 (Marks: 1) - Please choose one

Bezier curve is the ideal standard for representing the ----- piecewise polynomial curves.

- ▶ None of the given
- ▶ Non complex
- ▶ Most complex
- ▶ **More complex (Page 338)** ✓

Question No: 15 (Marks: 1) - Please choose one

Keep polygon orientations consistent to make sure that when viewed from the outside, all the polygons on the surface are oriented in the **same** direction.

- ▶ None of the given
- ▶ Neither
- ▶ Different
- ▶ **Same (page 345) ✓**

Question No: 16 (Marks: 1) - Please choose one

The ----- is most simple example that exhibits the property self similarity.

- ▶ Mosse
- ▶ **Fern (Page 355) ✓**
- ▶ None of the given
- ▶ Thohar

Question No: 17 (Marks: 1) - Please choose one

A common mistake people make when creating three-dimensional graphics is to start thinking too soon that the final image appears on a flat, two-dimensional screen. Avoid thinking about which pixels need to be drawn, and instead try to visualize ----- space.

- ▶ Multi-dimensional
- ▶ One-dimensional
- ▶ Two-dimensional
- ▶ **Three-dimensional (Page 371) ✓**

Question No: 18 (Marks: 1) - Please choose one

Which of the following properties of rational Bezier curves fails if the weight assigned to a control point is negative?

- ▶ End-point interpolation
- ▶ Variation Diminishing
- ▶ Symmetry
- ▶ **Convex-Hull [Click here 4 detail](#) ✓**

Question No: 19 (Marks: 1) - Please choose one

In the Phong reflection model, there are 3 constants (a, b, c) which are used to describe the qualities of which of the following phenomena?

- ▶ **The attenuation of a point light source with distance ✓**
- ▶ The size (in each dimension) which the light is assumed to have
- ▶ The amount to perturb reflection vectors as they are mirrored across the normal
- ▶ The material reaction to ambient, diffuse and specular light (respectively)

Question No: 20 (Marks: 1) - Please choose one

The Phong reflection model simplifies light-matter interactions into (essentially) 4 vectors and a number of constants. Which piece of the Phong model is responsible for giving spheres their bright white spots?

- ▶ **Specular** ✓
- ▶ Diffuse
- ▶ Ambient

Question No: 21 (Marks: 1) - Please choose one

When you hit a surface in ray tracing, generally shadow rays are tested against all objects in a scene. If these rays come back saying they hit an object in the scene, which of the following do you do?

- ▶ add all components (i.e. ambient, diffuse and specular) from that light source to the object.
- ▶ add all EXCEPT the ambient light from that light source to the object (i.e. diffuse and specular)
- ▶ **add only the ambient light from that light source to the object** ✓
- ▶ add none of the light from that light source to the object

Question No: 22 (Marks: 1) - Please choose one

The ColorSpace tool is a handy tool that we can use to interactively add two colours together to see the effects of the various strategies for handling oversaturated colours.

- ▶ False
- ▶ **True (Page 235)** ✓

Question No: 23 (Marks: 1) - Please choose one

A polygon is usually defined by a sequence of ----- and Edges.

- ▶ Ending lines
- ▶ Points
- ▶ **Vertices (Page 248)** ✓
- ▶ Edges

Question No: 24 (Marks: 1) - Please choose one

Which of the following properties of Bezier curves guarantees that a line passes through the control polygon as many times or more times than the line passes through the Bezier curve itself?

- ▶ End-point interpolation
- ▶ **Variation Diminishing** ✓
- ▶ Symmetry
- ▶ Convex-Hull

Question No: 25 (Marks: 1) - Please choose one

Parity is a concept used to determine which _____ lie within a polygon. (Choose best suitable answer)

- ▶ Edge
- ▶ Vertices
- ▶ **Pixel (Page 80) mid**
- ▶ None of the given

Question No: 26 (Marks: 1) - Please choose one

The actual filling process in boundary filling algorithm begins when a point _____ of the figure is selected.

- ▶ Outside the boundary
- ▶ **Inside the boundary (Page 102) mid**
- ▶ At boundary
- ▶ None of the given

Question No: 27 (Marks: 1) - Please choose one

Weiler-Atherton Polygon Clipping technique modifies the vertex-processing procedures for window boundaries so that _____ polygons are displayed correctly.

- ▶ Convex
- ▶ **Concave (Page 155)**
- ▶ Complex
- ▶ None of the given

Question No: 28 (Marks: 1) - Please choose one

If a line connecting any two points within a polygon does not intersect any edge, then it will be a _____ polygon.

- ▶ **Convex (Page 79)**
- ▶ Concave
- ▶ Complex
- ▶ None of the given

Question No: 29 (Marks: 1) - Please choose one

_____ can be defined as a mapping of point $P(x, y, z)$ onto its image $P'(x', y', z')$ in the view plane which constitutes the display surface.

- ▶ Mapping plane
- ▶ Three Coordinate Planes
- ▶ View plane
- ▶ **Projection (Page 193)**

Question No: 30 (Marks: 1) - Please choose one

The reflected light wave turns out to be a / an _____ case since light is reflected at the same angle as the incident wave (when the surface is smooth and uniform, as we'll assume for now).

- ▶ Unknown
- ▶ **Simple (Page 296)**
- ▶ Complex
- ▶ Abnormal

FINAL TERM EXAMINATION
Spring 2010
CS602- Computer Graphics

Question No: 1 (Marks: 1) - Please choose one

Computer graphics and computer vision are _____ disciplines.

Related (Page 24) mid

Interrelated

Same

Different

Question No: 2 (Marks: 1) - Please choose one

OpenGL has become the industry's most widely used and supported _____ graphics application programming interface (API), bringing thousands of applications to a wide variety of computer platforms.

2-Dimensional

3-Dimensional

2-Dimensional and 3-Dimensional (Page 301) ✓

Question No: 3 (Marks: 1) - Please choose one

----- sets the global idle callback to be 'func' so a GLUT program can perform background processing tasks or continuous animation when window system events are not being received.

glutIdle function (Page 313) ✓

glutKeyboardFunc

glutReshapeFunc

glutDisplayFunc

Question No: 4 (Marks: 1) - Please choose one

$X^2/a^2 - y^2/b^2 = 1$ is an equation of

Hyperbola (Page 70)

Parabola ($4px = y^2$)

None of given

Ellipse ($X^2/a^2 + y^2/b^2 = 1$)

Question No: 5 (Marks: 1) - Please choose one

To modify the object shape, shearing transformation cannot be used.

False (Page 192)

True

Question No: 6 (Marks: 1) - Please choose one

Rotation is performed around a fixed point called _____.

Pivot point rotation (Page 119)

Question No: 7 (Marks: 1) - Please choose one

Computer graphics is very helpful in producing graphical representations for scientific visualization and analysis

True (Page 9)

False

Question No: 8 (Marks: 1) - Please choose one

Save a line with both endpoints inside all clipping boundaries.

Trivial Reject

Trivial Accept (Page 142)

None of given

Question No: 9 (Marks: 1) - Please choose one

Dark lights are nothing more than lights in which one or more of the color values are _____.

Unknown

Negative (Page 235) ✓

Positive

Zero

Question No: 10 (Marks: 1) - Please choose one

A series of _____ computer operations convert an object's three-dimensional coordinates to pixel positions on the screen. Transformations, which are represented by matrix multiplication, include modeling, viewing, and projection operations. Such operations include rotation, translation, scaling, reflecting, orthographic projection, and perspective projection.

Three (Page 371) ✓

- Two
- Four
- Ten

Question No: 11 (Marks: 1) - Please choose one

At a physical surface, our eye's perception of the color depends on the distribution of photon energies that arrive and trigger our _____ cells.

- Eye
- Retina

Cone (Page 398) ✓

Question No: 12 (Marks: 1) - Please choose one

This projection technique has the direction of projection perpendicular to the viewing plane, but the viewing direction is NOT perpendicular to one of the principle faces.

Orthographic Parallel Projection

Axometric Parallel Projection (Page 194) ✓

Oblique Parallel Projection

Question No: 13 (Marks: 1) - Please choose one

In the Phong Reflection model, _____ light is the same everywhere.

Ambient ✓

- Diffuse
- Specular
- Emissive

Question No: 14 (Marks: 1) - Please choose one

A plane is two dimensional since in order to uniquely define any point on its surface we require _____ numbers.

Two (Page 356) ✓

- Three
- Four
- Five

Question No: 15 (Marks: 1) - Please choose one

In perspective projection, for your view to come out correctly, you will also want the _____ to pass through the middle of the screen.

X axis

Y axis

Z axis (Page 200) ✓

None

Question No: 16 (Marks: 1) - Please choose one

Neither floating-point nor signed integer values are clamped to the range _____ before updating the current color.

0 , -1.0

-1 , 1

1 , -1

0, 1 (Page 321) ✓

Question No: 17 (Marks: 1) - Please choose one

An object's _____ determine its orientation relative to the light sources. For each vertex, OpenGL uses the assigned normal to determine how much light that particular vertex receives from each light source.

Unit

Normal (Page 400) ✓

None of given

Question No: 18 (Marks: 1) - Please choose one

Which was the oldest shading model?

Flat Shading (Page 245)

Phong Shading

Gouraud Shading ✓

Question No: 19 (Marks: 1) - Please choose one

Which of the following affine transforms does NOT affect vectors?

Scale

Rotation

Shear

Translation

Final Term MCQS and Quizzes

Question # 1 of 10 (Total Marks: 1) Select correct option:

We want our scene to look more realistic, we should use _____ lights.

Ambient (Page 282)

Point Parallel

Spot

None of the given

Question # 2 of 10 (Total Marks: 1) Select correct option:

This is a simple example of line clipping: the display window is the canvas and also the default -----
-, thus all line segments inside the canvas are drawn.

Clipping Rectangle (Page 141)

Clipping Circle

Clipping Polygon

Clipping Angle

Question # 3 of 10 (Total Marks: 1) Select correct option:

One problem with Gouraud shading is that the ----- intensities can never be greater than the intensities at the edges.

Triangles (Page 246)

Squares

Rectangles

Polygons

Question # 4 of 10 (Total Marks: 1) Select correct option:

There is more penetration of light in case of _____ surfaces.

Conductor (like metals)

Nonconductor (like dielectrics) (Page 235)

Both conductor and nonconductor

None of the given

Question # 5 of 10 (Total Marks: 1) Select correct option:

_____ lights should be avoided because they are not for real time environment.

Point

Parallel

Spot (Page 244)

None of the given

Question # 6 of 10 (Total Marks: 1) Select correct option:

The physical range of colors a device can display is called _____ of the device.

Sharpness

Gamut (Page 229)

Colouring

Colouring with Sharpness

Question # 7 of 10 (Total Marks: 1) Select correct option:

_____ is simply the calculation of color reflected by the surface.

Shading (Page 240)

Clamping

Scaling

None of the given

Question # 8 of 10 (Total Marks: 1) Select correct option:

When obtaining normals for a triangle, which of the following mathematical constructs is NOT used?

Vector normalization

Vector cross products

Vector dot products

Point-Point subtraction

Question # 9 of 10 (Total Marks: 1) Select correct option:

Loosely, the alpha component of the RGBA quad represents the _____ of a surface.

Opaqueness (Page 227)

Light
Darkness
Shine

Question # 10 of 10 (Total Marks: 1) Select correct option:

An algorithm that clips a polygon must deal with many ----- cases. The case is particularly noteworthy in that the concave polygon is clipped into ----- isolate polygons.

Similar, three

Different, two (Page 146)

Different, three
Similar, two

Question # 1 of 10 (Total Marks: 1) Select correct option:

_____ lighting is not dependent on any source.

Ambient [Click here for detail](#)

Diffuse
Specular
Emissive

Question # 2 of 10 (Total Marks: 1) Select correct option:

In order to get a more realistic representation of lighting, we'll need to understand how light passes through a medium and how hitting the boundary layer at the ----- of two media can affect light's properties.

Intersection (Page 296)

Union
Endpoints
Edges

Question # 3 of 10 (Total Marks: 1) Select correct option:

Lambertian shading was used mostly back when computers weren't fast enough to do _____ in real time.

Phong shading
Processing
Shading

Gouraud shading (Page 245)

Question # 4 of 10 (Total Marks: 1) Select correct option:

In Perspective Projection the point of View (POV) must lie on the _____.

All axis

Z axis (Page 200)

X axis

Y axis

Question # 5 of 10 (Total Marks: 1) Select correct option:

If we want any object to glow, we should use _____ lights.

Ambient

Diffuse

Specular

Emissive (Page 240)

Question # 6 of 10 (Total Marks: 1) Select correct option:

There are not many different ways of representing the intensity of a particular color element.

True

False (Page 276)

Question # 7 of 10 (Total Marks: 1) Select correct option:

In Perspective Projection the screen plane must be parallel to the_____.

Y-Z plane

X-Y plane (Page 200)

Z-Y plane

X-Z plane

Question # 8 of 10 (Total Marks: 1) Select correct option:

_____ light is reflected in all directions from surface.

Ambient

Diffuse (page 239)

Specular

Emissive

Question # 9 of 10 (Total Marks: 1) Select correct option:

A space curve can be confined to a plane.

Yes

No (Page 331)

Question # 10 of 10 (Total Marks: 1) Select correct option:

To convert the information in the A matrix into that required for the P matrix, we do some simple matrix algebra, First we have $UA=UNP$ then Simply $A = \text{-----}$

UP

NP (Page 333)

UN

None

Question # 1 of 10 (Total Marks: 1) Select correct option:

Perspective projection is specified with the function glFrustum().

Yes (Page 376)

No

Question # 2 of 10 (Total Marks: 1) Select correct option:

Choose a camera lens or adjust the zoom

projection transformation (Page 372)

viewport transformation

modeling transformation'

viewing transformation

Question # 3 of 10 (Total Marks: 1) Select correct option:

Using a lighting model based upon the Blinn Phong model means that we'll always get a uniform specular highlight based upon the color of the ----- light and material, which means that all reflections based on this model, will be reminiscent of plastic.

Union

Refracting

Intersection

Reflecting (Page 296)

Question # 4 of 10 (Total Marks: 1) Select correct option:

Refractive index is a function of temperature, mostly due to density changes in materials with changes in temperature.

True (Page 300)

False

Question # 5 of 10 (Total Marks: 1) Select correct option:

Length L depends on the angle alpha and the z coordinate of the point to be projected and L can be represented by -----.

$z * 1 / \tan(\alpha)$ (Page 198)

$z * L^2$

$z * 1 / \tan(\beta)$

$z * 1 / \tan(\gamma)$

Question # 6 of 10 (Total Marks: 1) Select correct option:

The traditional approach in real-time computer graphics has been to calculate lighting at a vertex as a sum of the _____ light.

Ambient

Ambient, diffuse, and specular (Page 281)

Specular

Diffuse, and specular

Question # 7 of 10 (Total Marks: 1) Select correct option:

Another way to define a space curve by using intermediate points and the tangents at each end for making the curve

Yes

No (Page 334)

Question # 8 of 10 (Total Marks: 1) Select correct option:

An independent consortium, the OpenGL Architecture Review Board, guides the OpenGL specification. With broad industry support, OpenGL is the only truly open, vendor-neutral, ----- graphics standard.

Tertiary

Binary

Single platform

Multiplatform (Page 301)

Question # 9 of 10 (Total Marks: 1) Select correct option:

glutReshapeWindow requests a change in the size of the current window. The width and height parameters are size extents in pixels. The width and height must be ----- values.

Neutral

Negative

Positive (Page 311)

None of the given

Question # 10 of 10 (Total Marks: 1) Select correct option:

A space curve is not confined to a plane. It is free to twist through space. To define a space curve we must use parametric functions that are -----.

Binary polynomials

Mono polynomials

Quadratic polynomials

Cubic polynomials (Page 331)

Question # 1 of 10 (Total Marks: 1) Select correct option:

Refractive index is a function of temperature, mostly due to changes in ----- of materials with changes in temperature. A simple correction can be applied in most circumstances to allow us to use a value

given at one temperature at another.

Density (Page 300)

pressure
nature
volume

Question # 2 of 10 (Total Marks: 1) Select correct option:

If we assign a different value to the parametric variable for the intermediate point, then we obtain different values for the coefficients. This, in turn, means that a different curve is produced, although it passes through the ----- three points.

isolate
different

same (Page 328)

none

Question # 3 of 10 (Total Marks: 1) Select correct option:

The attenuation formula is $f = \frac{1}{C + Ld + Qd^2}$, where C, L and Q are the constant, linear and quadratic attenuation factors and d is the distance between the vertex being lit and the light source.

$\frac{1}{C + Ld + Qd^2}$ [Click here for detail](#)

$\frac{1}{C + Ld + Qd}$
 $\frac{1}{C + L + d + Qd^2}$
 $\frac{1}{Cd + Ld + Qd^2}$

Question # 4 of 10 (Total Marks: 1) Select correct option:

Bezier curve is tangent to the lines connecting _____.

First two points
Last two points

First two points and last two point (Page 340)

None of the given

Question # 5 of 10 (Total Marks: 1) Select correct option:

End points and an intermediate point on the curve, then we now ----- quantities that we can express in terms of these coefficients (3 points x 3 coordinates each), and we can use these three points to define a unique curve.

Six
Three
Two

Nine (Page 326)

Question # 6 of 10 (Total Marks: 1) Select correct option:

Choose a camera lens or adjust the zoom

projection transformation (Page 372)

viewport transformation
modeling transformation
viewing transformation

Question # 7 of 10 (Total Marks: 1) Select correct option:

_____ OpenGL function is used for aiming and positioning the camera towards the object

glLoadIdentity() (Page 375)

gluLookAt()

glFrustum()

None of Above

Question # 8 of 10 (Total Marks: 1) Select correct option:

A parametric curve is one whose defining equations are given in terms of a -----, common, independent variable called the parametric variable.

Triple

Double

Single (Page 325)

None of the given

Question # 9 of 10 (Total Marks: 1) Select correct option:

The reflection coefficients are in the ----- range and are specified as part of the material property. However, they are strictly empirical and since they simply adjust the overall intensity of the material color, the material color values are usually adjusted so the color intensity varies rather than using a reflection coefficient.

[0, 10]

[0, 1] (Page 281)

[0, 5]

[0, 2]

Question # 10 of 10 (Total Marks: 1) Select correct option:

To ensure a smooth transition from one section of a piecewise _____ to the next, we can impose various continuity conditions at the connection points

non parametric curve

parametric curve

polygon vector (not confirm) (Page 245)

None of the these

Question # 1 of 10 (Total Marks: 1) Select correct option:

The curve is always contained within the _____ of the control points

Tangents

Convex Hull (Page 340)

Subdivision

None of Above

Question # 2 of 10 (Total Marks: 1) Select correct option:

Projection can be defined as a mapping of point $P(x, y, z)$ onto its image $P'(x', y', z')$ in the -----, which constitutes the display surface. The mapping is determined by a projection line called the projector that passes through P and intersects the -----.

Two Coordinate Planes

View plane or projection plan (Page 193)

Three Coordinate Planes

Mapping plane

Question # 3 of 10 (Total Marks: 1) Select correct option:

Determine how large we want the final photograph to be - for example, we might want it enlarged

projection transformation

viewport transformation (Page 372)

modeling transformation

viewing transformation

Question # 4 of 10 (Total Marks: 1) Select correct option:

Ambient light is the light that comes from -----directions, thus all surfaces are illuminated equally regardless of orientation. However, this is a big hack in traditional lighting calculations since "real" ambient light really comes from the light reflected from the "environment."

All (Page 281)

Opposite

Same

Four different

Question # 5 of 10 (Total Marks: 1) Select correct option:

Silhouette edges occur when dot product of surface normal vector and the view vector is _____.

Zero (Page 345)

One

Both zero and one

Question # 6 of 10 (Total Marks: 1) Select correct option:

If the current matrix (according to `glMatrixMode`) is multiplied by the translation matrix, with the product replacing the current matrix. That is, if M is the current matrix and T is the translation matrix, then M is replaced with -----.

$M-T$

$M+T$

M/T

$M*T$ (Page 317)

Question # 7 of 10 (Total Marks: 1) Select correct option:

Arrange the scene to be photographed into the desired composition

projection transformation

viewport transformation

modeling transformation (Page 317)

viewing transformation

Question # 8 of 10 (Total Marks: 1) Select correct option:

In the forms of texture mapping, Image to world space and world space to image, each suffers from different problems related to minification and magnification. Which of the two shows the following problem: When the texture is larger than the screen space it maps to, many texture units (texels) are never sampled?

Image to world space

World space to image

X-axis

Y-axis

Question # 9 of 10 (Total Marks: 1) Select correct option:

Imagine a curve in three-dimensional space, each point on the curve has a unique set of coordinates: a specific x value, y value, and z value. Each coordinate is controlled by a ----- parametric equation.

Opposite

Similar

Separate (Page 325)

Question # 10 of 10 (Total Marks: 1) Select correct option:

We allow the parametric variable to take on values only in the interval -----.

$-1 \leq u \leq 0$

$0 \leq u \leq 2$

$0 \leq u \leq 1$ (Page 326)

$-1 \leq u \leq 1$

Question # 1 of 10 (Total Marks: 1) Select correct option:

Bezier curve can represent the more complex piecewise _____ curve.

Polynomial (Page 338)

Exponential

Cubic

None of above

Question # 2 of 10 (Total Marks: 1) Select correct option:

A fractal is generally a property called _____.

Fractal Dimension

Self-similarity (Page 355)

Koch Curve

None of above

Question # 3 of 10 (Total Marks: 1) Select correct option:

Normalized cross product of two vectors on that surface provides normal vector

Yes (Page 347)

No

Question # 4 of 10 (Total Marks: 1) Select correct option:

Every point on a curve has a straight line associated with it called the _____

State line

tangent line (Page 334)

curved line

None of the given

Question # 5 of 10 (Total Marks: 1) Select correct option:

The value returned is a unique small integer identifier for the window. The range of allocated identifiers starts at -----, This window identifier can be used when calling glutSetWindow.

Three

Two

One (Page 308)

Zero

Question # 6 of 10 (Total Marks: 1) Select correct option:

Curve and surface equations can be expressed in either a parametric or a non parametric form.

True [click here for detail](#)

False

Question # 7 of 10 (Total Marks: 1) Select correct option:

Bernstein polynomial functions are the basic functions of _____ curves.

NURBS

Bezier (Page 342)

Both NURBS and Bazier

None of the given

Question # 8 of 10 (Total Marks: 1) Select correct option:

Geometric patterns that is repeated at ever smaller scales to produce irregular shapes and surfaces are called _____

Geometric patterns

Fractals (Page 352)

Animated components

Segments

Question # 9 of 10 (Total Marks: 1) Select correct option:

The Phong reflection model simplifies light-matter interactions into (essentially) 4 vectors and a number of constants. Each piece of the Phong model uses different vectors and constants. Which portion does NOT include taking a dot product?

Ambient

Diffuse

Specular

Question # 10 of 10 (Total Marks: 1) Select correct option:

This projection technique does NOT have the direction of projection perpendicular to the viewing plane.

a) Orthographic Parallel Projection

b) Axonometric Parallel Projection

c) Oblique Parallel Projection (Page 197)

Question # 1 of 10 (Total Marks: 1) Select correct option:

In OpenGL, there are several different matrices. We have discussed two of them in class. Which one of the below would be used in conjunction with a glRotatef function call?

a) GL_MODELVIEW

b) GL_PROJECTION

Question # 2 of 10 (Total Marks: 1) Select correct option:

In OpenGL, there are several different matrices. We have discussed two of them in class. Which one of the below would be used in conjunction with glFrustum?

a) GL_MODELVIEW

b) GL_PROJECTION

Question # 3 of 10 (Total Marks: 1) Select correct option:

Which of the following is the order that geometry operations are performed in OpenGL (where we read the order from left to right)?

a) GL_PROJECTION GL_MODELVIEW Perspective division

b) GL_MODELVIEW GL_PROJECTION Perspective division

c) Perspective division GL_PROJECTION GL_MODELVIEW

d) GL_MODELVIEW Perspective division GL_PROJECTION

e) GL_PROJECTION Perspective division GL_MODELVIEW

Question # 4 of 10 (Total Marks: 1) Select correct option:

Which of the following is NOT true about quaternions?

a) They are made up of 4 numbers

- b) They should always be normalized to length 1
- c) They can be used to represent all affine transforms**
- d) They can be used to define the rotation of an object

Question # 5 of 10 (Total Marks: 1) Select correct option:

In class, we discussed three forms of shading for “Utah” graphics. Which was the first to use per vertex normals?

a) Flat Shading (Page 245)

- b) Phong Shading
- c) Gouraud Shading

Question # 6 of 10 (Total Marks: 1) Select correct option:

Given any implicit equation, which of the following is true for all (x, y, z) that make the equation exactly zero?

- a) All those points are inside the object defined by the implicit equation
- b) All those points are on the surface of the object defined by the implicit equation** [Click here 4 detail](#)
- c) All those points are outside the object defined by the implicit equation
- d) You can't know anything without knowing what the implicit equation is

Question # 7 of 10 (Total Marks: 1) Select correct option:

When solving ray-sphere intersections using the implicit equation for a sphere, you must solve the quadratic equation. Which of the following do you know if the $B^2 - 4AC$ (i.e. the part under the square root) is negative?

- a) The ray intersects the sphere at a negative t... discard this result
- b) The ray intersects the sphere at a positive t... continue to the solution
- c) The ray does not intersect the sphere... discard this result**
- d) The ray begins inside the sphere... this is a special case

Question # 8 of 10 (Total Marks: 1) Select correct option:

_____ sets the global idle call back to be 'func' so a GLUT program can perform background processing tasks or continuous animation when window system events are not being received.

glutIdleFunc (Page 313)

glutMainLoop
glutDisplayFunc
glutReshapeFunc

Question # 9 of 10 (Total Marks: 1) Select correct option:

A tangent vector certainly defines the slope at one end of the curve, but a vector has characteristics of.....

direction
magnitude

both direction and magnitude (Page 336)

None of the given

Question # 10 of 10 (Total Marks: 1) Select correct option:

The degree of a Bezier curve is equal to $n-1$, where n is the number of control points

Yes (Page 339)

No

Question # 1 of 10 (Total Marks: 1) Select correct option:

Bit mask to select a window with multisampling support. If multisampling is not available, a ----- window will automatically be chosen.

Non-multisampling (Page 310)

Multisampling

Mono-multisampling

Di-multisampling

Question # 2 of 10 (Total Marks: 1) Select correct option:

OpenGL is well structured with an intuitive design and logical commands. Efficient OpenGL routines typically result in applications with fewer lines of code than those that make up programs generated using other graphics libraries or packages. In addition, OpenGL drivers ----- information about the underlying hardware, freeing the application developer from having to design for specific hardware features.

Encapsulate (Page 302)

Shows

Hibernates

None of the given

Question # 3 of 10 (Total Marks: 1) Select correct option:

With similar expressions for $y(u)$ and $z(u)$. Again the a , b , c and d terms are constant coefficients. As we did with Equation for a plane curve, we combine the $x(u)$, $y(u)$, and $z(u)$ expressions into a single vector equation $P(u) = \text{-----}$.

$Au^2+bu+cu+d$

$Au^4+bu^3+cu^2+d$

Au^3+bu^2+cu+d

Au^3+bu^2+cu+d (Page 331)

Question # 4 of 10 (Total Marks: 1) Select correct option:

Each number that makes up a matrix is called an _____ of the matrix.

Element (Page 106)

Variable

Value

Component

Question # 5 of 10 (Total Marks: 1) Select correct option:

The matrix generated by gluPerspective is multiplied by the current matrix, just as if glMultMatrix were called with the generated matrix. To load the perspective matrix onto the current matrix stack instead, precede the call to gluPerspective with a call to -----.

glRotated

gluPerspective (Page 318)

glTranslated

glLoadIdentity

Question # 6 of 10 (Total Marks: 1) Select correct option:

Which one of the following step is not involved to write pixel using video BIOS services.

Setting desired video mode

Using BIOS service to set color of a screen pixel

Calling BIOS interrupt to execute the process of writing pixel.

Using OpenGL service to set color of a screen pixel (Page 45)

Question # 7 of 10 (Total Marks: 1) Select correct option:

Shadow mask methods can display a _____ range of colors.

Small

Wide (Page 28)

Random

Crazy

Question # 8 of 10 (Total Marks: 1) Select correct option:

Using Cohen-Sutherland line clipping, it is impossible for a vertex to be labeled 1111.

True

False

Question # 9 of 10 (Total Marks: 1) Select correct option:

Intensity of the electron beam is controlled by setting _____ levels on the control grid, a metal cylinder that fits over the cathode.

Amplitude

Current

Voltage (Page 26)

Electron

Question # 10 of 10 (Total Marks: 1) Select correct option:

The circle and ellipse are symmetric across 8 octants.

☐ True

☒ False (Page 60)

Question # 1 of 10 (Total Marks: 1) Select correct option:

UV light is used in Plasma Panel displays to excite phosphor.

☒ True [Click here for detail](#)

☐ False

Question # 2 of 10 (Total Marks: 1) Select correct option:

Which of the following is not true about matrices?

☐ $A + B = B + A$

☐ $a(A + B) = aA + aB$

☒ $(AT)T = AT$

☐ $A + (B + C) = (A + B) + C$

Question # 3 of 10 (Total Marks: 1) Select correct option:

According to Odd Parity Rule, a point is inside the polygon, if:

☐ Line from an outside point to this point does not cross the edges odd number of times

☐ Line from any point to this point crosses the edges odd number of times

☒ Line from an outside point to this point crosses the edges odd number of times (Page 80)

☐ Line from this point to any point outside the polygon intersects any edge

Question # 4 of 10 (Total Marks: 1) Select correct option:

As opposed to direct memory access method, BIOS routines provide an easier and faster method of drawing pixels on screen.

☐ True

☒ False (Page 47)

Question # 5 of 10 (Total Marks: 1) Select correct option:

When a point $P(x,y)$ is rotated by θ the coordinates of transformed point P' are given as:

☒ $x' = x \cos(\theta) - y \sin(\theta), y' = x \sin(\theta) + y \cos(\theta)$ (Page 181)

☐ $x' = y \cos(\theta) - x \sin(\theta), y' = y \sin(\theta) + x \cos(\theta)$

☐ $x' = x \cos(\theta) + y \sin(\theta), y' = x \sin(\theta) - y \cos(\theta)$

☐ $x' = y \cos(\theta) + x \sin(\theta), y' = y \sin(\theta) - x \cos(\theta)$

Question # 6 of 10 (Total Marks: 1) Select correct option:

The amount of time it takes to illuminate a specific location on phosphor coated screen is called Persistence.

☐ True

☒ False (Page 27)

Question # 7 of 10 (Total Marks: 1) Select correct option:

Incremental line drawing algorithm makes use of the equation of straight line.

☐ True

☒ False (Not sure)

Question # 8 of 10 (Total Marks: 1) Select correct option:

In matrix multiplication:

- ☐ The two matrices must be square
- ☒ **The number of rows of 1st matrix must be the same as the number of columns of the second.**
- ☐ The two matrices must either be row matrices or column matrices
- ☐ The number of columns of 1st matrix must be the same as the number of rows of the second.

Question # 9 of 10 (Total Marks: 1) Select correct option:

In Horizontal retrace, after completion of all the pixels in a scan line, the refreshing continues from the 1st pixel of the next scan line.

- ☐ True
- ☒ **False (Page 28)**

Question # 10 of 10 (Total Marks: 1) Select correct option:

Parity Rule is used to determine whether a pixel is inside a polygon or not.

- ☒ **True (Page 80)**
- ☐ False

Question # 1 of 10 (Total Marks: 1) Select correct option:

When dot product of two vectors equals zero, this implies that the two vectors are:
parallel to each other

orthogonal (perpendicular) to each other. (Page 177)

intersect each other
equal to each other

Question # 2 of 10 (Total Marks: 1) Select correct option:

In Pixmap exactly one bit is used to hold color value of each pixel.

- ☐ True
- ☒ **False (Page 28)**

Question # 3 of 10 (Total Marks: 1) Select correct option:

The Boundary Fill and Flood Fill algorithms:

- ☐ Must use 4-connected approach
- ☐ Must use 8-connected approach
- ☒ **May use 4-connected or 8-connected approach [click here for details](#)**
- ☐ Must not use recursive approach

Question #4 of 10 (Total Marks: 1) Select correct option:

To show 256 colors, the no of bits required for each pixel are

a. 8 (Page 39)

- b. 16
- c. 32
- d. 64

Question # 5 of 10 (Total Marks: 1) Select correct option:

25 * 80 resolution with 16 colors supports

a. Text mode (Page 43)

- b. Graphics mode
- c. Both
- d. None

Question # 6 of 10 (Total Marks: 1) Select correct option:

Two matrices are said to be equal, if they have

- a. same order
- b. same corresponding elements

c. Same order and same corresponding elements.

- d. Different elements.

Question # 7 of 10 (Total Marks: 1) Select correct option:

Two points are said to be collinear, if they lie on the

a. same line (page 53)

- b. different but parallel lines
- c. either on the same plane or two parallel planes
- d. different plane

Question # 8 of 10 (Total Marks: 1) Select correct option:

A Polygon is convex, if the line connecting:

- a. Any two points outside the polygon intersects its boundary

b. Any two points inside the polygon don't intersect any edges of the polygon. (Page 78)

- c. A point inside the boundary with any point outside, does not intersect the polygon boundary
- d. Any two vertices, intersects some edge of polygon.

Question # 9 of 10 (Total Marks: 1) Select correct option:

Polygon is complex, if the line connecting:

- a. Any two points outside the polygon intersects its boundary.
- b. Any two points inside the polygon intersects its boundary.
- c. A point inside the boundary with any point outside does not intersect the polygon boundary.

d. Any two vertices, intersects some edge of polygon. (Page 79)

Question # 10 of 10 (Total Marks: 1) Select correct option:

The equation of hyperbola centered at origin (if the transverse axis is along x -axis) can be given as:

- a. $x^2/b^2 + y^2/a^2 - 1 = 0$
- b. $x^2/b^2 + y^2/a^2 + 1 = 0$
- c. $x^2/a^2 - y^2/b^2 - 1 = 0$ [Click here for detail](#)
- d. $x^2/b^2 - y^2/a^2 - 1 = 0$

Question # 1 of 10 (Total Marks: 1) Select correct option:

Which one is not valid out code to perform trivial accept / reject test in line clipping:

- a. 1101
- b. 1001 (Page 143)
- c. 0101
- d. 0110

Question # 2 of 10 (Total Marks: 1) Select correct option:

Which one of the following is not the graphics library is use:

- a. FastGL
- b. OpenGL
- c. DirectX
- d. EasyGL (Page 42)

Question # 3 of 10 (Total Marks: 1) Select correct option:

In video text memory, _____ are used to display a character.

2 bytes (Page 43)

- 4 bytes
- 8 bytes
- 16 bytes

Question # 4 of 10 (Total Marks: 1) Select correct option:

In _____ algorithm(s), old color must be read before it is invoked.

Scan line filling

Flood filling (Page 104)

- Both scan line and flood filling
- Scan filling

Question # 5 of 10 (Total Marks: 1) Select correct option:

According to the architecture of raster graphics system, display processor memory will act as_____.

Video controller (Page 36)

System memory

Frame buffer

Video controller and System memory

Question # 6 of 10 (Total Marks: 1) Select correct option:

Global coordinate systems can be defined with respect to local coordinate system

True

False (Page 255)

Question # 7 of 10 (Total Marks: 1) Select correct option:

Cross product of two vectors result in a _____.

Magnitude

Vector (Page 347)

Scalar

Value

Question # 8 of 10 (Total Marks: 1) Select correct option:

Shortcoming of Sutherland-Hodgeman Algorithm is concave polygons may be displayed with extraneous lines

True (Page 155)

False

Question # 9 of 10 (Total Marks: 1) Select correct option:

In 2D transformations, rotations applied to a point P can be denoted as_____ (Where? represents theta).

$P' = R(\theta_1 + \theta_2) \cdot P$ (Page 119)

$P' = (R(\theta_1) + R(\theta_2)) \cdot P$

$P' = R(\theta_1 \times \theta_2) \cdot P$

$P' = R(\theta_1) \cdot P$

Question # 10 of 10 (Total Marks: 1) Select correct option:

A straight line can be moved to another location by applying _____ to each of the line endpoints and redrawing the line between the new coordinates.

Rotation

Translation (Page 118)

Reflection

Scaling factor

Question # 1 of 10 (Total Marks: 1) Select correct option:

If the values of scaling factors s_x and s_y are less than 1, then size of object will be _____.

Reduced (Page 121)

Remain same

Enlarged

Shear

Question # 2 of 10 (Total Marks: 1) Select correct option:

Locations can be translated or "transformed" from one coordinate system to the other.

True (Page 163)

False

Question # 3 of 10 (Total Marks: 1) Select correct option:

If the value of scaling factors s_x and s_y is greater than 1, then size of objects will be _____.

Reduced

Enlarged (Page 121)

Remain same

Shear

Question # 4 of 10 (Total Marks: 1) Select correct option:

A column matrix is also known as _____.

Column vector (Page 107)

Row vector

Vector

Simple Matrix

Question # 5 of 10 (Total Marks: 1) Select correct option:

Dot product of two vectors results in _____ quantity.

Scalar (Page 176)

Vectors

Magnitude

Value

Question # 6 of 10 (Total Marks: 1) Select correct option:

Set up your tripod and pointing the camera at the scene

projection transformation

viewport transformation

modeling transformation

viewing transformation (Page 372)

Question # 7 of 10 (Total Marks: 1) Select correct option:

_____ is based on characteristic size or scale

Fractal Geometry

Traditional Geometry

Euclidean Geometry (Page 359)

None of Above

Question # 8 of 10 (Total Marks: 1) Select correct option:

Bernstein polynomial functions are the basic functions of _____ curves.

NURBS

Bezier (Page 342)

Both NURBS and Bazier

None of the given

Question # 9 of 10 (Total Marks: 1) Select correct option:

Which of the following does NOT figure into the Field of View of a pinhole camera?

a) The direction of projection

b) The distance from the center of projection to the projection plane

c) The size of the projection plane

Question # 10 of 10 (Total Marks: 1) Select correct option:

When transforming a random Axis-Aligned Bounding Box defined by the points (nearx, neary, nearz) and (farx, fary, farz) to the standard orthographic viewing box, which affine transforms are used?

a) shear and translation

b) rotation and scale

c) scale and shear

d) translation and scale

Question # 1 of 10 (Total Marks: 1) Select correct option:

In class, we discussed the purpose of the front and back clipping planes in OpenGL. Which of the following was NOT a purpose for using clipping planes?

a) division by zero

b) objects behind the center of projection mapping onto the projection plane

c) avoiding the problems of infinite viewing volume size

Question # 2 of 10 (Total Marks: 1) Select correct option:

In class, we discussed how the image of the Double Eagle Tanker was obtained for the large poster in the main hall of Sitterson. It required rendering several perspective images using OpenGL. Which of the following was NOT a step required in that process?

a) handling projection planes non-orthogonal to the viewing direction

- b) cutting a single projection plane into many separate projection planes
- c) rotating the viewing direction to be the same as the $-z$ direction
- d) handling several different centers of projection**

Question # 3 of 10 (Total Marks: 1) Select correct option:

In “Utah” graphics, lights are simplified in order to approximate light/matter interaction with a minimum amount of work. Which of the following is NOT true about the simplifications made in “Utah” graphics lights?

- a) Light intensity and color are folded into one value.
- b) Lights are assumed to have zero size
- c) Spotlights cannot be handled**
- d) Soft shadows cannot be handled

Question # 4 of 10 (Total Marks: 1) Select correct option:

The Phong reflection model simplifies light-matter interactions into (essentially) 4 vectors and a number of constants. Which of the following is NOT a vector needed for the Phong reflection model?

- a) Surface Normal
- b) Direction to Viewer
- c) Direction to Material Center**
- d) Direction to Light

Question # 5 of 10 (Total Marks: 1) Select correct option:

In the Phong Reflection model, ambient light is the same everywhere.

- a) true**
- b) false

Question #6 of 10 (Total Marks: 1) Select correct option:

We discussed several global illumination algorithms in class. Which of the following listed can properly handle diffuse-diffuse reflections

- a) Ray Tracing
- b) Radiosity**
- c) Photon Mapping
- d) RenderMan

Question # 7 of 10 (Total Marks: 1) Select correct option:

We discussed several global illumination algorithms in class. Which of the following is generally characterized by shiny spheres and checkerboards?

- a) Ray Tracing**
- b) Radiosity
- c) Photon Mapping
- d) RenderMan

Question # 8 of 10 (Total Marks: 1) Select correct option:

We discussed several global illumination algorithms in class. Which of the following listed can properly handle caustics?

- a) Ray Tracing
- b) Radiosity
- c) Photon Mapping**
- d) RenderMan

Question # 9 of 10 (Total Marks: 1) Select correct option:

In radiosity a matrix is formed of size $m \times m$. Which of the following can be known if the (i, j) position in the matrix is zero?

- a) patch i is much larger than patch j
- b) patch i is much smaller than patch j
- c) patch i is visible from patch j
- d) patch i is not visible from patch j**

Question # 10 of 10 (Total Marks: 1) Select correct option:

When solving for ray-polygon intersections, after intersecting the ray with a plane, the dominant component of the plane normal is found. this is used to

- a) ignore any component other than the dominant when you project to 2D
- b) ignore the dominant component when you project to 2D**
- c) solve the inside-outside test only for that component

Question # 1 of 10 (Total Marks: 1) Select correct option:

If solving for inside-outside of a triangle using barycentric coordinates, and you have two barycentric coordinates solved ($bc_1 = .57$, $bc_2 = .62$) which of the following do you know?

- a) The point is outside the triangle**
- b) The point is on the boundary of the triangle
- c) The point is inside the triangle
- d) The hit point is on the “back face” of the triangle

Question # 2 of 10 (Total Marks: 1) Select correct option:

The majority of the execution time of a ray tracer is spent in ray-object intersection code.

- a) true**
- b) false

Question # 3 of 10 (Total Marks: 1) Select correct option:

Snell's law is used to calculate which one of the following rays?

- a) start rays
- b) shadow rays
- c) reflection rays

d) transmission rays

Question # 4 of 10 (Total Marks: 1) Select correct option:

You might as well recurse on reflection and transmission rays until you hit nothing. This always is bound to happen.

a) true

b) false

Question # 5 of 10 (Total Marks: 1) Select correct option:

Which of the following Ray Tracing additions works to remove jaggies alone?

Mkkg

b) jittering

Question # 6 of 10 (Total Marks: 1) Select correct option:

In the Pixar short “Geri’s Game”, the character Geri was created using which of the following techniques?

a) Fractals

b) Bump mapping

c) Environment mapping

d) Catmull-Clark Subdivision Surfaces

Question # 7 of 10 (Total Marks: 1) Select correct option:

In the Pixar short “Geri’s Game”, the trees in the background were created using which of the following techniques?

a) Fractals

b) Bump mapping

c) Environment mapping

d) Catmull-Clark Subdivision Surfaces

Question # 8 of 10 (Total Marks: 1) Select correct option:

The basic functions $f_i(u)$ in Bezier curve must be symmetric with respect to u and $(u-2)$

yes

no (Page 341)

Question # 9 of 10 (Total Marks: 1) Select correct option:

In the Pixar short “Geri’s Game”, Geri’s glasses seemed to bend the light as it passed through.

Which of the following techniques was used?

a) Fractals

b) Bump mapping

c) Environment mapping

d) Catmull-Clark Subdivision Surfaces

Question # 10 of 10 (Total Marks: 1) Select correct option:

A polygon is usually defined by a sequence of vertices and -----

Edges (Page 146)

Vertices

Points Ending lines

Question # 1 of 10 (Total Marks: 1) Select correct option:

Gouraud shading is the current de jure shading standard in accelerated ----- hardware.

2Dimensional

3Dimensional (Page 245)

Multidimensional

None

Question #2 of 10 (Total Marks: 1) Select correct option:

_____ is based on characteristic size or scale

Fractal Geometry (Fractal shapes are self similar and independent of size or scaling)

Traditional Geometry

Euclidean Geometry (Euclidean shapes normally have a few characteristic sizes or length scales) (Page 359)

None of Above

Question # 3 of 10 (Total Marks: 1) Select correct option:

Arrange the scene to be photographed into the desired composition

projection transformation

viewport transformation

modeling transformation (Page 372)

viewing transformation

Question # 4 of 10 (Total Marks: 1) Select correct option:

Which language API defines graphics operations independent of the operating system or computer hardware?

Additional hardware specific libraries are used to provide an interface between API and the hardware and between the user and the platform specific windowing system.

a. DirectX

b. Graphix Windowing Toolkit

c. CGI

d. OpenGL (Page 302)

Question # 5 of 10 (Total Marks: 1) Select correct option:

When transforming a random Axis-Aligned Bounding Box defined by the points (nearx, neary, nearz) and (farx, fary, farz) to the standard orthographic viewing box, which affine transforms are used?

a) shear and translation

b) rotation and scale

- c) scale and shear
d) translation and scale

Question # 6 of 10 (Total Marks: 1) Select correct option:

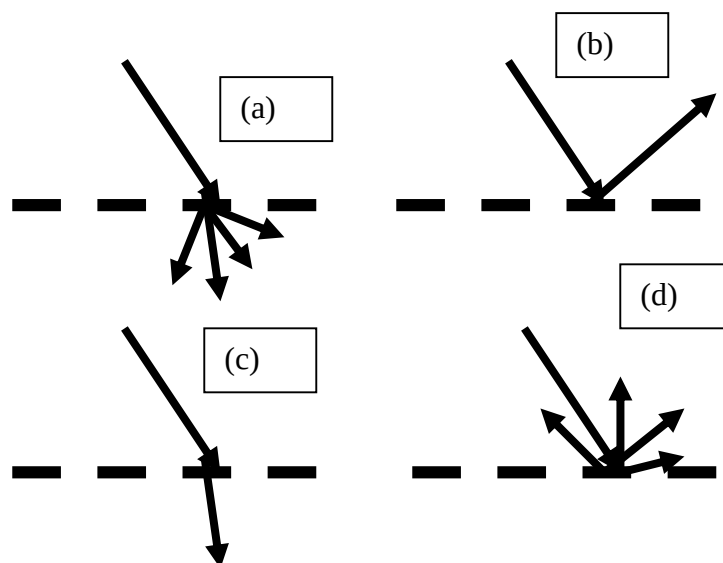
The _____ functions multiply the current matrix by a rotation matrix.

[glRotated](#) and [glRotatef](#) (Page 318)

Question # 7 of 10 (Total Marks: 1) Select correct option:

Match the pictures on the right with the corresponding term on the left. The arrows in the picture denote light rays. The dashed lines represent the material type to be considered. The key is in the interaction of the light rays with the material.

Specular
Diffuse
Transparent
Translucent



b,d,c,a [Click here for detail](#)
d,b,c,a
a,c,b,d
c,d,b,a

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Dot product of two vectors result in ----- quantity,

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Answer (Please select your correct option)

☒ Scalar  correct

☐ Vectors

☐ None of all

☐ Magnitude

Made by: Waqar Siddhu



Einstein, among other well known names in the world of science, His theory of relativity maintains that space and time are merely different aspects of the ----- thing.

Einstein, among other well known names in the world of science, made a special study of time in relation to his research in physics. His theory of relativity maintains that space and time are merely different aspects of the same thing.

VuAnswers.com

Answer (Please select your correct option)

☐ Non of the given

☐ Unknown

☐ Different

☒ Same

correct

Made by: Waqar Siddhu

OpenGL is built for compatibility across hardware and operating systems. This architecture makes it easy to port OpenGL programs from one system to another. While each operating system has ----- requirements.

OpenGL is built for compatibility across hardware and operating systems. This architecture makes it easy to port OpenGL programs from one system to another. While each operating system has unique requirements, the OpenGL code in many programs can be used as is. page 303

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Answer (Please select your correct option)

☐ None of the given

☐ Compatibility

☐ Same

☒ Unique

correct

Made by: Waqar Siddhu



The rate at which light intensity decreases at a greater distance is called as -----.

Light in the real world loses its intensity as the inverse square of the distance from the light source to the surface being illuminated. However, when put into practice, this seemed to drop off the light intensity in too abrupt a manner and then not to vary too much after the light was far away.

page 290

VuAnswers.com

Answer (Please select your correct option)



Lambertian Shading



Light Intensity



Light Decrement



Light Attenuation



correct

Made by: Waqar Siddhu

In order to get a more realistic representation of lighting, we'll need to understand how light passes through a medium and how hitting the boundary layer at the ----- of two media can affect light's properties.

To do this, we'll need to understand how light passes through a medium and how hitting the boundary layer at the intersection of two media can affect light's properties.

page 296

VuAnswers.com

Answer (Please select your correct option)

☐ Intersection

correct

☐ Union☐ Endpoints☐ Edges

Made by: Waqar Siddhu

What makes this really challenging to model is that the index of refraction for most materials is a function of the----- of the light. This means that not only is there a shift in the angle of refraction, but that the shift is different for differing -----of light.

What makes this really challenging to model is that the index of refraction for most materials is a function of the wavelength of the light. This means that not only is there a shift in the angle of refraction, but that the shift is different for differing wavelengths of light.

page 299

VuAnswers.com

Answer (Please select your correct option)

☐ Reflecting angle, Reflecting angle☐ Refracting angle, Refracting angle☐ Frequency, Frequency☒ Wavelength, Wavelength

correct

Made by: Waqar Siddhu



Refractive index is a function of temperature, mostly due to density changes in materials with changes in temperature.

Refractive index is a function of temperature, mostly due to density changes in materials with changes in temperature.

page 300

VuAnswers.com

Answer (Please select your correct option)

True



correct

False



Made by: Waqar Siddhu

NURBS stands for-----.

NURBS (Non Uniform
Rational Beta Splines)

page 225

VuAnswers.com

Answer (Please select your correct option)

☐ Non Universal Rational Binary Spline

☐ Non Uniform Rational Binary Splines

☒ Non Uniform Rational Beta Splines



correct

☐ Non Universal Rational Beta Splines

Made by: Waqar Siddhu



A space curve is not confined to a plane. It is free to twist through space. To define a space curve we must use parametric functions that are -----.

A space curve is not confined to a plane. It is free to twist through space. To define a space curve we must use parametric functions that are cubic polynomials.

page 331

VuAnswers.com

Answer (Please select your correct option)

☐ Binary polynomials

☐ Mono polynomials

☐ Quadratic polynomials

☒ Cubic polynomials



correct

Made by: Waqar Siddhu

With similar expressions for $y(u)$ and $z(u)$. Again the a , b , c and d terms are constant coefficients. As we did with Equation for a plane curve, we combine the $x(u)$, $y(u)$, and $z(u)$ expressions into a single vector equation $P(u)=$ -----.

With similar expressions for $y(u)$ and $z(u)$. Again the a , b , c and d terms are constant coefficients. As we did with Equation for a plane curve, we combine the $x(u)$, $y(u)$, and $z(u)$ expressions into a single vector equation :

page 331

$$p(u) = au^3 + bu^2 + cu + d$$

VuAnswers.com

Answer (Please select your correct option)

☐ $Au^4 + bu^3 + cu^2 + d$

☐ $Au^3 + bu^2 + cu^2 + d$

☒ $Au^3 + bu^2 + cu + d$

correct

☐ $Au^2 + bu + cu + d$

Made by: Waqar Siddhu

To convert the information in the A matrix into that required for the P matrix, we do some simple matrix algebra, First we have $UA=UNP$ then Simply

A=-----

To convert the information in the A matrix into that required for the P matrix, we do some simple matrix algebra, using Equations 9, 10 and 13. First we have
 $GP = UNP$ (14)

pag 333

$$A = NP$$

VuAnswers.com

Answer (Please select your correct option)

☐ None of the given

☐ UP

☒ NP

☐ UN

correct

Made by: Waqar Siddhu



Bezier curve is the ideal standard for representing the ----- piecewise polynomial curves.

Bezier curve is the ideal standard for representing the more complex piecewise polynomial curves.

page 238

VuAnswers.com

Answer (Please select your correct option)

☐ None of the given

☐ Non complex

☐ Most complex

☒ More complex



correct

Made by: Waqar Siddhu

For high-quality images, it's a good idea to subdivide more on the silhouette edges than in the interior. If the surface is to be rotated relative to the eye, this is tougher to do, since the silhouette edges keep moving. Silhouette edges occur where the ----- are perpendicular to the vector from the surface to the viewpoint.

For high-quality images, it's a good idea to subdivide more on the silhouette edges than in the interior. If the surface is to be rotated relative to the eye, this is tougher to do, since the silhouette edges keep moving. Silhouette edges occur where the normal vectors are perpendicular to the vector from the surface to the viewpoint.

pagw 345

VuAnswers.com

Answer (Please select your correct option)

☐ None of the given☐ Unit vectors☐ Tangent vectors☒ Normal vectors

correct

Made by: Waqar Siddhu

A twenty-sided approximation to a sphere doesn't look good unless the image of the sphere on the screen is quite -----, but there's an easy way to increase the accuracy of the approximation.

A twenty-sided approximation to a sphere doesn't look good unless the image of the sphere on the screen is quite small, but there's an easy way to increase the accuracy of the approximation.

VuAnswers.com

Answer (Please select your correct option)

☐ Medium

☐ Large

☒ Small

☐ None of the given

correct

Made by: Waqar Siddhu

A recursive ----- technique can be used for other types of surfaces. Typically, the recursion ends either if a certain depth is reached or if some condition on the curvature is satisfied.

A recursive subdivision technique such as the one described in Example 5 can be used for other types of surfaces. Typically, the recursion ends either if a certain depth is reached or if some condition on the curvature is satisfied

VuAnswers.com

Answer (Please select your correct option)

☐ None of the given

☐ Addition

☐ Multiplication

☒ Subdivision

correct

Made by: Waqar Siddhu

According to Webster's Dictionary a fractal is defined as being "derived from the Latin word ----- meaning broken, various extremely irregular curves or shapes that repeat themselves at any scale on which they are examined."

According to Webster's Dictionary a fractal is defined as being "derived from the Latin word fractus meaning broken, uneven: any of various extremely irregular curves or shapes that repeat themselves at any scale on which they are examined."

page 352

VuAnswers.com

Answer (Please select your correct option)

☒ Fractus

correct

☐ Fractul☐ Fratus☐ Fractul

Made by: Waqar Siddhu



The ----- is most simple example that exhibits the property self similarity.

The fern is typical of many plants in that it exhibits self similarity

page 255

VuAnswers.com

Answer (Please select your correct option)

Mosse



Fern



correct

Thohar



None of the given



Made by: Waqar Siddhu

The transformation process to produce the desired scene for viewing is analogous to taking a photograph with a -----.

The transformation process to produce the desired scene for viewing is analogous to taking a photograph with a camera.

page 372

VuAnswers.com

Answer (Please select your correct option)

☐ None of the given

☐ Rendering

☐ Transformation

☒ Camera

correct

Made by: Waqar Siddhu

Projection transformations, that construct a 4×4 matrix M , which is then multiplied by the coordinates of each vertex v in the scene to accomplish the transformation $v' = Mv$, Remember that vertices always have ----- coordinates, though in most cases w is 1 and for two-dimensional data z is 0.

To specify viewing, modeling, and projection transformations, we construct a 4×4 matrix M , which is then multiplied by the coordinates of each vertex v in the scene to accomplish the transformation (Remember that vertices always have four coordinates (x, y, z, w) , though in most cases w is 1 and for two-dimensional data z is 0.)

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Answer (Please select your correct option)

☐ One

page 373

☐ Two

☒ Four

correct

☐ Three

Made by: Waqar Siddhu

The viewing transformation can be specified, the current matrix is set to the ----- matrix with `glLoadIdentity()`. This step is necessary since most of the transformation commands multiply the current matrix by the specified matrix and then set the result to be the current matrix.

the current matrix is set to the identity matrix with `glLoadIdentity()`. This step is necessary since most of the transformation commands multiply the current matrix by the specified matrix and then set the result to be the current matrix.

pag 375

VuAnswers.com

Answer (Please select your correct option)

☐ Rectangular☐ Square☒ Identity

correct

☐ Normal

Made by: Waqar Siddhu



----- basic types of projections are provided for us by OpenGL, along with several corresponding commands for describing the relevant parameters in different ways.

Two basic types of projections are provided for us by OpenGL, along with several corresponding commands for describing the relevant parameters in different ways.

page 376

VuAnswers.com

Answer (Please select your correct option)

Five



Two



correct

Three



Four



Made by: Waqar Siddhu

Rendering a Lit Sphere having steps required to add lighting to our scene. Define ----- vectors for each vertex of all the objects. These vectors determine the orientation of the object relative to the light sources.

Rendering a Lit Sphere

These are the steps required to add lighting to our scene. Define NORMAL vectors for each vertex of all the objects. These NORMALS determine the orientation of the object relative to the light sources.

page 398

[VuAnswers.com](https://vuanswers.com)

Answer (Please select your correct option)

Unit

☐

Normal

☒

correct

Transformation

☐

None of given

☐**Made by: Waqar Siddhu**



Triangle fans, conceptually, look like the folding fans you see in shops.

Triangle fans, conceptually, look like the folding fans you see in Asian souvenir shops.

VuAnswers.com

Answer (Please select your correct option)

False



True



correct

Made by: Waqar Siddhu



Dark lights are nothing more than lights in which one or more of the color values are_____.

VuAnswers.com

Answer (Please select your correct option)

☐ Unknown

☒ Negative



correct

☐ Null

☐ Positive

Made by: Waqar Siddhu



Which is the oldest in given types of the shading?

nai pta

VuAnswers.com

Answer (Please select your correct option)

☐ Flat Shading

☐ Phong Shading

☐ None of all

☒ Gouraud Shading



Made by: Waqar Siddhu



_____ is used for circumference of a circle.

The perimeter C of a circle is called the circumference, and is given by
 $C = 2 \pi r$

page 59

VuAnswers.com

Answer (Please select your correct option)

☐ 2r

☐ $2 \pi c$

☐ 2π

☒ None of the given



correct

Made by: Waqar Siddhu



Both Boundary Filling and Flood filling algorithms are _____ than scan line filling algorithm.

nai pta

VuAnswers.com

Answer (Please select your correct option)

☐ None of the given

☐ Better

ya lag rha ha

☒ Worse



☐ Almost same

Made by: Waqar Siddhu



Save a line with both endpoints inside all clipping boundaries is called as _____.

Trivial Accept - save a line with both endpoints inside all clipping boundaries.

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VuAnswers.com

Answer (Please select your correct option)

☐ None of the given

☐ Total inside

☐ Trivial Reject

☒ Trivial Accept



correct

Made by: Waqar Siddhu



Tomography is the technique used in _____.

VuAnswers.com

Answer (Please select your correct option)

☐ X-rays photography



correct

☐ Pixel paint

☐ Entertainment

☐ Artis's paintbrush

Made by: Waqar Siddhu



_____ polygons are clipped correctly by Sutherland-Hodgeman Algorithm.

The Sutherland-Hodgeman clipping algorithm clips any polygon against a convex clip polygon.

248

[VuAnswers.com](https://vuanswers.com)

Answer (Please select your correct option)

- ☒ Convex  correct chek may b all but mention with this only
- ☐ Concave
- ☐ Complex
- ☐ All of the given

Made by: Waqar Siddhu

The axonometric projection is _____ where the direction of projection makes same angle with all three axes.

The most common axonometric projection is an isometric projection where the projection plane intersects each coordinate axis in the model coordinate system at an equal distance or the direction of projection makes equal angles with all of the three principal axes

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VuAnswers.com

Answer (Please select your correct option)

☐ Oblique projection☐ Trimetric projection☐ Diametric projection☒ Isometric projection

correct

Made by: Waqar Siddhu



Texture mapping is a technique for interpolating _____ over the triangle being rasterized.

Texture mapping is a technique for interpolating an image over the triangle being rasterized.

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VuAnswers.com

Answer (Please select your correct option)

☐ None of the given

☐ Colors in the image

☒ Image



correct

☐ Shape in image

Made by: Waqar Siddhu



We want our scene to look more realistic, we should use _____ lights.

VuAnswers.com

Answer (Please select your correct option)

☐ Point

☒ Parallel

☐ Spot

☐ None of the given

correct

Made by: Waqar Siddhu



_____ can be defined as a mapping of point $P(x, y, z)$ onto its image $P'(x', y', z')$ in the view plane which constitutes the display surface.

Projection can be defined as a mapping of point $P(x, y, z)$ onto its image $P'(x', y', z')$ in the projection plane or view plane, which constitutes the display surface

page 262

VuAnswers.com

Answer (Please select your correct option)

☐ Mapping plane☐ Three Coordinate Planes☐ View plane☒ Projection

correct

Made by: Waqar Siddhu

The reflected light wave turns out to be a / an _____ case since light is reflected at the same angle as the incident wave (when the surface is smooth and uniform, as we'll assume for now).

The reflected light wave turns out to be a simple case since light is reflected at the same angle as the incident wave (when the surface is smooth and uniform, as we'll assume for now).

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VuAnswers.com

Answer (Please select your correct option)

☐ Unknown☒ Simple

correct

☐ Complex☐ Abnormal

Made by: Waqar Siddhu



Computer graphics is very helpful in producing graphical representations for scientific visualization.

Computer graphics is very helpful in producing graphical representations for scientific visualization

page 9

VuAnswers.com

Answer (Please select your correct option)

False



True



correct

Made by: Waqar Siddhu

When light strike with the thick colored lacquer surface it perform the following steps.

Consider a thick colored lacquer surface. The lacquer itself is transparent, but suspended in the lacquer are reflective pigment off of which light gets reflected,

page 241

VuAnswers.com

Answer (Please select your correct option)

reflected



correct

bounced, altered



split, shifted



All of the given



Made by: Waqar Siddhu



Spotlights have ____ angles associated with them.

Spotlights have two angles associated with them.

page 244

VuAnswers.com

Answer (Please select your correct option)

☒ 2  correct

☐ 3

☐ 4

☐ 5

Made by: Waqar Siddhu



There are various types of transformations as we have seen, in case of 2D transformations, these include:

mid

repeat

VuAnswers.com

Answer (Please select your correct option)



Scaling



Rotation



Translation



All of the given

Made by: Waqar Siddhu

When we perform the rotation about Z-axis

$$x' = x \cos \theta$$

$$y' = y \sin \theta$$

The value of

VuAnswers.com

Answer (Please select your correct option)

☐ $\sin \theta$

☐ $\tan \theta$

☐ $\cos \theta$

☒ z

correct

✓ Pg 258

Made by: Waqar Siddhu

Computer animation is a form of-----

Answer (Please select your correct option)

VuAnswers.com

☐ Colour printer output

☒ Video graphics



☐ CAD/CAM

☐ LCD

Made by: Waqar Siddhu

TV series are made as simply as possible from the animation point of view. This approach is generally known as -----.

Answer (Please select your correct option)

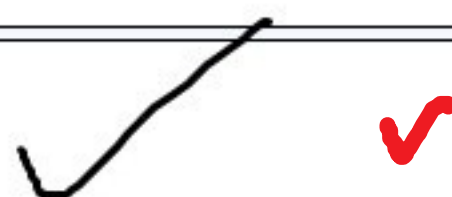
VuAnswers.com

☐ Low animation

☐ High resolution

☐ Full animation

☒ Limited animation



Made by: Waqar Siddhu

OpenGL is built for compatibility across hardware and operating systems. This architecture makes it easy to port OpenGL programs from one system to another. While each operating system has ----- requirements.

Answer (Please select your correct option)

VuAnswers.com

☐ Compatibility

☐ Same

☒ Unique

☐ None of the given

Made by: Waqar Siddhu

Coordinate systems are the measured frames of reference within which geometry is defined, manipulated and viewed.

Answer (Please select your correct option)

VuAnswers.com

True



False

Made by: Waqar Siddhu

glutReshapeWindow requests a change in the size of the *current window*. The width and height parameters are size extents in pixels. The width and height must be ----- values.

Answer (Please select your correct option)

VuAnswers.com

☐ Non of the given

☐ Neutral

☐ Negative

☒ Positive



Made by: Waqar Siddhu

The `glClearColor` function specifies the red, green, blue, and alpha values used by `glClear` to clear the color buffers. Values specified by `glClearColor` are clamped to the range -----.

Answer (Please select your correct option)

VuAnswers.com

☐ [0,-1]

☐ [-1,1]

☐ [.5,1]

☒ [0,1]



Made by: Waqar Siddhu

if we assign a different value to the parametric variable for the intermediate point, then we obtain different values for the coefficients. This, in turn, means that a different curve is produced, although it passes through the -----three points.

Answer (Please select your correct option)

VuAnswers.com

☐ None of the given

☐ Isolate

☐ Different

☒ Same



Made by: Waqar Siddhu

NURBS stands for-----.

Answer (Please select your correct option)

VuAnswers.com

☐ Non Universal Rational Binary Spline

☐ Non Uniform Rational Binary Splines

☒ Non Uniform Rational Beta Splines



☐ Non Universal Rational Beta Splines

Made by: Waqar Siddhu

A point is defined as $[ax, ay, az, a]^T$ where "a" can be ANY value.

Answer (Please select your correct option)

VuAnswers.com

True



False

Made by: Waqar Siddhu

An orthogonal set of vectors-----

Answer (Please select your correct option)

VuAnswers.com

- ☐ Must be a set of linearly independent vectors
- ☐ Must be a set of linearly dependent vectors
- ☒ Must be made up of the basis vectors (e_1 , e_2 , and e_3)
- ☐ Can be made up of any set of vectors



Made by: Waqar Siddhu

Bezier curve is the ideal standard for representing the ----- piecewise polynomial curves.

Answer (Please select your correct option)

VuAnswers.com

☐ None of the given

☐ Non complex

☐ Most complex

☒ More complex



Made by: Waqar Siddhu

Keep polygon orientations consistent to make sure that when viewed from the outside, all the polygons on the surface are oriented in the same direction.

Answer (Please select your correct option)

VuAnswers.com

☐ None of the given



☐ Neither

☐ Different

☒ Same



Made by: Waqar Siddhu

For high-quality images, it's a good idea to subdivide more on the silhouette edges than in the interior. If the surface is to be rotated relative to the eye, this is tougher to do, since the silhouette edges keep moving. Silhouette edges occur where the ----- are perpendicular to the vector from the surface to the viewpoint.

Answer (Please select your correct option)

VuAnswers.com

☐ None of the given

☐ Unit vectors

☐ Tangent vectors

☒ Normal vectors



Made by: Waqar Siddhu

According to Webster's Dictionary a fractal is defined as being "derived from the Latin word ----- meaning broken, various extremely irregular curves or shapes that repeat themselves at any scale on which they are examined."

Answer (Please select your correct option)

VuAnswers.com

☐ Fractus



☐ Frectul

☐ Fratus

☐ Fractul

Made by: Waqar Siddhu

The ----- is most simple example that exhibits the property self similarity.

Answer (Please select your correct option)

VuAnswers.com

☐ None of the given

☐ Thohar

☐ Mosse

☒ Fern



Made by: Waqar Siddhu

The transformation process to produce the desired scene for viewing is analogous to taking a photograph with a -----.

Answer (Please select your correct option)

VuAnswers.com

☐ None of the given

☐ Rendering

☐ Transformation

☒ Camera



Made by: Waqar Siddhu

The viewing and modeling transformations are combined to form the ----- matrix, which is applied to the incoming object coordinates to yield eye coordinates.

Answer (Please select your correct option)

VuAnswers.com

☐ Two

☐ Square

☐ Viewport

☒ Modelview



Made by: Waqar Siddhu

----- basic types of projections are provided for us by OpenGL, along with several corresponding commands for describing the relevant parameters in different ways.

Answer (Please select your correct option)

VuAnswers.com

Five

Two

Three

Four



Made by: Waqar Siddhu

At a physical surface, our eye's perception of the colour depends on the distribution of photon energies that arrive and trigger our ----- cells. Those photons come from a light source or combination of sources, some of which are absorbed and some are reflected by the surface.

Answer (Please select your correct option)

VuAnswers.com

☐ Lens

☐ Eye

☒ Cone

☐ Retina



Made by: Waqar Siddhu

In the Phong reflection model, there are 3 constants (a, b, c) which are used to describe the qualities of which of the following phenomena?

Answer (Please select your correct option)

VuAnswers.com

☒ The attenuation of a point light source with distance



☐ The size (in each dimension) which the light is assumed to have

☐ The amount to perturb reflection vectors as they are mirrored across the normal

☐ The material reaction to ambient, diffuse and specular light (respectively)

Made by: Waqar Siddhu

When obtaining normals for a triangle, which of the following mathematical constructs is NOT used?

Answer (Please select your correct option)

VuAnswers.com

☐ Point-Point subtraction

☐ Vector normalization

☐ Vector cross products

☒ Vector dot products



Made by: Waqar Siddhu

Dark lights are nothing more than lights in which one or more of the color values are_____.

Answer (Please select your correct option)

VuAnswers.com

☐ Unknown

☒ Negative



☐ Null

☐ Positive

Made by: Waqar Siddhu

The traditional approach in real-time computer graphics has been to calculate lighting at a vertex as a sum of the _____ light.

Answer (Please select your correct option)

VuAnswers.com

☐ Diffuse, and specular

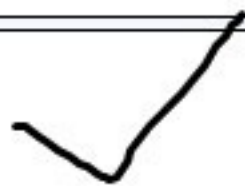
☐ Ambient

☒ Ambient, diffuse, and specular

☐ Specular



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Made by: Waqar Siddhu

Unlike ambient light, the intensity of diffuse light is directional and is a function of the angle of the incoming light and the surface. This type of shading is called Lambertian shading after Lambert's _____ law.

Answer (Please select your correct option)

VuAnswers.com

- ☐ Perpendicular
- ☒ Cosine  
- ☐ Sine
- ☐ Tangent

Made by: Waqar Siddhu

There are _____ basic types of polygon.

Answer (Please select your correct option)

VuAnswers.com

2



3



4



10



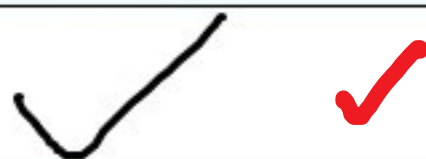
Made by: Waqar Siddhu

_____ polygons are basically concave polygons that may have self-intersecting edges.

Answer (Please select your correct option)

VuAnswers.com

☒ Complex



☐ None of the given

☐ Hybrid

☐ Convex

Made by: Waqar Siddhu

The boundary-fill method requires the coordinates of _____.

Answer (Please select your correct option)

VuAnswers.com

☐ Starting point

☐ Filling colour

☐ Boundary colour

☒ All of the given



Made by: Waqar Siddhu

A polygon is usually defined by a sequence of vertices and _____. (Choose best suitable option)

Answer (Please select your correct option)

VuAnswers.com

☐ Ending lines

☒ Edges

☐ Circles

☐ None of the given

Made by: Waqar Siddhu

A three-dimensional reflection can be performed relative to a selected reflection _____.

Answer (Please select your correct option)

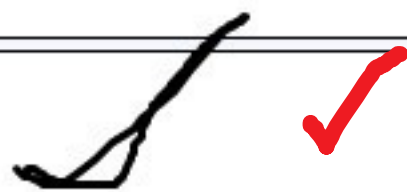
VuAnswers.com

☐ None of the given

☐ Plane

☐ Axis

☒ Both Axis and plane



Made by: Waqar Siddhu

_____ transformation produces shape distortions as if objects were composed of layers that are caused to slide over each other.

Answer (Please select your correct option)

VuAnswers.com

☐ Translation

☐ Reflection

☒ Shear

☐ Rotation

Made by: Waqar Siddhu

The _____ technique has the direction of projection perpendicular to the viewing plane, but the viewing direction is NOT perpendicular to one of the principle faces.

Answer (Please select your correct option)

VuAnswers.com

☐ Oblique Parallel Projection

☒ Axonometric Parallel Projection

☐ Orthographic Parallel Projection

☐ None of the given

Made by: Waqar Siddhu

We want our scene to look more realistic, we should use _____ lights.

Answer (Please select your correct option)

VuAnswers.com

☐ Point

☒ Parallel

☐ Spot

☐ None of the given

Made by: Waqar Siddhu

Lambertian shading was used mostly back when computers weren't fast enough to do _____ in real time.

Answer (Please select your correct option)

VuAnswers.com

☐ Gouraud shading



☐ Shading in which triangles painted with single solid color

☐ Processing

☐ None of the given

Made by: Waqar Siddhu

In case of _____ , we'll get dull and diffused view.

p 294

Answer (Please select your correct option)

VuAnswers.com

☐ Backscattering

☐ Forward scattering

☒ Both Backscattering and Forward scattering

☐ None of the given

Made by: Waqar Siddhu

The reflected light wave turns out to be a / an _____ case since light is reflected at the same angle as the incident wave (when the surface is smooth and uniform, as we'll assume for now).

Answer (Please select your correct option)

VuAnswers.com

☐ Unknown

☒ Simple

☐ Complex

☐ Abnormal

Made by: Waqar Siddhu

A line may have three forms with respect to it's _____.

Answer (Please select your correct option)

VuAnswers.com

☐ Slop



☐ Plan

☐ Points

☐ None of the given

Made by: Waqar Siddhu

If the value of scaling factors S_x and S_y is greater than 1, then size of objects will be _____.

Answer (Please select your correct option)

VuAnswers.com

☒ Reduced



☐ Enlarged

☐ Remain same

☐ None of the given

Made by: Waqar Siddhu

We maintain the saturation of color values by using

Answer (Please select your correct option)

VuAnswers.com

- ☐ Clamping
 - ☐ Scaling
 - ☐ Shifting
 - ☒ all of the given
- ✓ ✓

Made by: Waqar Siddhu

There are various types of transformations as we have seen, in case of 2D transformations, these include:

Answer (Please select your correct option)

VuAnswers.com

☐ Scaling

☐ Rotation

☐ Translation

☐ All of the given



Made by: Waqar Siddhu

In 24-bit display ____ number of different colors can be displayed.

p 227

Answer (Please select your correct option)

VuAnswers.com

☒ 256^3



☐ 256^7

☐ 256^5

☐ 256^6

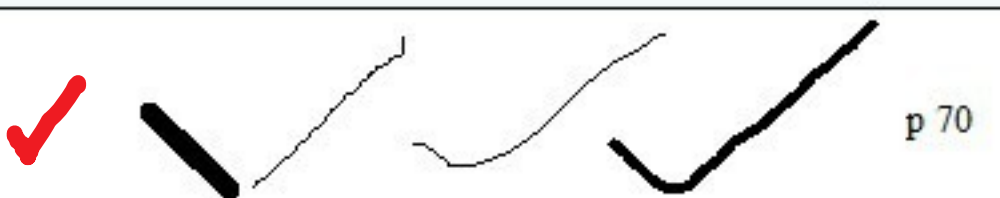
Made by: Waqar Siddhu

Each hyperbola consists of two -----

Answer (Please select your correct option)

VuAnswers.com

- ☐ Vertices
- ☐ Nodes
- ☒ Branches



p 70

Made by: Waqar Siddhu

Computer animation is a form of-----

Answer (Please select your correct option)

VuAnswers.com

☐ Colour printer output

☒ Video graphics

☐ CAD/CAM

☐ LCD

Made by: Waqar Siddhu

An independent consortium, the OpenGL Architecture Review Board, guides the OpenGL specification. With broad industry support, OpenGL is the only truly open, vendor-neutral, ----- graphics standard.

Answer (Please select your correct option)

VuAnswers.com

☐ Tertiary

☐ Binary

☐ Single platform

☒ Multiplatform



Made by: Waqar Siddhu

All OpenGL applications produce consistent visual display results on any OpenGL API-compliant hardware, ----- of operating system or windowing system.

Answer (Please select your correct option)

VuAnswers.com

☒ Regardless



☐ With respect

☐ Kernel

☐ Libraries

Made by: Waqar Siddhu

In order to get a more realistic representation of lighting, we'll need to understand how light passes through a medium and how hitting the boundary layer at the ----- of two media can affect light's properties.

Answer (Please select your correct option)

VuAnswers.com

☒ Intersection



☐ Union

☐ Endpoints

☐ Edges

Made by: Waqar Siddhu

Using a lighting model based upon the Blinn Phong model means that we'll always get a uniform specular highlight based upon the colour of the ----- light and material, which means that all reflections based on this model, will be reminiscent of plastic.

Answer (Please select your correct option)

VuAnswers.com

☐ Union

☐ Refracting

☐ Intersection

☒ Reflecting



Made by: Waqar Siddhu

The value returned is a unique small integer identifier for the window. The range of allocated identifiers starts at ----- . This window identifier can be used when calling glutSetWindow.

Answer (Please select your correct option)

VuAnswers.com

☐ Three

☐ Two

☒ One

☐ Zero



Made by: Waqar Siddhu

_____ sets the global idle call back to be 'func' so a GLUT program can perform background processing tasks or continuous animation when window system events are not being received.

Answer (Please select your correct option)

VuAnswers.com

☒ glutIdleFunc

☐ glutMainLoop

☐ glutDisplayFunc

☐ glutReshapeFunc

Made by: Waqar Siddhu

A space curve is not confined to a plane. It is free to twist through space. To define a space curve we must use parametric functions that are -----.

Answer (Please select your correct option)

VuAnswers.com

☐ Binary polynomials

☐ Mono polynomials

☐ Quadratic polynomials

☒ Cubic polynomials



Made by: Waqar Siddhu

A point is defined as $[ax, ay, az, a]^T$ where "a" can be ANY value.

Answer (Please select your correct option)

VuAnswers.com

True



False

Made by: Waqar Siddhu

Given the vector $[1, 2, 3, 0]^T$ and the point $[3, 2, 1, 1]^T$, the equation for the plane defined by them is

Answer (Please select your correct option)

VuAnswers.com

☐ $1x + 2y + 3z + 10 = 0$

☐ $3x + 2y + 1z + 10 = 0$

☐ $1x + 2y + 3z - 10 = 0$

☐ $3x + 2y + 1z - 10 = 0$

Made by: Waqar Siddhu

Which of the following affine transforms does NOT affect vectors?

Answer (Please select your correct option)

VuAnswers.com

☐ Shear

☐ Rotation

☐ Scale

☒ Translation

Made by: Waqar Siddhu

Bezier curve is the ideal standard for representing the ----- piecewise polynomial curves.

Answer (Please select your correct option)

VuAnswers.com

☐ None of the given

☐ Non complex

☐ Most complex

☒ More complex



Made by: Waqar Siddhu

Keep polygon orientations consistent to make sure that when viewed from the outside, all the polygons on the surface are oriented in the same direction.

Answer (Please select your correct option)

VuAnswers.com

☐ Neither

☐ Different

☒ Same

☐ None of the given

Made by: Waqar Siddhu

Unless the surface is to be drawn only once, you should probably save the calculated vertex and normal coordinates so that the calculations don't need to be repeated each time that the -----
- is drawn.

Answer (Please select your correct option)

VuAnswers.com

☐ Rectangle

☐ Polygon

☒ Surface



☐ Triangle

Made by: Waqar Siddhu

A recursive ----- technique can be used for other types of surfaces. Typically, the recursion ends either if a certain depth is reached or if some condition on the curvature is satisfied.

Answer (Please select your correct option)

VuAnswers.com

☐ None of the given

☐ Addition

☐ Multiplication

☒ Sub division



Made by: Waqar Siddhu

Fractal are geometric patterns that is repeated at ever smaller scales to produce ----- shapes and surfaces that can not be represented by classical geometry.

Answer (Please select your correct option)

VuAnswers.com

☐ None of the given

☐ Linear

☐ Regular

☒ Irregular



Made by: Waqar Siddhu

According to Webster's Dictionary a fractal is defined as being "derived from the Latin word ----- meaning broken, various extremely irregular curves or shapes that repeat themselves at any scale on which they are examined."

Answer (Please select your correct option)

VuAnswers.com

☐ Fractus



☐ Frectul

☐ Fratus

☐ Fractul

Made by: Waqar Siddhu

A correspondence must be established between the transformed coordinates and screen ----- . This is known as a viewport transformation.

Answer (Please select your correct option)

VuAnswers.com

☐ Edges

☐ Vertices

☐ Coordinates

☐ Pixels



Made by: Waqar Siddhu

The transformation process to produce the desired scene for viewing is analogous to taking a photograph with a -----.

Answer (Please select your correct option)

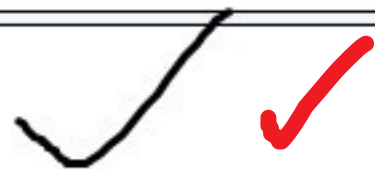
VuAnswers.com

☐ None of the given

☐ Rendering

☐ Transformation

☐ Camera



Made by: Waqar Siddhu

OpenGL applies the projection matrix to yield ----- coordinates. This transformation defines a viewing volume.

Answer (Please select your correct option)

VuAnswers.com

☐ Three

☐ Four

☒ Clip

☐ Normal

Made by: Waqar Siddhu

----- basic types of projections are provided for us by OpenGL, along with several corresponding commands for describing the relevant parameters in different ways.

Answer (Please select your correct option)

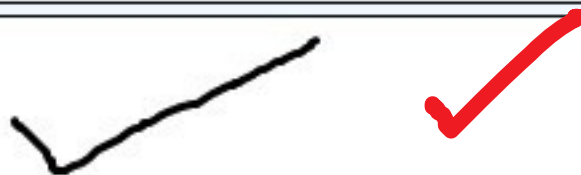
VuAnswers.com

Five

Two

Three

Four



Made by: Waqar Siddhu

Which of the following describes the purpose of the ϵ (epsilon) portion of the rendering equation?

Answer (Please select your correct option)

VuAnswers.com

☐ Emitted light from p' towards p

☐ Intensity of light from p' to p

☐ Geometry function (0 if p' not visible from p , $1/r^2$ otherwise)

☐ Reflection of light from p'' across p' to p

Made by: Waqar Siddhu

In the Phong reflection model, there are 3 constants (a, b, c) which are used to describe the qualities of which of the following phenomena?

Answer (Please select your correct option)

VuAnswers.com

☒ The attenuation of a point light source with distance



☐ The size (in each dimension) which the light is assumed to have

☐ The amount to perturb reflection vectors as they are mirrored across the normal

☐ The material reaction to ambient, diffuse and specular light (respectively)

Made by: Waqar Siddhu

Plasma-panel Displays use a gas mixture and phosphorus coating for showing display.

Answer (Please select your correct option)

VuAnswers.com

False



True

Made by: Waqar Siddhu

A straight line can be moved to another location by applying _____ to each of the line endpoints and redrawing the line between the new coordinates.

Answer (Please select your correct option)

VuAnswers.com

☐ Rotation

☒ Translation

☐ Reflection

☐ Scaling factor

Made by: Waqar Siddhu

Because clipping against one edge is independent of all others,so it is impossible to arrange the clipping stages in a pipeline.

Answer (Please select your correct option)

VuAnswers.com

True



False



Made by: Waqar Siddhu

Tomography is the technique used in _____.

Answer (Please select your correct option)

VuAnswers.com

☐ X-rays photography



☐ Pixel paint

☐ Entertainment

☐ Artis's paintbrush

Made by: Waqar Siddhu

In _____ algorithm, old color must be read before it is invoked.

mid

Answer (Please select your correct option)

VuAnswers.com

☐ Scan line filling

☐ Flood fill



☐ Both scan line and flood fill

☐ None of the given

Made by: Waqar Siddhu

In Trivial acceptance/reject test there are four bits of nine regions, Bit3 represents condition _____.

Answer (Please select your correct option)

VuAnswers.com

☐ Outside half plane of left edge, to the left of left edge $X < X_{min}$

☒ Outside half plane of right edge, to the right of right edge $X > X_{max}$

p 143



☐ Outside half plane of bottom edge, below bottom edge $Y < Y_{min}$

☐ Outside half plane of top edge, above top edge $Y > Y_{max}$

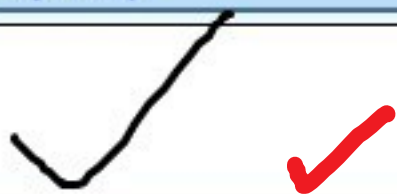
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The dot product of two vectors A and B is _____, iff the angle between them is less than 90 or greater than 270 degrees.

Answer (Please select your correct option)

VuAnswers.com

☒ Greater than zero (0)



☐ Less than zero (0)

☐ Equal to Zero (0)

☐ None of the given

Made by: Waqar Siddhu

We want our scene to look more realistic, we should use _____ lights.

Answer (Please select your correct option)

VuAnswers.com

☐ Point

☒ Parallel

☐ Spot

☐ None of the given

Made by: Waqar Siddhu

Lambertian shading was used mostly back when computers weren't fast enough to do _____ in real time.

Answer (Please select your correct option)

VuAnswers.com

☒ Gouraud shading



☐ Shading in which triangles painted with single solid color

☐ Processing

☐ None of the given

Made by: Waqar Siddhu

Because clipping against _____ edge / edges is independent of all others, it is possible to arrange the clipping stages in a pipeline.

Answer (Please select your correct option)

VuAnswers.com

☐ Four

☒ One

☐ Two

☐ Three

Made by: Waqar Siddhu

In case of _____, we'll get dull and diffused view.

Answer (Please select your correct option)

VuAnswers.com

☐ Backscattering

☐ Forward scattering

☒ Both Backscattering and Forward scattering

☐ None of the given



Made by: Waqar Siddhu

A line may have three forms with respect to it's _____.

Answer (Please select your correct option)

VuAnswers.com

☐ Slop



☐ Plan

☐ Points

☐ None of the given

Made by: Waqar Siddhu

We can draw the circle using _____.

Answer (Please select your correct option)

VuAnswers.com

☐ Pentane

☐ Hexanes

☐ Trident

☐ Octants



Made by: Waqar Siddhu

We maintain the saturation of color values by using

Answer (Please select your correct option)

VuAnswers.com

- ☐ Clamping
 - ☐ Scaling
 - ☐ Shifting
 - ☐ all of the given
- ✓ ✓

Made by: Waqar Siddhu

There are various types of transformations as we have seen, in case of 2D transformations, these include:

Answer (Please select your correct option)

VuAnswers.com

☐ Scaling

☐ Rotation

☐ Translation

☒ All of the given



Made by: Waqar Siddhu

When we perform the rotation about Z-axis

$$x' = x \cos \theta$$

$$y' = y \sin \theta$$

The value of

VuAnswers.com

Answer (Please select your correct option)

☐ $\sin \theta$

☐ $\tan \theta$

☐ $\cos \theta$

☒ z



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$$x' = x \cos \theta$$

$$y' = y \sin \theta$$

The value of

$$z' = \underline{\hspace{1cm}}$$

Answer (Please select your correct option)

VuAnswers.com

☐ $\sin \theta$

☐ $\tan \theta$

☐ $\cos \theta$

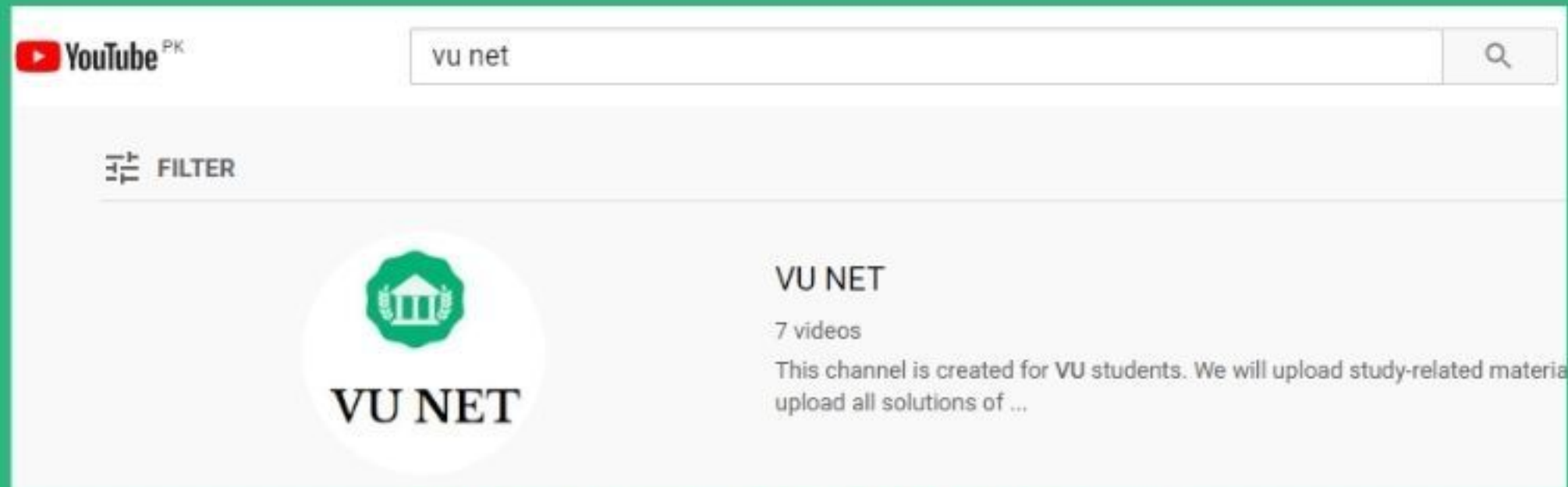
☒ z

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Made by: Waqar Siddhu

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CS602-COMPUTER GRAPHICS

FINAL TERM MCQS

Prepared by: JUNAID MALIK

AL-JUNAID TECH INSTITUTE



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Ph# 0304-1659294
Email: junaidfazal08@gmail.com

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ALL answers are verified if found any mistake then Correct ACCORDINGLY

1. Computer graphics and computer vision are _____ disciplines.
 - Related
 - Interrelated
 - Same
 - **Different** ✓
2. OpenGL has become the industry's most widely used and supported _____ graphics application programming interface (API), bringing thousands of applications to a wide variety of computer
 - platforms.
 - Dimensional
 - Dimensional
 - **2 Dimensional and 3-Dimensional** ✓
3. _____ sets the reshape callback for the *current window*.
 - glutIdle function
 - glutKeyboardFunc
 - **glutReshapeFunc** ✓
 - glutDisplayFunc
4. _____ sets the global idle callback to be 'func' so a GLUT program can perform background processing tasks or continuous animation when window system events are not being received.
 - **glutIdle function** ✓
 - glutKeyboardFunc
 - glutReshapeFunc
 - glutDisplayFunc
5. $X^2/a^2 - y^2/b^2 = 1$ is an equation of
 - **Hyperbola** Page no : 70
 - Parabola
 - None of given
 - Ellipse

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6. To modify the object shape, shearing transformation cannot be used.

➤ False Page no : 124 ✓

➤ True

7. Rotation is performed around a fixed point called _____.

➤ Pivot

➤ Point

➤ rotation Page no : 114 ✓

8. Computer graphics is very helpful in producing graphical representations for scientific visualization and analysis

➤ True ✓

➤ False

9. Save a line with both endpoints inside all clipping boundaries.

➤ Trivial Reject

➤ Trivial Accept Page no : 137 ✓

➤ None of given

10. Dark lights are nothing more than lights in which one or more of the color values are _____. Unknown

➤ Negative Page no : 230 ✓

➤ Positive Zero

11. A series of _____ computer operations convert an object's three-dimensional coordinates to pixel positions on the screen. Transformations, which are represented by matrix multiplication, include modeling, viewing, and projection operations. Such operations include rotation, translation, scaling, reflecting, orthographic projection, and perspective projection.

➤ Three Page no : 366 ✓

➤ Two

➤ Four

➤ Ten

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12. At a physical surface, our eye's perception of the color depends on the distribution of photon energies that arrive and trigger our _____ cells.

➤ Eye Retina

➤ Cone Page no : 393 ✓

12.This projection technique has the direction of projection perpendicular to the viewing plane, but the viewing direction is NOT perpendicular to one of the principle faces.

Orthographic Parallel Projection

➤ Axonometric Parallel Projection Page no : 189 ✓

➤ Oblique Parallel Projection

13.The Phong reflection model simplifies light-matter interactions into (essentially) 4 vectors and a number of constants. Which piece of the Phong model is responsible for giving spheres their bright white spots?

➤ Specular Page no : 234 ✓

➤ Ambient

➤ Diffuse

14.In the Phong Reflection model, _____ light is the same everywhere.

➤ Ambient ✓

➤ Diffuse

➤ Specular

➤ Emissive

15.A plane is two dimensional since in order to uniquely define any point on its surface we require _____ numbers.

➤ Two ✓

➤ Three

➤ Four

➤ Five

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16. In perspective projection, for your view to come out correctly, you will also want the _____ to pass through the middle of the screen.

- X axis
- Y axis
- Z axis Page no : 195 ✓
- None

17. Neither floating-point nor signed integer values are clamped to the range _____ before updating the current color.

- 0, -1.0
- -1, 1
- 1, -1

0, 1 Page no : 316 ✓

18. Bezier curve is the ideal standard for representing the _____ piecewise polynomial curves. Most complex

- Less complex
- None of given
- More complex Page no : 333 ✓

19. An object's _____ determine its orientation relative to the light sources. For each vertex, OpenGL uses the assigned normal to determine how much light that particular vertex receives from each light source.

- Unit
- Normal Page no : 395 ✓
- None of given

21. Which of the following affine transforms does NOT affect vectors?

- Scale
- Rotation
- Shear
- Translation Page no : 113

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22) This projection technique does NOT have the direction of projection perpendicular to the viewing plane.

- Orthographic Parallel Projection
- Axonometric Parallel Projection
- Oblique Parallel Projection Page no : 189 ✓

23) This projection technique has the direction of projection perpendicular to the viewing plane, and the viewing direction is perpendicular to one of the principle faces.

- Orthographic Parallel Projection Page no : 189 ✓
- Axonometric Parallel Projection
- Oblique Parallel Projection

24) In OpenGL, there are several different matrices. We have discussed two of them in class. Which one of the below would be used in conjunction with a `glRotatef` function call?

- GL_MODELVIEW Page no : 388 ✓
- GL_PROJECTION

25) In OpenGL, there are several different matrices. We have discussed two of them in class. Which one of the below would be used in conjunction with `glFrustum`?

- GL_MODELVIEW
- GL_PROJECTION Page no : 369 ✓

26) Which of the following is the order that geometry operations are performed in OpenGL (where we read the order from left to right)?

- GL_PROJECTION GL_MODELVIEW Perspective division
- GL_MODELVIEW GL_PROJECTION Perspective division ✓
- Perspective division GL_PROJECTION GL_MODELVIEW
- GL_MODELVIEW Perspective division GL_PROJECTION
- GL_PROJECTION Perspective division GL_MODELVIEW

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27) The Phong reflection model simplifies light-matter interactions into (essentially) 4 vectors and a number of constants. Each piece of the Phong model uses different vectors and constants. Which portion does NOT include taking a dot product?

- Ambient Page no : 234 ✓
- Diffuse
- Specular

28) Which of the following is NOT a modern application for Computer Graphics---

- ► Stop-motion animation Page no : 6
- ► Computer Aided Geometric Design
- ► Video Games
- ► Scientific Visualization

29) Both Boundary Filling and Flood filling algorithms are non-recursive techniques,

- False Page no : 97
- True

30) TV series are made as simply as possible from the animation point of view. This approach is generally known as -----.

- Full animation
- Limited animation Page no : 423 ✓
- Low animation
- High resolution

31) An eight frame run cycle that ----- frame/frames to each step gives a fast and vigorous dash. At this speed the successive leg positions are quite widely separated and may need dry brush or speed lines to make the movement flow.

- ► Two
- ► One
- ► Three

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- ► Four Page no :432 ✓

32) ----- Reflection is the effect of reflecting light toward the direction from which it came, no matter the orientation of the surface.

- ► Forward scattering
- ► Diffuse Lambertian
- ► Backscattering

- ► Retro Page no : 288 ✓

33)What makes this really challenging to model is that the index of refraction for most materials is a function of the----- of the light. This means that not only is there a shift in the angle of refraction, but that the shift is different for differing -----of light.

- ► Reflecting angle, Reflecting angle
- ► Refracting angle, Refracting angle
- ► Frequency, Frequency

- ► Wavelength, Wavelength Page no : 293 ✓

34) The reflected light wave turns out to be a -----case since light is reflected at the same angle as the incident wave (when the surface is smooth and uniform, as we'll assume for now).

- ► Abnormal
- ► Complex

- ► Simple Page no : 291 ✓

- ► Unknown

35) Tessellation can be adaptive to the _____ degree of curvature of a surface.

- ► Local Page no : 165 ✓

- ► Static
- ► Global
- ► Variable

36) _____ sets the reshape call back for the *current window*. The reshape callback is triggered when a window is reshaped.

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- ► glutMainLoop
- ► glutIdleFunc
- ► glutReshapeFunc Page no : 307 ✓
- ► glutDisplayFunc

37) Signed integer colour components, when specified, are linearly mapped to floating-point values such that the most positive representable value maps to 1.0, and the most negative representable value maps to ----- . Floating-point values are mapped directly.

- ► -1.0
- ► 0.0
- ► 2.0
- ► 1.0 Page no : 315 ✓

38) NURBS stands for-----

- ► Non Universal Rational Binary Spline
- ► Non Uniform Rational Binary Splines
- ► Non Uniform Rational Beta Splines Page no :320 ✓
- ► Non Universal Rational Beta Splines

39) An orthogonal set of vectors-----

- ► Must be a set of linearly independent vectors ✓
- ► Must be a set of linearly dependent vectors
- ► Must be made up of the basis vectors (e_1 , e_2 , and e_3)
- ► Can be made up of any set of vectors

40) Bezier curve is numerically the ----- of all the polynomial-based curves used in these applications.

- ► None of the given
- ► Most stable
- ► Less stable
- ► Most unstable

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40. Bezier curve is the ideal standard for representing the ----- piecewise polynomial curves.

- ► None of the given
- ► Non complex
- ► Most complex
- ► More complex repeated Question No: 15

41) Keep polygon orientations consistent to make sure that when viewed from the outside, all the polygons on the surface are oriented in the ____ direction.

- ► None of the given
- ► Neither
- ► Different
- ► Same Page no : 340

42)The ----- is most simple example that exhibits the property self-similarity.

- ► Mosse
- ► Fern Page no : 350
- ► None of the given
- ► Thohar

43) A common mistake people make when creating three-dimensional graphics is to start thinking too soon that the final image appears on a flat, two-dimensional screen. Avoid thinking about which pixels need to be drawn, and instead try to visualize ----- space.

- ► Multi-dimensional
- ► One-dimensional
- ► Two-dimensional
- ► Three-dimensional Page no: 366

44)Which of the following properties of rational Bezier curves fails if the weight assigned to a control point is negative?

- ► End-point interpolation

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- ► Variation Diminishing
- ► Symmetry
- ► Convex-Hull page no : 335

45) In the Phong reflection model, there are 3 constants (a, b, c) which are used to describe the qualities of which of the following phenomena?

- ► The attenuation of a point light source with distance Page no : 285
- ► The size (in each dimension) which the light is assumed to have
- ► The amount to perturb reflection vectors as they are mirrored across the normal
- ► The material reaction to ambient, diffuse and specular light (respectively)

46) The Phong reflection model simplifies light-matter interactions into (essentially) 4 vectors and a number of constants. Which piece of the Phong model is responsible for giving spheres their bright white spots?

- ► Specular repeated
- ► Diffuse
- ► Ambient

48) When you hit a surface in ray tracing, generally shadow rays are tested against all objects in a scene. If these rays come back saying they hit an object in the scene, which of the following do you do?

- ► add all components (i.e. ambient, diffuse and specular) from that light source to the object.
- ► add all EXCEPT the ambient light from that light source to the object (i.e. diffuse and specular)
- ► add only the ambient light from that light source to the object
- ► add none of the light from that light source to the object

49) The Color Space tool is a handy tool that we can use to interactively add two colours together to see the effects of the various strategies for handling oversaturated colours.

- ► False

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- ► True page no : 230

50) A polygon is usually defined by a sequence of ----- and Edges.

- ► Ending lines
- ► Points
- ► Vertices Page no : 243
- ► Edges

51) Which of the following properties of Bezier curves guarantees that a line passes through the control polygon as many times or more times than the line passes through the Bezier curve itself?

- ► End-point interpolation
- ► Variation Diminishing
- ► Symmetry
- ► Convex-Hull

52) Parity is a concept used to determine which _____ lie within a polygon. (Choose best suitable answer)

- ► Edge
- ► Vertices
- ► Pixels Page no : 80
- ► None of the given

53) The actual filling process in boundary filling algorithm begins when a point _____ of the figure is selected.

- Outside the boundary
- Inside the boundary
- At boundary
- None of the given

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54) Weiler-Atherton Polygon Clipping technique modifies the vertex-processing procedures for window boundaries so that _____ polygons are displayed correctly.

- ► Convex
- ► Concave Page no : 245
- ► Complex
- ► None of the given

55) If a line connecting any two points within a polygon does not intersect any edge, then it will be a _____ polygon.

- ► Convex Page no : 78
- ► Concave
- ► Complex
- ► None of the given

56) _____ can be defined as a mapping of point $P(x, y, z)$ onto its image $P'(x', y', z')$ in the view plane which constitutes the display surface.

- ► Mapping plane
- ► Three Coordinate Planes
- ► View plane Repeated
- ► Projection

57) The reflected light wave turns out to be a / an _____ case since light is reflected at the same angle as the incident wave (when the surface is smooth and uniform, as we'll assume for now).

- ► Unknown
- ► Simple Page no: 291
- ► Complex
- ► Abnormal

58) Which of the following is NOT true about quaternions?

- They are made up of 4 numbers
- They should always be normalized to length 1
- They can be used to represent all affine transforms
- They can be used to define the rotation of an object

1) In class, we discussed three forms of shading for "Utah" graphics. Which was the first to use per vertex normals?

- Flat Shading
- Phong Shading

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- Gouraud Shading Page no : 240

59) Given any implicit equation, which of the following is true for all (x, y, z) that make the equation exactly zero?

- All those points are inside the object defined by the implicit equation
- All those points are on the surface of the object defined by the implicit equation Page no :205
- All those points are outside the object defined by the implicit equation
- You can't know anything without knowing what the implicit equation is

60) When solving ray-sphere intersections using the implicit equation for a sphere, you must solve the quadratic equation. Which of the following do you know if the $B^2 - 4AC$ (i.e. the part under the square root) is negative?

- The ray intersects the sphere at a negative t... discard this result
- The ray intersects the sphere at a positive t... continue to the solution
- The ray does not intersect the sphere... discard this result Page no : 265
- The ray begins inside the sphere... this is a special case

61) _____ sets the global idle call back to be 'func' so a GLUT program can perform background processing tasks or continuous animation when window system events are not being received.

- glutIdleFunc
- glutMainLoop
- glutDisplayFunc
- glutReshapeFunc

62) A space curve can be confined to a plane.

- True
- False Page no : 326

63) A tangent vector certainly defines the slope at one end of the curve, but a vector has characteristics of.....

- direction
- magnitude
- both direction and magnitude Page no : 331
- None of the given

64)

We allow the parametric variable to take on values only in the interval -----

-.
-

- $-1 \leq u \leq 0$
- $0 \leq u \leq 2$

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• $0 \leq u \leq 1$ Page no : 321

• $-1 \leq u \leq 1$

65) The degree of a Bezier curve is equal to $n-1$, where n is the number of control points

• Yes Page no : 334

• No

66) Bernstein polynomial functions are the basic functions of _____ curves.

• NURBS

• Bezier Page no : 337

• Both NURBS and Bezier

• None of the given

67)

A parametric curve is one whose defining equations are given in terms of a -----, common, independent variable called the parametric variable.

• Triple

• Double

• Single Page no : 320

• None of the given

68)

Bit mask to select a window with multisampling support. If multisampling is not available, a ----- window will automatically be chosen.

• Non-multisampling Page no : 305

• Multisampling

• Mono-multisampling

• Di-multisampling

69)

Bezier curve is tangent to the lines connecting _____.

• First two points

• Last two points

• First two points and last two points

• None of the given

70) OpenGL is well structured with an intuitive design and logical commands.

Efficient OpenGL routines typically result in applications with fewer lines of code than those that make up programs generated using other graphics libraries or packages. In addition, OpenGL drivers ----- information about the underlying hardware, freeing the application developer from having to design for specific hardware features.

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- Encapsulate Page no : 297

- Shows
- Hibernates
- None of the given

71)

A space curve is not confined to a plane. It is free to twist through space. To define a space curve we must use parametric functions that are -----.

- Binary polynomials
- Mono polynomials
- Quadratic polynomials
- Cubic polynomials Page no : 326

72) End points and an intermediate point on the curve, then we now -----
-- quantities that we can express in terms of these coefficients (3 points x 3 coordinates each), and we can use these three points to define a unique curve.

- Six
- Three
- Two
- Nine Page no : 321

73) Bezier curve can represent the more complex piecewise _____ curve.

- Polynomial Page no : 33
- Exponential
- Cubic
- None of above

74) Curve and surface equations can be expressed in either a parametric or a non parametric form.

- True
- False Page no : 333

75) Using a lighting model based upon the Blinn Phong model means that we'll always get a uniform specular highlight based upon the colour of the ----- light and material, which means that all reflections based on this model, will be reminiscent of plastic.

- Union
- Refracting Page no : 291
- Intersection
- Reflecting

76) If the current matrix (according to glMatrixMode) is multiplied by the translation matrix, with the product replacing the current matrix. That is, if M is the

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current matrix and T is the translation matrix, then M is replaced with -----
--.

- M-T
- Page 21 of 22
- M+T
- M/T
- **M*T**

77) With similar expressions for $y(u)$ and $z(u)$. Again the a, b, c and d terms are constant coefficients. As we did with Equation for a plane curve, we combine the $x(u)$, $y(u)$, and $z(u)$ expressions into a single vector equation $P(u) = \text{-----}$.

- $Au^2+bu+cu+d$
- $Au^4+bu^3+cu^2+d$
- Au^3+bu^2+cu+d
- **Au^3+bu^2+cu+d Page no : 326**

78) The matrix generated by `gluPerspective` is multiplied by the current matrix, just as if `glMultMatrix` were called with the generated matrix. To load the perspective matrix onto the current matrix stack instead, precede the call to `gluPerspective` with a call to -----.

- `glRotated`
- `gluPerspective`
- `glTranslated`
- **`glLoadIdentity` Page no : 313**

79) Each number that makes up a matrix is called an _____ of the matrix.

- **Element Page no : 101**
- Variable
- Value
- Component

80) Which one of the following step is not involved to write pixel using video BIOS services.

- Setting desired video mode
- Using BIOS service to set color of a screen pixel
- Calling BIOS interrupt to execute the process of writing pixel.
- **Using OpenGL service to set color of a screen pixel**

81) Shadow mask methods can display a _____ range of colors.

- Small
- **Wide Page no : 29**
- Random

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- Crazy

82) Using Cohen-Sutherland line clipping, it is impossible for a vertex to be labeled 1111.

- True

- False

83) Intensity of the electron beam is controlled by setting _____ levels on the control grid, a metal cylinder that fits over the cathode.

- Amplitude
- Current
- Voltage Page no : 26
- Electron

84) Which of the following is NOT a modern application for Computer Graphics---

- ► Stop-motion animation (Page 6)
- ► Computer Aided Geometric Design
- ► Video Games
- ► Scientific Visualization

85) Both Boundary Filling and Flood filling algorithms are non-recursive techniques,

- ► False
- ► True

86) TV series are made as simply as possible from the animation point of view. This approach is generally known as -----.

- ► Full animation
- ► Limited animation (Page 428)
- ► Low animation
- ► High resolution

87) An eight frame run cycle that ----- frame/frames to each step gives a fast and vigorous dash. At this speed the successive leg positions are quite widely separated and may need dry brush or speed lines to make the movement flow.

- ► Two
- ► One
- ► Three
- ► Four (Page 437)

88) ----- reflection is the effect of reflecting light toward the direction from which it came, no matter the orientation of the surface.

- ► Forward scattering
- ► Diffuse Lambertian

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- ► Backscattering
- ► Retro (Page 293)

89) What makes this really challenging to model is that the index of refraction for most materials is a function of the----- of the light. This means that not only is there a shift in the angle of refraction, but that the shift is different for differing -----of light.

- ► Reflecting angle, Reflecting angle
- ► Refracting angle, Refracting angle
- ► Frequency, Frequency
- ► Wavelength, Wavelength (Page 229)

90) The reflected light wave turns out to be a -----case since light is reflected at the same angle as the incident wave (when the surface is smooth and uniform, as we'll assume for now).

- ► Abnormal
- ► Complex
- ► Simple (Page 296)
- ► Unknown

91) Tessellation can be adaptive to the _____ degree of curvature of a surface.

- ► Local (Page 170)
- ► Static
- ► Global
- ► Variable

92) _____ sets the reshape callback for the *current window*. The reshape callback is triggered when a window is reshaped.

- ► glutMainLoop
- ► glutIdleFunc
- ► glutReshapeFunc (Page 312)
- ► glutDisplayFunc

93) Signed integer color components, when specified, are linearly mapped to floating-point values such that the most positive representable value maps to 1.0, and the most negative representable value maps to ----- . Floating-point values are mapped directly.

- ► -1.0
- ► 0.0
- ► 2.0
- ► 1.0 (Page 320)

94) NURBS stands for-----.

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- ► Non Universal Rational Binary Spline
- ► Non Uniform Rational Binary Splines
- ► Non Uniform Rational Beta Splines (Page 325)
- ► Non Universal Rational Beta Splines

95) An orthogonal set of vectors-----

- ► Must be a set of linearly independent vectors
- ► Must be a set of linearly dependent vectors
- ► Must be made up of the basis vectors (e_1 , e_2 , and e_3) click here 4 details
- ► Can be made up of any set of vectors

96) Bezier curve is numerically the ----- of all the polynomial-based curves used in these applications.

- ► None of the given
- ► Most stable (Page 338)
- ► Less stable
- ► Most unstable

97) Bezier curve is the ideal standard for representing the ----- piecewise polynomial curves.

- ► None of the given
- ► Non complex
- ► Most complex
- ► More complex (Page 338)

98) Keep polygon orientations consistent to make sure that when viewed from the outside, all the polygons on the surface are oriented in the same direction.

- ► None of the given
- ► Neither
- ► Different
- ► Same (page 345)

99) The ----- is most simple example that exhibits the property self-similarity.

- ► Mosse
- ► Fern (Page 355)
- ► None of the given
- ► Thohar

100) A common mistake people make when creating three-dimensional graphics is to start thinking too soon that the final image appears on a flat, two-dimensional screen. Avoid thinking about which pixels need to be drawn, and instead try to visualize ----- space.

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- ► Multi-dimensional
- ► One-dimensional
- ► Two-dimensional
- ► Three-dimensional (Page 371)

101) Which of the following properties of rational Bezier curves fails if the weight assigned to a control point is negative?

- ► End-point interpolation
- ► Variation Diminishing
- ► Symmetry
- ► Convex-Hull

102) We want our scene to look more realistic, we should use _____ lights.

- Ambient (Page 282)
- Point Parallel
- Spot
- None of the given

103) This is a simple example of line clipping: the display window is the canvas and also the default -----, thus all line segments inside the canvas are drawn.

- Clipping Rectangle (Page 141)
- Clipping Circle
- Clipping Polygon
- Clipping Angle

104) One problem with Gouraud shading is that the ----- intensities can never be greater than the intensities at the edges.

- Triangles (Page 246)
- Squares
- Rectangles
- Polygons

105) There is more penetration of light in case of _____ surfaces.

- Conductor (like metals)
- Nonconductor (like dielectrics) (Page 235)
- Both conductor and nonconductor
- None of the given

106) _____ lights should be avoided because they are not for real time environment.

- Point
- Parallel

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- **Spot (Page 244)**

- None of the given

107) The physical range of colors a device can display is called _____ of the device.

- Sharpness

- **Gamut (Page 229)**

- Colouring
- Colouring with Sharpness

108) _____ is simply the calculation of color reflected by the surface.

- **Shading (Page 240)**

- Clamping
- Scaling
- None of the given

109) When obtaining normals for a triangle, which of the following mathematical constructs is NOT used?

- Vector normalization
- Vector cross products

- **Vector dot products**

- Point-Point subtraction

110) Loosely, the alpha component of the RGBA quad represents the _____ of a surface.

- **Opaqueness (Page 227)**

- Light
- Darkness
- Shine

111) An algorithm that clips a polygon must deal with many ----- cases. The case is particularly noteworthy in that the concave polygon is clipped into ----- isolate polygons.

- Similar, three

- **Different, two (Page 146)**

- Different, three
- Similar, two

112) _____ lighting is not dependent on any source.

- **Ambient**

- Diffuse
- Specular
- Emissive

113)

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In order to get a more realistic representation of lighting, we'll need to understand how light passes through a medium and how hitting the boundary layer at the ----- of two media can affect light's properties.

- Intersection (Page 296)

- Union
- Endpoints
- Edges

114) Lambertian shading was used mostly back when computers weren't fast enough to do _____ in real time.

- Phong shading
- Processing
- Shading
- Gouraud shading (Page 245)

115)

In Perspective Projection the point of View (POV) must lie on the _____.

- All axis
- Z axis (Page 200)
- X axis
- Y axis

116) If we want any object to glow, we should use _____ lights.

- Ambient
- Diffuse
- Specular
- Emissive (Page 240)

117) There are not many different ways of representing the intensity of a particular color element.

- True
- False (Page 276)

118) In Perspective Projection the screen plane must be parallel to the _____.

- Y-Z plane
- X-Y plane (Page 200)
- Z-Y plane
- X-Z plane

119) _____ light is reflected in all directions from surface.

- Ambient
- Diffuse (page 239)

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- Specular
- Emissive

120) A space curve can be confined to a plane.

- Yes
- No (Page 331)

121) To convert the information in the A matrix into that required for the P matrix, we do some simple matrix algebra, First we have $UA=UNP$ then Simply $A = \text{-----}$

- UP
- NP (Page 333)
- UN
- None

122) Perspective projection is specified with the function `glFrustum()`.

- Yes (Page 376)
- No

122) Choose a camera lens or adjust the zoom

- projection transformation (Page 372)
- viewport transformation
- modeling transformation'
- viewing transformation

123) Using a lighting model based upon the Blind Phong model means that we'll always get a uniform specular highlight based upon the color of the ----- light and material, which means that all reflections based on this model, will be reminiscent of plastic.

- Union
- Refracting
- Intersection
- Reflecting (Page 296)

124) Refractive index is a function of temperature, mostly due to density changes in materials with changes in temperature.

- True (Page 300)
- False

125)

Length L depends on the angle alpha and the z coordinate of the point to be projected and L can be represented by -----.

- $z * 1 / \tan(\alpha)$ (Page 198)
- $z * L^2$
- $z * 1 / \tan(\beta)$

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- $z * 1 / \tan(\gamma)$

126) The traditional approach in real-time computer graphics has been to calculate lighting at a vertex as a sum of the _____ light.

- Ambient
- Ambient, diffuse, and specular (Page 281)
- Specular
- Diffuse, and specular

127)

Another way to define a space curve by using intermediate points and the tangents at each end for making the curve

- Yes
- No (Page 334)

128) An independent consortium, the OpenGL Architecture Review Board, guides the OpenGL specification. With broad industry support, OpenGL is the only truly open, vendor-neutral, ----- graphics standard.

- Tertiary
- Binary
- Single platform
- Multiplatform (Page 301)

129) `glutReshapeWindow` requests a change in the size of the current window. The width and height parameters are size extents in pixels. The width and height must be ----- values.

- Neutral
- Negative
- Positive (Page 311)
- None of the given

130) A space curve is not confined to a plane. It is free to twist through space. To define a space curve we must use parametric functions that are -----.

- Binary polynomials
- Mono polynomials
- Quadratic polynomials
- Cubic polynomials (Page 331)

131)

Refractive index is a function of temperature, mostly due to changes in ----- of materials with changes in temperature. A simple correction can be applied in most circumstances to allow us to use a value given at one temperature at another.

- Density (Page 300)

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- pressure
- nature
- volume

132) If we assign a different value to the parametric variable for the intermediate point, then we obtain different values for the coefficients. This, in turn, means that a different curve is produced, although it passes through the----- three points.

- isolate
- different
- same (Page 328)
- none

133) The attenuation formula is $f = \frac{1}{C + Ld + Qd^2}$, where C, L and Q are the constant, linear and quadratic attenuation factors and d is the distance between the vertex being lit and the light source.

- $1/(C + Ld + Qd^2)$
- $1/(C + Ld + Qd)$
- $1/(C + L + d + Qd^2)$
- $1/(Cd + Ld + Qd^2)$

144) Bezier curve is tangent to the lines connecting _____.

- First two points
- Last two points
- First two points and last two point (Page 340)
- None of the given

145) End points and an intermediate point on the curve, then we now ----- quantities that we can express in terms of these coefficients (3 points x 3 coordinates each), and we can use these three points to define a unique curve.

- Six
- Three
- Two
- Nine (Page 326)

146) Choose a camera lens or adjust the zoom

- projection transformation (Page 372)
- viewport transformation
- modeling transformation
- viewing transformation

148) _____ OpenGL function is used for aiming and positioning the camera towards the object

- `glLoadIdentity()` (Page 375)

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- gluLookAt()
- glFrustum()
- None of Above

149) A parametric curve is one whose defining equations are given in terms of a ---
-----, common, independent variable called the parametric variable.

- Triple
- Double
- Single (Page 325)
- None of the given

150) The reflection coefficients are in the ----- range and are specified as part of the material property. However, they are strictly empirical and since they simply adjust the overall intensity of the material color, the material color values are usually adjusted so the color intensity varies rather than using a reflection coefficient.

- [0, 10]
- [0, 1] (Page 281)
- [0, 5]
- [0, 2]

151) To ensure a smooth transition from one section of a piecewise _____ to the next, we can impose various continuity conditions at the connection points on parametric curve parametric curve

- polygon vector (not confirm) (Page 245)
- None of the these

152) The curve is always contained within the _____ of the control points

- Tangents
- Convex Hull (Page 340)
- Subdivision
- None of Above

Question # 1

A space curve can be confined to a plane.

- Yes
- No (Page 331)

Question # 2

To convert the information in the A matrix into that required for the P matrix, we do some simple matrix

algebra, First we have $UA=UNP$ then Simply $A =$ -----

- UP
- NP (Page 333)

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- UN
- None
- 16

Question #3

Perspective projection is specified with the function glFrustum().

- Yes (Page 376)
- No

Question # 4

Choose a camera lens or adjust the zoom

- projection transformation (Page 372)
- viewport transformation
- modeling transformation'
- viewing transformation

Question # 5

Using a lighting model based upon the Blinn Phong model means that we'll always get a uniform specular highlight based upon the color of the ----- light and material, which means that all reflections based on this model, will be reminiscent of plastic.

- Union
- Refracting
- Intersection
- Reflecting (Page 296)

Question # 6

Refractive index is a function of temperature, mostly due to density changes in materials with changes in temperature.

- True (Page 300)
- False

Question # 7

Length L depends on the angle alpha and the z coordinate of the point to be projected and L can be represented by -----.

- $z * 1 / \tan(\alpha)$ (Page 198)
- $z * L^2$
- $z * 1 / \tan(\beta)$
- $z * 1 / \tan(\gamma)$
- 17

Question # 8

The traditional approach in real-time computer graphics has been to calculate lighting at a vertex as a sum of the _____ light.

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- Ambient
- Ambient, diffuse, and specular (Page 281)
- Specular
- Diffuse, and specular

Question # 9

Another way to define a space curve by using intermediate points and the tangents at each end for making the curve

- Yes
- No (Page 334)

Question # 10

An independent consortium, the OpenGL Architecture Review Board, guides the OpenGL specification. With broad industry support, OpenGL is the only truly open, vendor-neutral, ----- graphics standard.

- Tertiary
- Binary
- Single platform
- Multiplatform (Page 301)

Question # 11

glutReshapeWindow requests a change in the size of the current window. The width and height parameters are size extents in pixels. The width and height must be ----- values.

- Neutral
- Negative
- Positive (Page 311)
- None of the given

Question # 12

A space curve is not confined to a plane. It is free to twist through space. To define a space curve we must use parametric functions that are -----.

- Binary polynomials
- Mono polynomials
- Quadratic polynomials
- Cubic polynomials (Page 331)

Question # 13

Refractive index is a function of temperature, mostly due to changes in ----- of materials with changes in temperature. A simple correction can be applied in most circumstances to allow us to use a value given at one temperature at another.

- Density (Page 300)

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- pressure
- nature
- volume

Question # 14

If we assign a different value to the parametric variable for the intermediate point, then we obtain different values for the coefficients. This, in turn, means that a different curve is produced, although it passes through the ----- three points.

- isolate
- different
- same (Page 328)
- none

Question #15

Bezier curve is tangent to the lines connecting _____.

- First two points
- Last two points
- First two points and last two point (Page 340)
- None of the given

Question # 16

End points and an intermediate point on the curve, then we now ----- quantities that we can express in terms of these coefficients (3 points x 3 coordinates each), and we can use these three points to define a unique curve.

- Six
- Three
- Two
- Nine (Page 326)
- 19

Question # 17

Choose a camera lens or adjust the zoom

- projection transformation (Page 372)
- viewport transformation
- modeling transformation
- viewing transformation

QuestionZ#18

_____ OpenGL function is used for aiming and positioning the camera towards the object

- glLoadIdentity() (Page 375)

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- gluLookAt()
- glFrustum()
- None of Above

Question # 19

A parametric curve is one whose defining equations are given in terms of a -----, common, independent variable called the parametric variable.

- Triple
- Double
- Single (Page 325)
- None of the given

Question # 20

The reflection coefficients are in the ----- range and are specified as part of the material property. However, they are strictly empirical and since they simply adjust the overall intensity of the material color, the material color values are usually adjusted so the color intensity varies rather than using a reflection coefficient.

- [0, 10]
- [0, 1] (Page 281)
- [0, 5]
- [0, 2]

Question # 21

To ensure a smooth transition from one section of a piecewise _____ to the next, we can impose various continuity conditions at the connection points

- non parametric curve
- parametric curve
- polygon vector (not confirm) (Page 245)
- None of the these

Question # 22

The curve is always contained within the _____ of the control points

- Tangents
- Convex Hull (Page 340)
- Subdivision
- None of Above

Question # 23

Projection can be defined as a mapping of point $P(x, y, z)$ onto its image $P'(x', y', z')$ in the -----, which constitutes the display surface. The mapping is determined by a projection line called the projector that passes through P and intersects the -----.

- Two Coordinate Planes

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- View plane or projection plan (Page 193)

- Three Coordinate Planes
- Mapping plane

Question # 24

Determine how large we want the final photograph to be - for example, we might want it enlarged

- projection transformation
- viewport transformation (Page 372)
- modeling transformation
- viewing transformation

Question # 25

Ambient light is the light that comes from -----directions, thus all surfaces are illuminated equally regardless of orientation. However, this is a big hack in traditional lighting calculations since "real" ambient light really comes from the light reflected from the "environment."

- All (Page 281)
- Opposite
- Same
- Four different

Question # 26

Silhouette edges occur when dot product of surface normal vector and the view vector is _____.

- Zero (Page 345)
- One
- Both zero and one

Question # 27

If the current matrix (according to `glMatrixMode`) is multiplied by the translation matrix, with the product replacing the current matrix. That is, if M is the current matrix and T is the translation matrix, then M is replaced with -----.

- $M-T$
- $M+T$
- M/T
- $M*T$ (Page 317)

Question # 28

Arrange the scene to be photographed into the desired composition

- projection transformation
- viewport transformation
- modeling transformation (Page 317)

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- viewing transformation

Question # 29

In the forms of texture mapping, Image to world space and world space to image, each suffers from different problems related to magnifications and magnification. Which of the two shows the following problem: When the texture is larger than the screen space it maps to, many texture units (texels) are never sampled?

- Image to world space
- World space to image
- X-axis
- Y-axis

Question # 31

Imagine a curve in three-dimensional space, each point on the curve has a unique set of coordinates: a specific x value, y value, and z value. Each coordinate is controlled by a ----- parametric equation.

- Opposite
- Similar
- Separate (Page 325)

Question # 32

We allow the parametric variable to take on values only in the interval -----.

- $-1 \leq u \leq 0$
- $0 \leq u \leq 2$
- $0 \leq u \leq 1$ (Page 326)
- $-1 \leq u \leq 1$

Question # 33

Bezier curve can represent the more complex piecewise ----- curve.

- Polynomial (Page 338)
- Exponential
- Cubic
- None of above

Question # 34

A fractal is generally a property called -----.

- Fractal Dimension
- Self-similarity (Page 355)
- Koch Curve
- None of above

Question # 35

Normalized cross product of two vectors on that surface provides normal vector

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- Yes (Page 347)

- No

Question # 36

Every point on a curve has a straight line associated with it called the _____
State line

- tangent line (Page 334)

- curved line
- None of the given

Question # 36

The value returned is a unique small integer identifier for the window. The range of allocated identifiers starts at ----- . This window identifier can be used when calling glutSetWindow.

- Three
- Two
- One (Page 308)
- Zero

Question # 37

Curve and surface equations can be expressed in either a parametric or a non parametric form.

- True
- False

Question # 38

Bernstein polynomial functions are the basic functions of _____ curves.

- NURBS
- Bezier (Page 342)
- Both NURBS and Bazier
- None of the given

Question # 39

Geometric patterns that is repeated at ever smaller scales to produce irregular shapes and surfaces are called _____

- Geometric patterns
- Fractals (Page 352)
- Animated components
- Segments

Question # 40

In order to get a more realistic representation of lighting, we'll need to understand how light passes through a medium and how hitting the boundary layer at the -----
----- of two media can affect light's properties.

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- Intersection (Page 296)

- Union
- Endpoints
- Edges

Question # 41

_____ sets the global idle call back to be 'func' so a GLUT program can perform backgroundprocessing tasks or continuous animation when window system events are not being received.

- glutIdleFunc (Page 313)

- glutMainLoop
- glutDisplayFunc
- glutReshapeFunc

Question # 42

A tangent vector certainly defines the slope at one end of the curve, but a vector has characteristics of.....

- direction
- magnitude
- both direction and magnitude (Page 336)
- None of the given

Question # 43

The degree of a Bezier curve is equal to $n-1$, where n is the number of control points

- Yes (Page 339)
- No

Question # 44

Bit mask to select a window with multisampling support. If multisampling is not available, a -----window will automatically be chosen.

- Non-multisampling (Page 310)

- Multisampling
- Mono-multisampling
- Di-multisampling

Question # 45

OpenGL is well structured with an intuitive design and logical commands. Efficient OpenGL routines typically result in applications with fewer lines of code than those that make up programs generated using other graphics libraries or packages. In addition, OpenGL drivers ----- information about the underlying hardware, freeing the application developer from having to design for specific hardware features.

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- Encapsulate (Page 302)

- Shows
- Hibernates
- None of the given

Question # 46

With similar expressions for $y(u)$ and $z(u)$. Again the a , b , c and d terms are constant coefficients. As we did with Equation for a plane curve, we combine the $x(u)$, $y(u)$, and $z(u)$ expressions into a single vector equation

$P(u) = \text{-----}$.

- $Au^2 + bu + cu + d$
- $Au^4 + bu^3 + cu^2 + d$
- $Au^3 + bu^2 + cu + d$
- $Au^3 + bu^2 + cu + d$ (Page 331)

Question # 48

The matrix generated by `gluPerspective` is multiplied by the current matrix, just as if `glMultMatrix` were called with the generated matrix. To load the perspective matrix onto the current matrix, stack instead, precede the call to `gluPerspective` with a call to -----.

- `glRotated`
- `gluPerspective` (Page 318)
- `glTranslated`
- `glLoadIdentity`

Question # 49

The basic functions $f_i(u)$ in Bezier curve must be symmetric with respect to u and $(u-2)$

- yes
- no (Page 341)

Question # 50

Arrange the scene to be photographed into the desired composition

- projection transformation
- viewport transformation
- modeling transformation (Page 372)
- viewing transformation

Question No: 51

NURBS stands for-----.

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- Non Universal Rational Binary Spline
- Non Uniform Rational Binary Splines
- **Non Uniform Rational Beta Splines (Page 325)**
- Non Universal Rational Beta Spline
- Question No: 1
- Which of the following is NOT a modern application for Computer Graphics-----

•

- **► Stop-motion animation (Page 6)**

- ► Computer Aided Geometric Design
- ► Video Games
- ► Scientific Visualization

- Question No: 52
- Both Boundary Filling and Flood filling algorithms are non-recursive techniques,

- **► False**

- ► True

Question No: 53

TV series are made as simply as possible from the animation point of view. This approach is generally known as -----.

- ► Full animation
- **► Limited animation (Page 431)**
- ► Low animation
- ► High resolution

Question No: 54

An eight frame run cycle that ----- frame/frames to each step gives a fast and vigorous dash. At this speed the successive leg positions are quite widely separated and may need dry brush or speed lines to make the movement flow.

- Two
- One

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► Three

► Four (Page 440)

Question No: 55

----- Reflection is the effect of reflecting light toward the direction from which it came, no matter the orientation of the surface.

- Forward scattering
- Diffuse Lambertian
- Backscattering
- Retro (Page 296)

Question No: 56

What makes this really challenging to model is that the index of refraction for most materials is a function of the----- of the light. This means that not only is there a shift in the angle of refraction, but that the shift is different for differing -----of light.

- Reflecting angle, Reflecting angle
- Refracting angle, Refracting angle
- Frequency, Frequency
- Wavelength, Wavelength (Page 301)

Question No: 57

The reflected light wave turns out to be a -----case since light is reflected at the same angle as the incident wave (when the surface is smooth and uniform, as we'll assume for now).

- Abnormal
- Complex
- Simple (Page 299)
- Unknown

Question No: 58

Tessellation can be adaptive to the _____ degree of curvature of a surface.

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▶ Local (Page 173)

- ▶ Static
- ▶ Global
- ▶ Variable

Question No: 59

_____ sets the reshape callback for the *current window*. The reshape callback is triggered when a window is reshaped.

- ▶ glutMainLoop
- ▶ glutIdleFunc
- ▶ glutReshapeFunc (Page 315)
- ▶ glutDisplayFunc

Question No: 60

Signed integer color components, when specified, are linearly mapped to floating-point values such that the most positive representable value maps to 1.0, and the most negative representable value maps to ----- . Floating-point values are mapped directly.

- ▶ -1.0
- ▶ 0.0
- ▶ 2.0
- ▶ 1.0 (Page 323)

Question No: 61

NURBS stands for-----.

- ▶ Non Universal Rational Binary Spline
- ▶ Non Uniform Rational Binary Splines
- ▶ Non Uniform Rational Beta Splines (Page 328)
- ▶ Non Universal Rational Beta Splines

Question No: 62

An orthogonal set of vectors-----

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- ▶ Must be a set of linearly independent vectors
- ▶ Must be a set of linearly dependent vectors
- ▶ **Must be made up of the basis vectors (page no 254)**
- ▶ Can be made up of any set of vectors

Question No: 63

Bezier curve is numerically the ----- of all the polynomial-based curves used in these applications.

- ▶ None of the given
- ▶ **Most stable (Page 341)**
- ▶ Less stable
- ▶ Most unstable

Question No: 64

Bezier curve is the ideal standard for representing the ----- piecewise polynomial curves.

- ▶ None of the given
- ▶ Non complex
- ▶ Most complex
- ▶ **More complex(Page 341)**

Question No: 65

- Keep polygon orientations consistent to make sure that when viewed from the outside, all the polygons on the surface are oriented in the same direction.
 - ▶ None of the given
 - ▶ Neither
 - ▶ Different
 - ▶ **Same (page 347)**

Question No: 66

The ----- is simplest example that exhibits the property self-similarity.

- ▶ Moses

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▶ Fern (Page 358)

- ▶ None of the given
- ▶ Thohar

Question No: 66

A common mistake people make when creating three-dimensional graphics is to start thinking too soon that the final image appears on a flat, two-dimensional screen. Avoid thinking about which pixels need to be drawn, and instead try to visualize ----- space.

- ▶ Multi-dimensional
- ▶ One-dimensional
- ▶ Two-dimensional
- ▶ Three-dimensional (Page 374)

Question No: 68

Which of the following properties of rational Bezier curves fails if the weight assigned to a control point is negative?

- ▶ End-point interpolation
- ▶ Variation Diminishing
- ▶ Symmetry
- ▶ Convex-Hull

Question No: 69

Actually, each component is a rational Bézier curve. We have made it very clear that all weights must be non-negative. If some of them are negative, the strong convex hull property or even the convex hull property will not hold.

Question No: 70

In the Phone reflection model, there are 3 constants (a, b, c) which are used to describe the qualities of which of the following phenomena?

- ▶ The attenuation of a point light source with distance
- ▶ The size (in each dimension) which the light is assumed to have
- ▶ The amount to perturb reflection vectors as they are mirrored across

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the normal

- ▶ The material reaction to ambient, diffuse and specular light (respectively)

Question No: 71

The Phong reflection model simplifies light-matter interactions into (essentially) 4 vectors and a number of constants. Which piece of the Phong model is responsible for giving spheres their bright white spots?

- ▶ Specular
- ▶ Diffuse
- ▶ Ambient

Question No: 72

When you hit a surface in ray tracing, generally shadow rays are tested against all objects in a scene. If these rays come back saying they hit an object in the scene, which of the following do you do?

- ▶ add all components (i.e. ambient, diffuse and specular) from that light source to the object.
- ▶ add all EXCEPT the ambient light from that light source to the object (i.e. diffuse and specular)
- ▶ add only the ambient light from that light source to the object
- ▶ add none of the light from that light source to the object

Question No: 73

The Color Space tool is a handy tool that we can use to interactively add two colours together to see the effects of the various strategies for handling oversaturated colours.

- ▶ False
- ▶ True (Page 238)

Question No: 74

A polygon is usually defined by a sequence of ----- and Edges.

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- ▶ Ending lines
- ▶ Points
- ▶ Vertices (Page 149)
- ▶ Edges

Question No: 75

Which of the following properties of Bezier curves guarantees that a line passes through the control polygon as many times or more times than the line passes through the Bezier curve itself?

- ▶ End-point interpolation
- ▶ Variation Diminishing
- ▶ Symmetry
- ▶ Convex-Hull

Question No: 76

Parity is a concept used to determine which _____ lie within a polygon.
(Choose best suitable answer)

- ▶ Edge
- ▶ Vertices
- ▶ Pixel (Page 80)
- ▶ None of the given

Question No: 77

The actual filling process in boundary filling algorithm begins when a point _____ of the figure is selected.

- ▶ Outside the boundary
- ▶ Inside the boundary (Page 105)
- ▶ At boundary
- ▶ None of the given

Question No: 78

Clipping technique modifies the vertex-processing procedures for window boundaries so that _____ polygons are displayed correctly.

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- ▶ Convex
- ▶ Concave (Page 158)
- ▶ Complex
- ▶ None of the given

Question No: 79

If a line connecting any two points within a polygon does not intersect any edge, then it will be a _____ polygon.

- ▶ Convex (Page 79)
- ▶ Concave
- ▶ Complex
- ▶ None of the given

Question No: 80

_____ can be defined as a mapping of point $P(x, y, z)$ onto its image $P'(x', y', z')$ in the view plane which constitutes the display surface.

- ▶ Mapping plane
- ▶ Three Coordinate Planes
- ▶ View plane
- ▶ Projection (Page 265)

Question No: 81

The reflected light wave turns out to be a / an _____ case since light is reflected at the same angle as the incident wave (when the surface is smooth and uniform, as we'll assume for now).

- ▶ Unknown
- ▶ Simple (Page 299)
- ▶ Complex
- ▶ Abnormal

Question No: 82

Computer graphics and computer vision are _____ disciplines.

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- **Related (Page 24)**

- Interrelated
- Same
- Different

Question No: 83

OpenGL has become the industry's most widely used and supported _____ graphics application programming interface (API), bringing thousands of applications to a wide variety of computer platforms.

- 2-Dimensional
- 3-Dimensional
- **2-Dimensional and 3-Dimensional (Page 301)**

Question No: 84

----- sets the global idle callback to be 'func' so a GLUT program can perform background processing tasks or continuous animation when window system events are not being received.

- **glutIdle function (Page 313)**
- glutKeyboardFunc
- glutReshapeFunc
- glutDisplayFunc

Question No: 85

$X^2/a^2 - y^2/b^2 = 1$ is an equation of

- **Hyperbola (Page 70)**
- Parabola ($4px = y^2$)
- None of given
- Ellipse ($X^2/a^2 + y^2/b^2 = 1$)

Question No: 86

To modify the object shape, shearing transformation cannot be used.

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- False (Page 195)

- True

Question No: 87

Rotation is performed around a fixed point called _____.

- Pivot point rotation (Page 122)

Question No: 88

Computer graphics is very helpful in producing graphical representations for scientific visualization and analysis

- True (Page 9)

- False

Question No: 89

Save a line with both endpoints inside all clipping boundaries.

- Trivial Reject
- Trivial Accept (Page 145)
- None of given

Question No: 90

Dark lights are nothing more than lights in which one or more of the color values are _____.

- Unknown
- Negative (Page 238)
- Positive
- Zero

Question No: 91

A series of _____ computer operations convert an object's three-dimensional coordinates to pixel positions on the screen. Transformations, which are represented by matrix multiplication, include modeling, viewing, and projection

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operations. Such operations include rotation, translation, scaling, reflecting, orthographic projection, and perspective projection.

- **Three (Page 374)**

- Two
- Four
- Ten

Question No: 92

At a physical surface, our eye's perception of the color depends on the distribution of photon energies that arrive and trigger our _____ cells.

- Eye
- Retina
- **Cone (Page 401)**

Question No: 93

- In the Phong Reflection model, _____ light is the same everywhere.
- **Ambient**
- Diffuse
- Specular
- Emissive

Question No: 94

This projection technique has the direction of projection perpendicular to the viewing plane, but the viewing direction is NOT perpendicular to one of the principle faces.

- **Orthographic Parallel Projection(Page 197)**
- Axonometric Parallel Projection
- Oblique Parallel Projection

Question No: 95

A plane is two dimensional since in order to uniquely define any point on its surface we require _____ numbers.

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- Two (Page 359)

- Three
- Four

Question No: 96

In perspective projection, for your view to come out correctly, you will also want the _____ to pass through the middle of the screen.

- X axis
- Y axis
- Z axis (Page 203)
- None

Question No: 97

Neither floating-point nor signed integer values are clamped to the range _____ before updating the current color.

- 0 , -1.0
- -1 , 1
- 1 , -1
- 0, 1 (Page 324)

Question No: 98

An object's _____ determine its orientation relative to the light sources. For each vertex, OpenGL uses the assigned normal to determine how much light that particular vertex receives from each light source.

- Unit
- Normal (Page 403)
- None of given

Question No: 99

Which was the oldest shading model?

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- Flat Shading (Page 249)

- Phong Shading
- Gouraud Shading

Question No: 100

Which of the following affine transforms does NOT affect vectors?

- Scale
- Rotation
- Shear
- Translation

Question No: 101

We want our scene to look more realistic, we should use _____ lights.

- Ambient (Page 285)
- Point Parallel
- Spot
- None of the given

Question No: 102

This is a simple example of line clipping: the display window is the canvas and also the default -----, thus all line segments inside the canvas are drawn.

- Clipping Rectangle (Page 144)
- Clipping Circle
- Clipping Polygon
- Clipping Angle

Question No: 103

One problem with Gouraud shading is that the ----- intensities can never be greater than the intensities at the edges.

- Triangles (Page 249)

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- Squares
- Rectangles
- Polygons

Question No: 104

There is more penetration of light in case of _____ surfaces.

- Conductor (like metals)
- Nonconductor (like dielectrics) (Page 243)
- Both conductor and nonconductor
- None of the given

Question No: 105

_____ Lights should be avoided because they are not for real time environment.

- Point
- Parallel
- Spot (Page 247)
- None of the given

Question No: 106

The physical range of colors a device can display is called _____ of the device.

- Sharpness
- Gamut (Page 232)
- Colouring
- Colouring with Sharpness

Question No: 107

_____ is simply the calculation of color reflected by the surface.

- Shading (Page 243)
- Clamping

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- Scaling
- None of the given

Question No: 108

When obtaining normal for a triangle, which of the following mathematical constructs is NOT used?

- Vector normalization
- Vector cross products
- Vector dot products
- Point-Point subtraction

Question No: 109

Loosely, the alpha component of the RGBA quad represents the _____ of a surface.

- Opaqueness (Page 239)
- Light
- Darkness
- Shine

Question No: 110

An algorithm that clips a polygon must deal with many ----- cases. The case is particularly note worthy in that the concave polygon is clipped into -----
- isolate polygons

- Similar, three
- Different, two (Page 146)
- Different, three
- Similar, two

Question No: 111

The circle and ellipse are symmetric across 8 octants.

- True

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- False (Page 60)

Question No: 112

UV light is used in Plasma Panel displays to excite phosphor.

- True
- False

Question No: 113

Which of the following is not true about matrices?

- $A + B = B + A$
- $a(A + B) = aA + aB$
- $(AT)T = AT$
- $A + (B + C) = (A + B) + C$

Question No: 114

According to Odd Parity Rule, a point is inside the polygon, if:

- Line from an outside point to this point does not cross the edges odd number of times
- Line from any point to this point crosses the edges odd number of times
- Line from an outside point to this point crosses the edges odd number of times (Page 80)
- Line from this point to any point outside the polygon intersects any edge

Question No: 115

As opposed to direct memory access method, BIOS routines provide an easier and faster method of drawing pixels on screen.

- True
- False (Page 47)

Question No: 116

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When a point $P(x,y)$ is rotated by θ the coordinates of transformed point P' are given as:

- $x' = x \cos(\theta) - y \sin(\theta), y' = x \sin(\theta) + y \cos(\theta)$ (Page 181)
- $x' = y \cos(\theta) - x \sin(\theta), y' = y \sin(\theta) + x \cos(\theta)$
- $x' = x \cos(\theta) + y \sin(\theta), y' = x \sin(\theta) - y \cos(\theta)$
- $x' = y \cos(\theta) + x \sin(\theta), y' = y \sin(\theta) - x \cos(\theta)$

Question No: 117

The amount of time it takes to illuminate a specific location on phosphor coated screen is called Persistence.

- True
- False (Page 27)

Question No: 118

Incremental line drawing algorithm makes use of the equation of straight line.

- True
- False (page 54)

Question No: 119

In matrix multiplication:

- The two matrices must be square
- The number of rows of 1st matrix must be the same as the number of columns of the second.
- The two matrices must either be row matrices or column matrices
- The number of columns of 1st matrix must be the same as the number of rows of the second.

Question No: 120

In Horizontal retrace, after completion of all the pixels in a scan line, the refreshing continues from the 1st pixel of the next scan line.

- True

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- False(Page 28)

Question No: 121

Parity Rule is used to determine whether a pixel is inside a polygon or not.

- True (Page 80)
- False

Question No: 122

When dot product of two vectors equals zero, this implies that the two vectors are:

- parallel to each other
- orthogonal (perpendicular) to each other. (Page 177)
- intersect each other
- equal to each other

Question No: 123

In Pixmap exactly one bit is used to hold color value of each pixel.

- True
- False (Page 28)

Question No: 124

The Boundary Fill and Flood Fill algorithms:

- Must use 4-connected approach
- Must use 8-connected approach
- May use 4-connected or 8-connected approach
- Must not use recursive approach

Question No: 125

To show 256 colors, the no of bits required for each pixel are

- 8 (Page 39)
- 16
- 32

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- 64

Question No: 126

25 * 80 resolution with 16 colors supports

- **Text mode (Page 43)**

- Graphics mode
- Both
- None

Question No: 127

Two matrices are said to be equal, if they have

- same order
- same corresponding elements
- **Same order and same corresponding elements.(page 11)**
- Different elements.

Question No: 128

Two points are said to be collinear, if they lie on the

- **same line (page 53)**
- different but parallel lines
- either on the same plane or two parallel planes
- different plane

Question No: 129

A Polygon is convex, if the line connecting:

- Any two points outside the polygon intersects its boundary
- **Any two points inside the polygon don't intersect any edges of the polygon.
(Page 171)**
- A point inside the boundary with any point outside, does not intersect the polygon boundary
- Any two vertices, intersects some edge of polygon.

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Question No: 130

Polygon is complex, if the line connecting:

- Any two points outside the polygon intersects its boundary.
- Any two points inside the polygon intersects its boundary.
- A point inside the boundary with any point outside does not intersect the polygon boundary.
- Any two vertices, intersects some edge of polygon. (Page 79)

Question No: 131

The equation of hyperbola centered at origin (if the transverse axis is along x -axis) can be given as:

- $x^2/b^2 - y^2/a^2 = 1$
- $x^2/b^2 + y^2/a^2 = 1$
- $x^2/a^2 - y^2/b^2 = 1$
- $x^2/b^2 - y^2/a^2 = 1$

Question No: 132

Which one is not valid out code to perform trivial accept / reject test in line clipping:

- 1101
- 1001 (Page 143)
- 0101
- 0110

Question No: 133

Which one of the following is not the graphics library is use:

- FastGL
- OpenGL
- DirectX
- EasyGL (Page 47)

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Question No: 134

In video text memory, _____ are used to display a character.

- 2 bytes (Page 43)
- 4 bytes
- 8 bytes
- 16 bytes

Question No: 135

In _____ algorithm(s), old color must be read before it is invoked.

- Scan line filling
- Flood filling (Page 104)
- Both scan line and flood filling
- Scan filling

Question No: 136

According to the architecture of raster graphics system, display processor memory will act as _____.

- Video controller (Page 36)
- System memory
- Frame buffer
- Video controller and System memory

Question No: 137

Global coordinate systems can be defined with respect to local coordinate system

- True
- False (Page 258)

Question No: 138

Cross product of two vectors results in a _____.

- Magnitude

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- Vector (Page 117) cross product of 2 vectors is a vector.

- Scalar
- Value

Question No: 139

Shortcoming of Sutherland-Hodgeman Algorithm is concave polygons may be displayed with extraneous lines

- True (Page 158)
- False

Question No: 140

In 2D transformations, rotations applied to a point P can be denoted as _____ (Where? Represents theta).

- $P' = R(\theta_1 + \theta_2) \cdot P$ (Page 127)
- $P' = (R(\theta_1) + R(\theta_2)) \cdot P$
- $P' = R(\theta_1 \times \theta_2) \cdot P$
- $P' = R(\theta_1) \cdot P$

Question No: 141

A straight line can be moved to another location by applying _____ to each of the line endpoints and redrawing the line between the new coordinates.

- Rotation
- Translation (Page 121)
- Reflection
- Scaling factor

Question No: 142

Locations can be translated or "transformed" from one coordinate system to the other.

- True (Page 166)
- False

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Question No: 143

If the value of scaling factors s_x and s_y is greater than 1, then size of objects will be _____.

- Reduced
- Enlarged (Page 121)
- Remain same
- Shear

Question No: 144

A column matrix is also known as _____.

- Column vector (Page 110)
- Row vector
- Vector
- Simple Matrix

Question No: 145

Dot product of two vectors results in _____ quantity.

- Scalar (Page 118)
- Vectors
- Magnitude
- Value

Question No: 146

Set up your tripod and pointing the camera at the scene

- projection transformation
- viewport transformation
- modeling transformation
- viewing transformation (Page 375)

Question No: 147

_____ is based on characteristic size or scale

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- Fractal Geometry
- Traditional Geometry
- Euclidean Geometry (Page 362)
- None of Above

Question No: 148

Bernstein polynomial functions are the basic functions of _____ curves.

- NURBS
- Bezier (Page 342)
- Both NURBS and Bazier
- None of the given

Question No: 149

Which of the following does NOT figure into the Field of View of a pinhole camera?

- The direction of projection
- The distance from the center of projection to the projection plane
- The size of the projection plane

Question No: 150

When transforming a random Axis-Aligned Bounding Box defined by the points (nearx, neary, nearz) and (farx, fary, farz) to the standard orthographic viewing box, which affine transforms are used?

- shear and translation
- rotation and scale
- scale and shear
- translation and scale

Question No: 150

In class, we discussed the purpose of the front and back clipping planes in OpenGL. Which of the following was NOT a purpose for using clipping planes?

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- division by zero
- objects behind the center of projection mapping onto the projection plane
- avoiding the problems of infinite viewing volume size

Question No: 152

In “Utah” graphics, lights are simplified in order to approximate light/matter interaction with a minimum amount of work. Which of the following is NOT true about the simplifications made in “Utah” graphics lights?

- Light intensity and color are folded into one value.
- Lights are assumed to have zero size
- Spotlights cannot be handled
- Soft shadows cannot be handled

Question No: 153

In the Phong Reflection model, ambient light is the same everywhere.

- true
- false

Question No: 154

We discussed several global illumination algorithms in class. Which of the following listed can properly handle diffuse-diffuse reflections

- Ray Tracing
- Radiosity
- Photon Mapping
- RenderMan

Question No: 155

We discussed several global illumination algorithms in class. Which of the following listed can properly handle caustics?

- Ray Tracing
- Radiosity

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- Photon Mapping
- RenderMan

Question No: 156

In radiosity a matrix is formed of size $n \times n$. Which of the following can be known if the (i, j) position in the matrix is zero?

- patch i is much larger than patch j
- patch i is much smaller than patch j
- patch i is visible from patch j
- patch i is not visible from patch j

Question No: 157

When solving for ray-polygon intersections, after intersecting the ray with a plane, the dominant component of the plane normal is found. this is used to

- ignore any component other than the dominant when you project to 2D
- ignore the dominant component when you project to 2D
- solve the inside-outside test only for that component

Question No: 158

If solving for inside-outside of a triangle using barycentric coordinates, and you have two barycentric coordinates solved ($bc1 = .57$, $bc2 = .62$) which of the following do you know?

- The point is outside the triangle
- The point is on the boundary of the triangle
- The point is inside the triangle
- The hit point is on the “back face” of the triangle

Question No: 159

The majority of the execution time of a ray tracer is spent in ray-object intersection code.

- true

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- false

Question No: 160

In the Pixar short “Geri’s Game”, the trees in the background were created using which of the following techniques?

- Fractals
- Bump mapping
- Environment mapping
- Catmull-Clark Subdivision Surfaces

Question No: 161

The basic functions $f_i(u)$ in Bezier curve must be symmetric with respect to u and $(u-2)$

- yes
- no (Page 344)

Question No: 162

In the Pixar short “Geri’s Game”, Geri’s glasses seemed to bend the light as it passed through. Which of the following techniques was used?

- Fractals
- Bump mapping
- Environment mapping
- Catmull-Clark Subdivision Surfaces

Question No: 163

A polygon is usually defined by a sequence of vertices and -----

- Edges (Page 149)
- Vertices
- Points Ending lines

Question No: 164

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Gouraud shading is the current de jure shading standard in accelerated ----- hardware.

- 2Dimensional
- 3Dimensional (Page 248)
- Multidimensional
- None

Question No: 165

_____ is based on characteristic size or scale

- Fractal Geometry (Fractal shapes are self similar and independent of size or scaling)
- Traditional Geometry
- Euclidean Geometry (Euclidean shapes normally have a few characteristic sizes or lengthscales) (Page 361)
- None of Above

Question No: 166

Arrange the scene to be photographed into the desired composition

- projection transformation
- viewport transformation
- modeling transformation (Page 375)
- viewing transformation

Question No: 167

Which language API defines graphics operations independent of the operating system or computer hardware? Additional hardware specific libraries are used to provide an interface between API and the hardware and between the user and the platform specific windowing system.

- DirectX
- Graphic Windowing Toolkit
- CGI

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- OpenGL (Page 305)

Question No: 168

When transforming a random Axis-Aligned Bounding Box defined by the points (nearx, neary, nearz) and (farx, fary, farz) to the standard orthographic viewing box, which affine transforms are used?

- shear and translation
- rotation and scale
- scale and shear
- Translation and scale

Question No: 169

Shadow mask methods can display a _____ range of colors.

- Small
- Wide (Page 28)
- Random
- Crazy

Question No: 170

Each number that makes up a matrix is called an _____ of the matrix.

- Element (Page 109)
- Variable
- Value
- Component